Payroll Taxes and Market Power

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Abstract

This paper provides comprehensive examination of a historically large quasi-experimental payroll tax cut that affected a small share of Brazilian firms due to plausibly exogenous legal variation. Difference-in-differences estimates report that while capital decreases, a payroll tax reduction causes an increase in employment, wages and profits. Responses are stronger for small firms. In terms of mechanisms, 65% of the employment effect arises from scaling plant size rather than input substitution. The study reveals that consumers pay 75% of payroll taxes, while firm owners and workers pay 25%. I develop and estimate a novel model of factor demand that quantifies the role of product and labor market power on the efficiency and incidence of taxation. Taxing market power rents reduces the efficiency costs of taxation in 36%, supporting proposals for relatively higher tax rates to larger firms. However, this comes with distributional implications: market power shifts the incidence away from workers towards firm owners and consumers.

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1 Introduction

Payroll taxes are responsible for 25% of total Government revenue (OECD 2019)[1]. Predicated on the basic idea that the most elastic side bears more of the tax, prior work has focused on comparing aggregate labor demand versus supply. Underlying this logic is the assumption that incidence is determined by market forces. Despite mounting evidence documenting firms' wage setting power (see Card et al. 2018 for a review), economists haven't incorporated yet the role of firms into the study of tax incidence. This paper asks what is the role of imperfect competition in the incidence of payroll taxation?

In this paper, I provide novel evidence on the incidence of payroll taxation by studying a quasi-experimental historically large labor cost reduction of 15 p.p. (figure 1), in the context of a payroll tax cut program in Brazil. Due to plausibly exogenous legal requirements, the policy affected only a small (even within local labor markets) share of firms.[2] This setting allows me to make fewer assumptions on general equilibrium effects, in order to estimate firm and worker level responses. I complement this appealing policy variation with a novel dataset that combines matched employer-employee and anonymized tax records on the universe of formal firms. The data provides a comprehensive laboratory of the Brazilian economy, allowing me to follow workers and firms before and after the tax change. The richness of the data and unique tax reform, allows me to provide comprehensive evaluation on the tax incidence among a holistic set of outcomes, and across a rich set of firms and workers' characteristics.

The main empirical strategy ultimately compares firms that received the tax cut with similar ones that didn't. Since eligible firms self-select themselves into treatment[3], I instrument the regressions using plausibly exogenous sector eligibility rules. Prior to the reform, firms in eligible and non-eligible sectors exhibit similar trends and levels in the labor market outcomes of interest. I find that the payroll tax cut causes a sharp expansion on firms' employment, with limited effects on earnings. The employment analysis is leveraged at the firm level to capture the effect of the tax reform on businesses. I find a 9% employment increase, which is mostly driven by firms with less market power[4]. This result is consistent

[1] Figure D.1 shows that in the US, payroll tax has received growing importance over the last decades, and now is the second most important Government funding source, accounting for more than 30% of total tax collection.

[2] Less than 1.5% of total firms (or 110,000 firms) at the peak of its implementation.

[3] Large share of eligible firms do not take-up this advantageous benefit.

[4] The result is invariant to multiple definitions of market power such as market share in the local labor market, or firm size.

with a broader literature that finds Government subsidies being more effective to boost employment on small business (Zwick and Mahon 2017; Criscuolo et al. 2019; Howell 2017; Bronzini and Iachini 2014). The setup of the Brazilian payroll tax reform is appropriate to connect with Industrial Policies because both of them are targeted at the firm, rather than worker level.

Given the underlying payroll tax variation induced by the reform, the implied elasticity of employment with respect to labor cost is -0.71. The large employment effect doesn't affect the between occupation sorting of workers, but exacerbates within firm earnings inequality. Figure 3 shows that whereas average earnings for workers at the bottom percentiles face no gains, there are substantial gains (of 14%) at the top of the distribution. To analyze the pass through to earnings and minimize the contamination from workers' turn over, I follow the displacement literature (Jacobson, LaLonde, and Sullivan 1993, Lachowska, Mas, and Woodbury 2020) to build a sample of stable incumbent workers, who are assigned to treatment based on their pre-reform employers. At the worker level, I find an average 1.5% increase in earnings, which is a sharp zero in the short run and a 3% significant effect in the long run. There is no significant difference across multiple workers' characteristics, such as tenure, gender and race. However, the difference is largely significative on workers' occupation. While workers in high skill occupations benefit from a 6% pass-through to earnings, low skill occupations experience a zero effect.

The identifying assumption is that conditional on fixed effects, eligibility is uncorrelated with time-varying unobserved determinants of employment and wage growth. I provide evidence that the identification assumption holds, and the estimates are robust to a wide variety of approaches. There are two main threats to identification. First, as in standard difference-in-differences, the design is compromised if parallel trends do not hold. This would be violated if the Government selects eligibility in a way that anticipates trends on the outcomes of interest. Second, the results would be biased if there were strategic selection into eligible sectors.

The formal and standard test for parallel trends is evaluating the statistical significance of pre-trends. I show not only that the pre-trends aren't statistically indistinguishable from zero in any of the outcomes, but also that eligibility is balanced in levels. Eligibility is not correlated with firms and workers characteristics in the pre-reform period, and results are robust to multiple estimation methods. As an alternative identifying strategy, I leverage a matching difference-in-differences to show that the results are qualitatively similar to the main empirical design when I match treated firms to never treated one, based on pre-reform characteristics. I also provide a more heuristic argument to highlight the arbitrary aspect of the eligibility criteria. Table 6, presents a non-exhaustive list of eligible versus non-eligible sectors that appear remarkably similar. For example, hotels are eligible, but motels are not.

The second threat is about strategic selection into eligible sectors. I show that results are robust to eligibility assignment in the pre-reform period. Also, as a robustness check, I restrict the sample to firms that have never changed sectors and results are similar. I noticed from this exercise that very few firms actually change sectors, which is consistent with the fact that in order to change sectors firms need to provide a long list of supporting documents to multiple agencies proving that their core activity has been indeed changed. I focus on the firms that have ever experienced a sector change, and show that there is not a trend of switching towards eligible sectors. All of these together is reassuring that the main results are not driven by firms self selecting into eligible sectors.

Third, in the context of developing economies, it is important to provide careful analysis on the role of informality (Ulyssea 2018; Haanwinckel and Soares 2021). One might suspect that the employment result is mechanically driven by formalization of existing informal workers, rather than an additional rise in employment caused by the reform. I perform multiple tests to show that this is not the case. First, I use the Brazilian regional diversity in terms of informality (measured by the national Census) to provide evidence that the employment expansion is not driven by highly informal areas. Second, I explore transition patterns from formal employment to non-employment/ informality to arrive at the same conclusion. The non-informality driven result is consistent with survey evidence[5] showing that most of the Brazilian informality is concentrated in self employment[6]. Along the same line, there is empirical evidence in Brazil and other developing countries that informal labor markets are segregated[7], and formalization decisions go beyond a simple cost-benefit analysis dictated by the labor cost (Alcaraz, Chiquiar, and Salcedo 2015; Dalberto and Cirino 2018).

Beyond the employment and wage effects emphasized by prior work, the comprehensiveness of the data and rare policy variation allows me to investi-

[5]Pesquisa Nacional por Amostra de Domicílios (PNAD) is a household survey administered by the Brazilian Census Bureau (IBGE).

[6]Compared to informal employees, the incentives for formalization in the self-employment case is less dependent on payroll taxes, and more considerative of other variables such as: licenses to operate, costs relative to opening and maintaining a firm, other payroll taxes, legal liabilities, sanitary and security regulations.

[7]For a comprehensive analysis on the underlying forces of informality, refer to Perry 2007.

gate other margins of adjustment, in order to provide a complete and novel assessment of the payroll tax incidence. Consistent with an optimal behavior of substituting to cheaper inputs, I find that capital usage decreases by 3%. Profits increase by 25%, and as expected in a setting with imperfect competition in the output market, revenues decrease by 2%. All this together with the previous labor market effect, accounts for an overall incidence of 45% borne by workers, 22% by firm owners, 33% by consumers, and 62% efficiency gain from eliminating the deadweight loss associated with the payroll tax.

The positive worker's earnings effect after a firm level shock provides evidence that there is labor market power (refer to figure 7 for intuition). However, there are multiple theoretical frameworks consistent with imperfect labor competition that would offer divergent explanations for the underlying mechanisms. Empirically, the wage pass-through for incumbent workers and new hires is quantitatively similar, which is inconsistent with standard bargaining models (Hall and Milgrom 2008; Pissarides 2009), and models with imperfect substitutability between incumbents and new hires (Kline et al. 2019). Dynamic wage posting models (as in Burdett and Mortensen 1998 ;Postel–Vinay and Robin 2002) provide a series of testable predictions: (i) more pass-through to poached workers vs hired from non-employment; (ii) greater wage growth to new hires; (iii) more wage pass-through at more productive firms. All of these predictions fail to find support in the data.

To interpret the empirical findings, I develop a simple model in which firms have labor market power due to workers idiosyncratic preferences (Manning 2011; Card et al. 2018), and monopolistic power in the output market[8] (Hamermesh 1996; Criscuolo et al. 2019). After empirically testing (and failing to reject) the theoretical framework, I use it to quantify mechanisms through which firms respond to the tax reform. The monopolistic competition limits plant expansion because firms have price setting power. The labor market power exacerbates the pass-through to wages, as firms face a positive sloped labor supply curve. I show that my model generalizes existing ones (Harasztosi and Lindner 2019; Curtis et al. 2021), and they align in the limit case, when the elasticity of labor supply goes to infinity.

I identify the model based on the reduced form estimates from the quasi-experimental policy variation, and rely on a perfectly fitted Classical Minimum Distance (CMD) approach to estimate the parameters of interest. All the esti-

[8]The market power is determined by a single elasticity, which means that the model is general enough to accommodate perfectly competitive markets, if the elasticity goes to infinity.

mates are in line with the literature (figure 8). The capital-labor elasticity of substitution (σ_{KL} = 1.06) is slightly above one, and supports recent theories that rationalize the recent rise of capital shares (Piketty 2013; Piketty and Zucman 2014). The labor supply elasticity faced by the firm (ϵ = 2.7) is remarkably similar to Lagos 2019. It implies that Brazilian workers leave on the table 27% of their marginal revenue product of labor. The output demand elasticity is 0.65, which suggests that overall output demand is inelastic. However, it doesn't mean that individual firms can raise prices without output implications. In particular, small firms face a demand elasticity of 2.08 (SE 1.71), which implies that there would be substantial output loss if they raise prices.

Overall, half of the employment boost is due to scale effect and the other half is due to substitution away from capital. The baseline result hides interesting heterogeneity. Large firms, due to their market power and influence on prices, are not able to respond to the reform by increasing scale. The scale effect for large firms is only 2% with an associated 10% price reduction. Small firms face a 18% employment increase, out of which 7% is due to scale and 11% due substitution. Based on the estimated parameters, I analyze alternative designs for the payroll tax subsidies. The first alternative policy is to target the benefit to firms with less market power, and I find that efficiency gains would increase 19%, and workers would pocket 87% more of the tax cut. The second alternative policy is to enforce perfect labor market competition, which boosts the efficiency gains by 25% and consumers would benefit 36% more.

Contributions. This paper builds on a large body of work that finds mixed evidence for the effects of payroll tax cuts on employment and wages. Part of the literature claims that workers bear the incidence based on empirical analysis that shows zero employment, and positive wage effect (Gruber 1997; Anderson and Meyer 2000). At the other extreme, recent studies find positive employment and zero earnings response (Saez, Schoefer, and Seim 2019; Kugler, Kugler, and Prada 2017). A third strand of work is in between, reporting partial pass through to earnings (Hamermesh 1979; Holmlund 1983; Cruces, Galiani, and Kidyba 2010; Kugler and Kugler 2009; Saez, Matsaganis, and Tsakloglou 2012). This paper also relates to recent studies that find mixed employment effect of the "Paycheck Protection Program (PPP)", relief plan implemented during the COVID-19 pandemic.[9] Despite major progress, there are natural limitations imposed on existent studies. First, payroll tax cuts are normally targeted at the worker level. This means that the labor cost difference between workers that slightly differ around the policy threshold, is discontinuously affected after the reform. The issue with this variation is that responses to the policy are confounded by pay equity norms (Dube, Giuliano, and Leonard 2019; Breza, Kaur, and Shamdasani 2018). Second, most reforms studied in the past were temporary and had a short time horizon. According to Chetty et al. 2011 adjustment costs and inertia can limit short run responses. I complement the literature by comparing short and long term effects. Third, existent work mostly relies on small payroll tax variation, which limits the ability of researchers to learn from the policy experiment. I address this limitation by evaluating a sizable tax reform that offered unprecedented large payroll tax reduction in the order of 20 percentage points.

Fourth, until recent years rich firm level administrative data was unavailable, which used to impose not only identification concerns, but also power issues, and restrictions on the set of outcomes/ heterogeneities that can be studied. I contribute to the literature, by providing evidence from a novel combination of matched employer-employee data and detailed anonymized tax records. Rich data allows this paper to extend existing analysis on employment and wages, to provide comprehensive incidence analysis that accounts for pass-through to profits, capital, and consumer price. Moreover, the setting allows me to contribute to understanding the impact of tax reforms on a broader set of firms and workers' characteristics. Prior work has studied heterogeneity at the worker level, but never at the firm level. For the specific case of the Brazilian reform, prior work was not able to observe actual treatment due to lack of firm level tax data. They attempted to proxy the policy variation based only on aggregated sector data. The identification concern with this approach is that it includes treated firms in the control group (firms in non-eligible sectors that manage to receive the benefit due to product eligibility criteria[10]), and control firms in the treated

[9] This strand of work rely on several interesting variation to infer causal effects: discontinuity in firm size eligibility (Chetty et al. 2020; Autor et al. 2022), exposure to bank system (Granja et al. 2020; Li, Watson, and LaJoie 2020), and timing of loans (Autor et al. 2020; Faulkender, Jackman, and Miran 2020; Bartik et al. 2020; Doniger and Kay 2021). The first identification concern to this literature is that only 0.4% of disbursement occurs above the threshold of 250 firms. Second, the timing variation on loans' disbursement is small, as firms quickly take them up.

[10] Firm level tax data shows that treatment due to product eligibility is present in virtually every sector of the Brazilian economy.

group (eligible firms that do not take-up the benefit).[11]

Fifth, for equality reasons, tax reforms normally affect a large share of firms and workers. This creates general equilibrium effects that threaten the SUTVA (stable unit treatment value assumption) identification condition. Spillovers to the control group compromises the validity of the empirical design, and also adds complexity to evaluate theoretical implications. For this reason, most of the payroll tax literature has stayed on the empirical side, which limits their ability to study mechanisms, design counterfactuals and identify parameters of interest. I extend the literature by studying a policy that affects a small share of firms (even at the local labor market level), and allows me to well-identify firm-specific responses, and directly connect reduced-form estimates to a novel structural model able to quantify the relative importance for margins of response such as scale and substitution. Based on the model and empirics, this paper contributes to the policy debate by proposing and measuring welfare-improving alternatives. In terms of the mixed findings in the payroll tax literature, this paper offers a novel idea to reconcile the debate: the role of firms' market power.

The empirical results in this paper suggests that Brazilian firms have labor market power, which is in line with a burgeoning strand of research (Card et al. 2018; Berger, Herkenhoff, and Mongey 2022; Lamadon, Mogstad, and Setzler 2022; Lagos 2019; Jäger and Heining 2022; Kline et al. 2019; Garin and Silvério 2019; Benmelech, Bergman, and Kim 2022; Burdett and Mortensen 1998; Azar, Berry, and Marinescu 2022). In a literature review, Manning 2021 argues that few papers[12] aim to directly estimate the labor supply curve faced by the firm,

[11] Dallava 2014 finds null employment effects for most of the sectors, and positive employment effects in only a few subsectors of the IT industry. Scherer 2015 finds 15% employment increase. This study focuses on small firms and seeks identification based on the fact that the "Simples" tax regime (tax tier for some small firms) is not eligible for the reform. The main drawback of this approach is twofold: (i) there is substantial migration between "Simples" and the regular tax regimes; (ii) treatment effect is measured on a subset of small firms that are more responsive than average, as I will show in a heterogeneity exercise. In concurrent work, Baumgartner, Corbi, and Narita 2022 found a 5% employment increase. They assume perfect take-up rate on eligible sectors and restrict their sample to a few sectors that they claim to not be affected by the product eligibility criteria. It turns out that both of these assumptions don't match tax data. On top of eliminating bias and increasing power, my evaluation is also more comprehensive in two other aspects: (1) I consider the vast majority of economic sectors in Brazil, while previous work is restricted to a few specific sectors. (2) I analyze the reform from its beginning until recent years, while other studies were restricted to the three initial years of the program, which can be concerning since treatment effects in some outcomes take time to kick in.

[12] Falch 2010; Staiger, Spetz, and Phibbs 2010 leverage plausibly exogenous variation on public workers' wages. These findings are valuable, but come with external validity concerns. Another popular approach to estimate the labor supply elasticity faced by the firm has been based on experimental evidence (Dal Bo, Finan, and Rossi 2013; Dube et al. 2020; Belot, Kircher, and Muller 2019).

because "it is hard to find plausibly exogenous variation to a single firm". I extend this literature by studying the unique quasi-experimental variation on labor demand for a small share of Brazilian firms. I estimate the labor supply elasticity faced by the firm and use it to measure the effect of labor market power on tax efficiency (adapting the Feldstein 1999 method to compute deadweight loss) and incidence. Prior literature does not accommodate the role of market power in the product and labor market simultaneously, to explicitly connect these competitive frictions to the incidence analysis.[13] For example, Berger, Herkenhoff, and Mongey 2022 use Monte Carlo simulations to predict the welfare and output impact of corporate tax in concentrated labor markets. My closed form incidence formulae contributes to the literature by allowing me to quantify channels through which imperfect competition drives tax incidence: scale, price and substitution.

This paper also advances the literature estimating elasticities of substitution between capital and labor (Karabarbounis and Neiman 2014; Raval 2019; Doraszelski and Jaumandreu 2018; Chirinko, Fazzari, and Meyer 2011; Caballero et al. 1995; Benzarti and Harju 2021; Oberfield and Raval 2021). In a meta-analysis, Gechert et al. 2022 criticizes prior work because of the use of cross-country variation, and omission of the first-order condition for capital. Papers that have addressed these concerns using local variation, optimality conditions for both inputs and modern econometric techniques (Harasztosi and Lindner 2019; Curtis et al. 2021) have suffered from three other shortcomings: isolating general equilibrium effects, lacking detailed capital data, and accounting for labor market power. The latter is an especially critical factor because as monopsonistic firms markdown wages, it changes the relative ratio of input's marginal products. I contribute to this body of work by estimating the elasticity of substitution in a setting that alleviates general equilibrium spillovers, with detailed balance sheet information at the firm level complemented by a model that incorporates the role of labor market power. To highlight the importance of labor market power, I also estimate the elasticity of substitution in a perfectly competitive labor market to compare the results.

Finally, there is an important strand of literature that studies firm level tax in-

[13] Card et al. 2018 conclude their paper explicitly mentioning this gap in the literature: "Finally, the idea that even highly advanced labor markets, like that of the United States, might be better characterized as imperfectly competitive opens a host of questions about the welfare implications of **industrial policies** and labor market institutions, such as the minimum wage, unemployment insurance, and **employment protection** (Katz et al. 1989; Acemoglu 2001; Coles and Mortensen 2016). Empirical work lags particularly far behind the theory in this domain. Additional evidence on how actual labor market policies affect firm and worker behavior is needed to assess the plausibility of these theoretical policy arguments."

centives for capital (Zwick and Mahon 2017); investments (Criscuolo et al. 2019); innovation (Howell 2017); R&D (Bronzini and Iachini 2014). These studies fit under the umbrella of Industrial Policies, and have unanimously documented greater employment responses to small firms. It is an open question what are the mechanisms driving this pattern. I contribute to this literature by first documenting the same pattern in Brazil. Second, I empirically reject two potential mechanisms, mean reversal and financial constraints. Finally, I pin down the mechanism able to rationalize the puzzle. It relies on a key ingredient: market power.

Structure The rest of the paper is organized as follows. In Section 2 I discuss the institutional background and the data. Section 3 presents the reduced form analysis. Section 4 develops the model. Section 5 takes the model to the data, estimates parameters of interest and provides interpretation for the results. Section 6 concludes.

2 Institutional Background and Data

2.1 Brazilian Payroll Tax System and the 2012 Reform

Similarly to most OECD countries, the Brazilian payroll taxes are designed to fund social security programs, such as retirement pensions and unemployment insurance. Tax rates are also similar to OECD's (refer to figure D.2 for cross-country comparison). However, the Brazilian payroll tax cut program has a few unprecedented features that are advantageous from an empirical perspective. First, the targeting was at the firm level. Second, the Brazilian reform offered a large tax reduction for benefited firms. Third, only a few firms were affected, isolating general equilibrium effects. Finally, the reform lasted for many years, which allows us to decompose short and long term effects.

Institutional Setting. The Brazilian payroll tax schedule has three components, and all of them are collected from firms. The main component is a 20% flat tax over the total wage bill. Secondly, there is an accident risk insurance component that varies between 1 to 3%[14]. The last layer of contribution is a 8 to 11% tax on wages, which is employee specific and can vary within workers of the same firm. All of these tax components are deposited in a social security fund that pools resources together. This implies that the public social security system does not

[14]This component is job specific, according to each job's accident exposure.

provide individual savings accounts, where resources are traceable and mapped to specific workers' benefits.

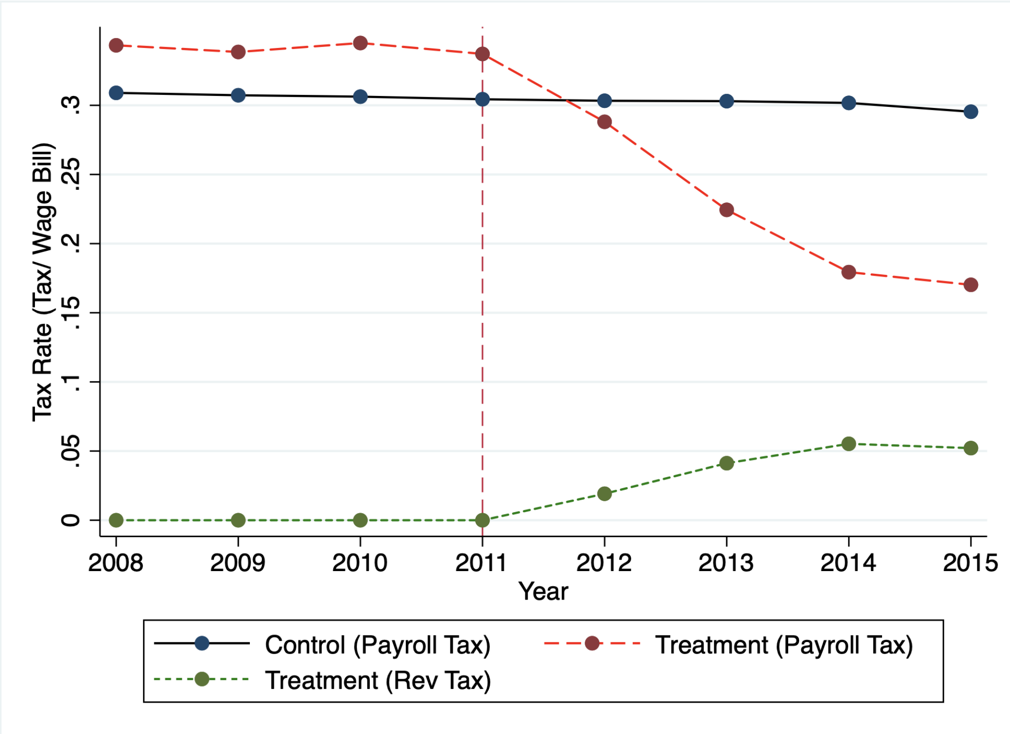
Tax Variation. On 14^{th} December, 2011 the Brazilian Congress enacted the payroll tax cut reform[15] that waived the main component of the payroll taxation, to a small share of sectors and products[16]. Treated firms faced a dramatic decrease in payroll tax rates, from 30 to 10 percentage points of the total wage bill. To provide slight compensation to the Government budget in face of this large drop in tax collection, the benefited firms were imposed to pay a small 1 to 2.5% taxes on net of exports gross revenue. Figure 1 compares the payroll and revenue tax variation. In the figure, taxes are divided by the total wage bill, so that all tax rates are comparable. It shows that the payroll tax drop is considerably larger than the revenue tax increase. This evidence showcases that the reform should be interpreted as a tax cut, as opposed to a tax substitution.[17] The reform didn't affect individuals' perception on the solvency of their retirement plans because the Federal Treasury committed to cover any deficits caused to the social security system.

[15]This reform was originally passed on an Executive decree (bill 540/2011), which was confirmed by Congress on 14^{th} December, 2011 through Law 12546/2011.

[16]Less than 1.5% of formal firms in the country are affected by this reform.

[17]The graph reports averages, but even when we look at outliers in the labor share, only 1% of firms wouldn't benefit from taking-up this benefit.

Figure 1: Tax Implication of the Reform



Note: This figure presents the evolution of tax rates for eventually treated vs control firms over the years. The blue line depicts payroll tax rates for control (never treated) firms, which slightly declined over the years, following global trends (OECD, 2019). The dashed red line represents the payroll tax rates for treated firms. The dashed green line presents the revenue tax rates that are substituted in once treatment takes place. Revenue tax rates are computed as a function of the total wage bill in order to facilitate comparisons.

Motivation. The tax reform was announced as an industrial policy aiming to increase competitiveness of domestic firms. The Government at the time had engaged in recurrent practices to subsidize specific corporations.[18] In section 2.2, I discuss potential confounding factors from concurrent policies and discuss appropriate remedies. I perform extensive empirical tests to recover the underlying rules that lead public policy to benefit certain firms, but not others. For example, I test (and reject) the hypothesis that becoming eligible to tax benefits is associated with more contributions to political campaigns. In sum, evidence suggests that the decision process to define eligible sectors was political, but those choices didn't affect the quasi-experimental variation, as it didn't anticipate sector specific trends on labor outcomes. Section 3.3 presents a set of tests that provide

[18]Those practices fit on a broader industrial policy umbrella, which are beyond the scope of tax benefits. One common initiative to this end, was offering subsidized interest rates to specific corporations through the use of public banks such as BNDES (*Banco Nacional de Desenvolvimento Econômico e Social*). "Plano Brasil Maior" was another example of the willingness of this Government to promote industrial policies that benefit specific firms.

details on the eligibility rules, and tests trends and balance across eligibility status.

Eligibility. The policy established a sector and product specific eligibility criteria for the payroll tax exemption. Additionally, the payroll tax cut did not affect firms in the "Simples" tax tier, which is a regime for small firms. This alternative tax system was created in 1996 and had two main goals: simplify tax rules[19] and reduce tax burden to small corporations. Given that said, firms in the "Simples" tax tier were not subjected to payroll taxes in the first place, and therefore were not affected by the payroll tax cut program.

Timing. The first tax bill outlining the policies and the eligible sectors were passed in December 2011, and implemented a few months immediately after (April 2012). The reform was initially outlined on an executive bill that skipped prior Congress discussion. This type of payroll tax cut has never been implemented previously in Brazil, so this was not an expected policy by employers and employees. The policy is still valid nowadays[20], and there is no expectation of being eliminated in the near future. There were several other tax bills including more sectors to the reform in 2013 and 2014.[21] To understand the relevance of the reform from a macro perspective, table 7 presents the number of firms, workers and sectors affected by the reform, and its respective share relative to the Brazilian economy. Table 7 is also useful to characterize the staggered rollout aspect of the program. It reports the cumulative values as all the cohorts (2012-2014) phase-in the program. At the peak of its implementation in 2014, the payroll tax cut has affected a relatively small share of Brazilian sectors (9%), firms (1.7%), and workers (6%).[22]

Industry Variation. Within broad defined sectors, the reform did not provide eligibility to all subsectors. For example, when the media industry became eligible to the reform, open television was contemplated while cable television was not. In the lodging industry, hotels are eligible, but motels are not. Similarly, table 6 provides multiple examples of similar subsectors in broad defined industries, where one of them became eligible and the other not. In the empirical anal-

[19] It aggreagates all tax payments in one gross revenue tax rate.

[20] As of September, 2022

[21] IT, Call Center and Hotels were added in 2012. Retail, Construction and Maintenance were added in 2013. And a final wave in 2014 added Transportation, Infra-structure and Media sectors.

[22] The fact that "Simples" firms are not eligible, and there is imperfect take-up in eligible sectors contribute to the share of firms being lower than the share of sectors. The fact that larger firms are more likely to take-up contributes to the share of workers being larger than the share of treated firms. Full list of eligible sectors can be found on table D.2.

ysis, I add 1-digit sector fixed effect to leverage variation within broad defined industries.

Product Eligibility. Product eligibility was defined based on the Mercosur Common Nomenclature (NCM). Most of the product eligible firms are in the manufacturing industry, but treatment due to NCM criterion is not restricted to the manufacturing sector. Indeed, the vast majority of sectors in the Brazilian economy contain firms treated due to the product NCM criteria.[23] Treatment due to the NCM eligibility criterion only allows for partial payroll tax waive, according to the share of eligible products in the firms' gross income.

Additional Changes. Over the years, 5 other tax bills[24] were passed promoting marginal changes to the program, such as modifying the revenue tax rates, or adding new sectors to the policy. One of the most relevant changes happened in December 2015 when the policy became less generous as the revenue tax rates increased from 1-2.5% to 1.5-4.5%[25]. At that moment, treatment assignment also became optional, which in practice was not a relevant change in the regime, since even in the early years of the reform take-up in eligible sectors was far from perfect. Indeed, the imperfect take-up rate is a central aspect of the reform that deserves more discussion.

Take-up. One might be puzzled that many eligible firms are not opting in generous Government benefits. Figure 12 shows that a substantial share of eligible firms do not take-up the benefit. To interpret this observational fact it is important to have in mind that the increase in revenue tax would surpass the payroll tax decrease for only 1% of eligible firms. Thus, the imperfect take-up cannot be rationalized through the lenses of a perfect tax optimization choice. There are a few facts that help to rationalize the imperfect take-up. First, the tax bills never mentioned any punishment to non-compliers. Possibly because from a legal point of view eligibility was seen as beneficial to firms. Based on the Brazilian tax code it is implausible for prosecutors to suit firms that don't opt in a supposedly beneficial tax system. Second, enrollment in the program was not automatic as in the Swedish case studied by Saez, Schoefer, and Seim 2019.[26]

In Brazil, firms have to self-report eligibility on Government provided soft-

[23]This can be precisely observed in the anonymized micro tax data.

[24]Law 12546/2011, Law 12715/2012, Law 12844/2013, Law 13161/2015, Law 13202/2015, and Law 13670/2018.

[25]Law 13.161/2015

[26]In Sweden firms filled the same tax forms before and after the reform. Once firms provided information on their employees, the Tax Authority was the one computing firms' tax benefits.

ware to enable tax exemptions[27], through separate tax forms. Figure 13 illustrates tax forms instructions and the set of information requested in the tax platform. Even though the tax substitution implied a net tax cut in most cases, empirical findings in other countries (Kleven and Waseem 2013; Janet, David, John, et al. 2006; Zwick 2021; Moffitt 2007) suggest that the operational filling process can lead to non responsiveness even in dominated tax regions[28]. In line with this view, figure 14 shows that take-up is monotonically increasing with firm size. This pattern is consistent with the fact that larger firms are more likely to have accounting support, be aware of tax benefits, and be able to pay for filling costs.

Spillover. Theoretical predictions on the consequences of tax variations, depend on whether the shock affects the market, or it is firm-specific. The fact that a very small share (less than 1.5%) of formal firms and workers benefited from the tax cut is indicative (but not conclusive) that the reform should be seen as a firm-specific variation. To advance in this understanding, I follow literature that has considered job switching patterns to define local labor markets (Felix 2021). This analysis shows that 67% of Brazilian job switchers stay in the same occupation and region, rather than the same industry. That said, I define the local labor market at the occupation x region cells.[29]. To evaluate spillovers within the local labor market, I provide a set of evidence supporting that the Brazilian reform was a firm, rather than market level shock.

First, I provide purely descriptive evidence that even at the local labor market level, the share of treated firms is small. Figure D.4 provides intuition that within a labor market there are eligible and non-eligible sectors. Within eligible sectors, there are unaffected firms that are either in the informal sector, or in ineligible tax tier ("Simples"), or decided not to take-up the benefit. Table D.1 walks through this logic and shows that conditional on having an eligible sector in the local labor market (LLM), less than 3% of firms in the LLM are affected. Even the share of affected workers is small, as table 8 describes.[30]

[27]Firms inform eligibility on block 0 and this enables block P where tax relevant information is input.

[28]It relates to an extensive body of work dedicated to understand rational attention (Hoopes, Reck, and Slemrod 2015), and other frictions that can rationalize low participation rates in public programs (Currie et al. 2001; Heckman and Smith 2004). Similarly, several papers study the role of tax salience (Chetty, Looney, and Kroft 2009; Chetty, Friedman, and Saez 2013; Finkelstein 2009)

[29]This definition uses the 2-digit occupation code from CBO, and the micro-region defined by the Brazilian Census Bureau (IBGE).

[30]Note that table 8 is conditional on formal workers on eligible tax tier. Adding to the denominator informal workers and workers in the "Simples" regime would bring the share of affected workers from 6% to approximately 3%.

Second, I run a spillover test with firms in the "Simples" tax regime (ineligible tax tier). These firms are ineligible for the payroll tax benefit, but they can operate in eligible industries. If the reform were to create a market level shock, then we should expect to see a negative employment effect in these firms compared to other "Simples" firms in non-eligible sectors. Figure 15 shows that this is not the case. It reports a small and not statistically different than zero spillover effect. It also shows that "Simples" firms in eligible vs ineligible sectors follow similar trends in the pre-reform period. In contrast, it shows that among firms in the eligible tax-tiers, there are substantial effects of being in the eligible sectors.

Finally, if there were a spillover effect we would expect to see relatively more wage pass-through in intensively treated local labor markets.[31] The reasoning is that spillovers affect worker's outside options, therefore generating more wage hikes. Figure D.3 shows that the wage effect for workers in high versus low intensively treated markets are not statistically different from each other. This evidence suggests that the driving force underlying the workers' earnings increase is not the market spillover. This supports the view that the Brazilian payroll tax reform should be seen as a firm-specific shock.

2.2 Data and Descriptive Statistics

I constructed two samples, one at the firm and other at the worker level, by combining anonymized tax and labor administrative data on the universe of formal firms operating in Brazil between 2008 and 2017. To exploit regional variation on informality, I merge the data with the 2010 Census. The advantage of this data is that it allows me to track firms and workers over time, which constitutes an ideal laboratory to understand the effect of payroll tax policies on detailed measures of labor market outcomes.

Labor Market Data. For labor market data I use Relação Anual de Informações Sociais (RAIS), which is the matched employer-employee data set administered by the Ministry of the Economy. This data provides firm and worker level information covering every formal labor contract since 1976. I restrict the analysis to the period between 2008 and 2017, which allows me to track firms before and after the implementation of the payroll tax program. At the firm level, RAIS

[31] I define treatment intensity by the share of treated firms, but the results are qualitatively the same if I define intensively treated markets based on the average or total amount of subsidy.

contains information on the tax regime[32], sector (at its most granular definition), total employment, wage bill, age and location. At the worker level, it contains variables regarding employment status, occupation, wage, race, gender, industry, municipality, as well as hiring and termination dates. Workers and firms are uniquely identified based on tax codes (PIS and CNPJ, respectively), which do not change over time. The main shortcoming in RAIS is the lack of information about informal and non-employed workers (see Dix-Carneiro 2014 for an overview of RAIS dataset).

I use other sources of administrative data to complement this dataset. The 2010 Census provides information that allows me to compute formalization rates at each of the 5,300 Brazilian municipalities. Finally, from the tax authority, I have firm level anonymized micro data on payroll tax, revenue tax, capital expenditure, and wage bill. This dataset is organized in two anonymized samples: one at the firm level, and the other at the worker level.

Firm Level Sample. To make the administrative data suitable to study the payroll tax reform in Brazil there are a few sample restrictions that are important to deal with specificities of the context. First, I exclude from the sample firms that have ever participated in the "Simples Nacional", which is a special tax tier not subjected to the payroll taxes studied in this paper.[33]. Firms in the "Simples" regime face a different tax tier which consolidates all tax liability in a single tax form with simplified and lower rates. Therefore, these firms are not eligible for the tax reform under analysis and neither are comparable to the firms in the regular tax tiers.

In terms of sector comprehensiveness, the sample encompasses 19 out of 21 one-digit sectors[34] of the Brazilian economy. The construction sector is excluded because the treatment assignment to this sector was problematic. The tax bill allowed construction firms to be treated in only certain of its construction sites, according to the site's license date. This makes some of the construction corporations being partially treated, and, therefore, even with the firm level tax data it is not possible to detect site level responses. Even if it was possible to observe the construction site level of granularity, this could be confounded by spillovers

[32]There is a simplified tax regime ("Simples Nacional") targeted to small firms that are not subjected to the payroll tax cut under analysis.

[33]The current gross revenue eligibility for the "Simples" regime is BRL 4.8 millions (around USD 1 million).

[34]Sectors are defined according to Classificação Nacional de Atividades Econômicas (CNAE), which is administered by the National Statistics Bureau (Instituto Brasileiro de Geografia e Estatística).

from non-treated sites within the same firm. Also, construction was at the epicenter of the "Car Wash" operation, a massive corruption scandal revealed in the decade of this study, which revealed that economic transactions on that sector were not responses to standard economic incentives of interest, but to illegal business negotiations.

The sector of retail and sales of motor vehicles was excluded to avoid lurking effects from other industrial policies, such as the program "Plano Brasil Maior". Important to notice that these two sectors are excluded at the one-digit (broadest) level, so it eliminates both eligible and non-eligible subsectors in these broad industries. The sample is not winsorized or balanced, but results are robust to these procedures (appendix C). The analysis with the balanced sample alleviates concerns with firms' attrition.

Worker Level Sample. To maintain consistency between the firm and worker level analysis, I keep the same sample restrictions presented before to ensure an equivalent set of employers in both data sets. I follow the displacement literature (Jacobson, LaLonde, and Sullivan 1993; Lachowska, Mas, and Woodbury 2020; and Szerman 2019), and impose a tenure restriction to track only workers that have been employed by the same employee for at least three years in the pre-reform period (2008-2011). This guarantees that results are driven by relatively stable employer-employee matches. This is important for two reasons. First, since we are interested in the incidence aspect, I want to minimize the turn over effect on wages, and focus on the pass-through interpretation. Second, since I assign workers to treatment based on the pre-reform period, it makes sense to consider workers with stable attachment to firms in order to classify them into treatment based on a meaningful employee-employer relationship. Results are qualitatively unchanged by the tenure sample requirement.

Descriptive Statistics. Table 8 reports descriptive statistics for the worker level sample in the baseline period (2008 to 2011). Each observation is a worker x year. Column (1) and (2) reports the pre-reform values for non-eligible and eligible workers, respectively. Column (3) pool this two groups together, and shows that on average worker's monthly earnings are $2,315 (approximately $450 USD[35]) on December 31st of each year. Workers' average age is 39 years old, 67% of the sample is white, 54% male, 70% has achieved high school or higher degree, and 27% are college educated. The table shows that in the baseline period, characteristics of eligible and non-eligible workers are similar. The one exception is gender, a

[35] As of the exchange rate in February, 1st, 2023.

variable that I will control for in all empirical specifications.

In the firm level sample there are 1,775,601 observations in the pre period (2008-2011). These firms are allocated in 19 one digit sectors that are broken down into 1,300 seven-digit industry codes (CNAE). Table 9 provides summary statistics for non-eligible (column 1), eligible (column 2), and average (column 3), in the pre-reform period (2008-2011). Column (4) reports the descriptive statistics for treated firms, in eligible sectors. As figure 14 highlights, conditional on eligibility, larger firms are more likely to take-up. This fact rationalizes why column (4) reports larger firm size compared to the other groups. Pooling eligible and non-eligible firms together, pre-reform employment on December 31^{st} of each year was 55.34 workers. The average payroll tax rate was 31.78%, and 89% of employment was in low skilled occupations.

3 Empirical Analysis

The payroll tax cut causes a sharp expansion on employment, with small but significant effects on long term wages. In this section, I present details about the main results, including heterogeneity analysis across firm size and workers characteristics.

3.1 Identification Strategy

The main empirical strategy is an event study instrumented by the sector eligibility. The design explores the staggered implementation of the program, and the fact that there is a large share of firms never eligible or treated. The IV is necessary to adjust for two margins of imperfect compliance: the imperfect take-up in eligible sectors; and the treatment in non-eligible sectors due to the product eligibility criteria. The standard IV assumptions: independence, exclusion, first stage, and monotonicity are explicitly stated in appendix A, and tested in section 3.3.

Concretely, the main specification combines the 2SLS framework with the event study set up, which provides two main advantages. First, it validates the identifying assumption by showing that the pre-reform coefficients of interest are not statistically different from zero. Second, it provides intuition about the dynamics of the treatment effect relative to the year prior to the policy implementation. This is particularly important as adjustment costs and inertia can potentially restrict immediate responses (Chetty et al. 2011).

Firm level. The first set of analysis are constructed on a sample at the firm x year level, where I estimate the following structural equation:

$$Y_{jt} = \sum_{k=-4,\neq -1}^{3} \beta_k D_{jt}^k + X_{jt}'\gamma + \alpha_j + \xi_{s1(j),t} + \epsilon_{jt} \tag{1}$$

where, X_{jt} are set of controls (e.g., education, gender, race, age and its square); $\xi_{s1(j),t}$ is 1-digit sector interacted with year fixed effect; α_j is the firm fixed effect; and k indexes the time relative to treatment. For each time t relative to treatment, there is one respective first stage equation. Thus, in total there are K first stage equations given by,

$$D_{jt}^k = \sum_{l=-4,\neq -1}^{3} \pi_{kl} \times \mathbb{I}(t = e_{s(j)} + l) \times L_{s(j)} + \alpha_j + \xi_{s1(j),t} + X_{jt}'\delta_k + \eta_{jt},$$
$$\forall k \in [-4, -2] \cup [0, 3] \tag{2}$$

where, $e_{s(j)}$ is the event date, in which firm j's sector becomes eligible; $L_{s(j)}$ indicates if firm j's sector is eventually eligible; and the remaining coefficients are the same as described before. Standard errors are conservatively clustered at the 5-digit industry-by-state level. Appendix A provides more details on the empirical model, underlying assumptions and reduced form equations. I also estimate an IV difference-in-differences model, where all the periods after the policy implementation are pooled to a single post period indicator. The first stage and structural equations are outlined in equations 3 and 4, respectively.

$$D_{jt} = \pi L_{s(j)t} + \alpha_j + \gamma_t + \xi_{s1(j),t} + X_{jt} + u_{jt} \tag{3}$$

where, D_{jt} indicates that firm j is treated in year t; $L_{s(j)t}$ indicates that firm j belongs to a sector that is eligible for treatment and that period t is after the starting eligibility date for sector s(j); X_{jt} are set of controls (e.g., education, gender, race, age and its square); $\xi_{s1,t}$ is 1-digit sector interacted with year fixed effect, α_j is the firm fixed effect. I cluster the standard errors at the level of the treatment variation (Bertrand, Duflo, and Mullainathan 2004 ; Cameron and Miller 2015). Because eligibility is defined at the industry level (mostly at the 7-digit) and there is state-level tax variation, standard errors are conservatively clustered at the 5-digit industry-by-state level.

The first stage coefficient π increases as the take-up rate on treated sectors increases, and deflates as there are more treatments occurring in non-treated sectors due to the NCM criteria. The associated reduced form is expressed in equation 4,

$$Y_{jt} = \delta L_{s(j)t} + \alpha_j + \gamma_t + \xi_{s1(j),t} + X_{jt} + u_{jt} \tag{4}$$

Worker level. The firm level sample can impose challenges to evaluate the earnings effect. For example, one might be concerned that at the firm level, the average earnings can be affected by compositional changes in the labor force. To address this concern, I take advantage of the granularity of the micro data, to estimate a similar model at the worker level. I assign workers' eligibility status ({0,1}) according to their pre-reform employer, and then evaluate individuals' outcomes regardless of the firms that they end up working for. Thus, $L_{s(i,t_0)}$ is equal to one if firm j's pre-reform sector eventually becomes eligible. Similarly, to the firm level specification, the pooled difference-in-differences model at the worker level is given by,

$$D_{it} = \pi L_{s(j_0)t} + \iota_i + \alpha_j(i,t) + \xi_{s1(i,t_0),t} + X_{it} + u_{it} \tag{5}$$

$$Y_{it} = \delta L_{s(j_0)t} + \iota_i + \alpha_j(i,t) + \xi_{s1(i,t_0),t} + X_{it} + u_{it} \tag{6}$$

where, i indexes workers, Y_{it} is workers' labor market outcome in year t; ι_i is the worker fixed effect; $\alpha_j(i,t)$ is the firm fixed effect; and the remaining variables and fixed effects are analogous to definitions in equations 3 and 4. Similarly to the firm level analysis, I also fit the event study model to the worker level sample. The structural and first stage equations are presented below,

$$Y_{it} = \sum_{k=-4,\neq -1}^{3} \beta_k D^k_{j(i,t_0)t} + X'_{it}\gamma + \iota_i + \alpha_{j(i,t)} + \xi_{s1(i,t_0),t} + \epsilon_{it} \tag{7}$$

$$\begin{aligned} D^k_{j(i,t_0)t} = & \sum_{l=-4,\neq -1}^{3} \pi_{kl} \times \mathbb{I}(t = e_{s(i,t_0)} + l) \times L_{s(i,t_0)} \\ & + \iota_i + \alpha_{j(i,t)} + \xi_{s1(i,t_0),t} + X'_{it}\delta_k + \eta_{it}, \forall k \in [-4,-2] \cup [0,3] \end{aligned} \tag{8}$$

where, $D^k_{j(i,t_0)t} = 1$, if $t = e_{j(i,t_0)} + k$; $e_{j(i,t_0)}$ is the year when the pre-reform firm enters treatment; $e_{s(i,t_0)}$ is the year when the pre-reform sector becomes eligible; and the remaining variables and fixed effects are the same as defined before. Standard errors are conservatively clustered at the 5-digits industry-by-state level.

Validity of design. Identification relies on the assumption that conditional on fixed effects, eligibility is uncorrelated with time-varying unobserved determinants of employment and wage growth. This implies that in the absence of the reform, outcomes for eligible and non-eligible firms would follow similar trends. There are two main threats to this design. First is the potential for Congress to anticipate sector-specific trends when determining eligibility rules. The second threat stems from the possibility that firms might strategically move to eligible sectors after the reform is announced. Section 3.3 provide several tests that mitigate these concerns.

3.2 Labor Market Outcomes

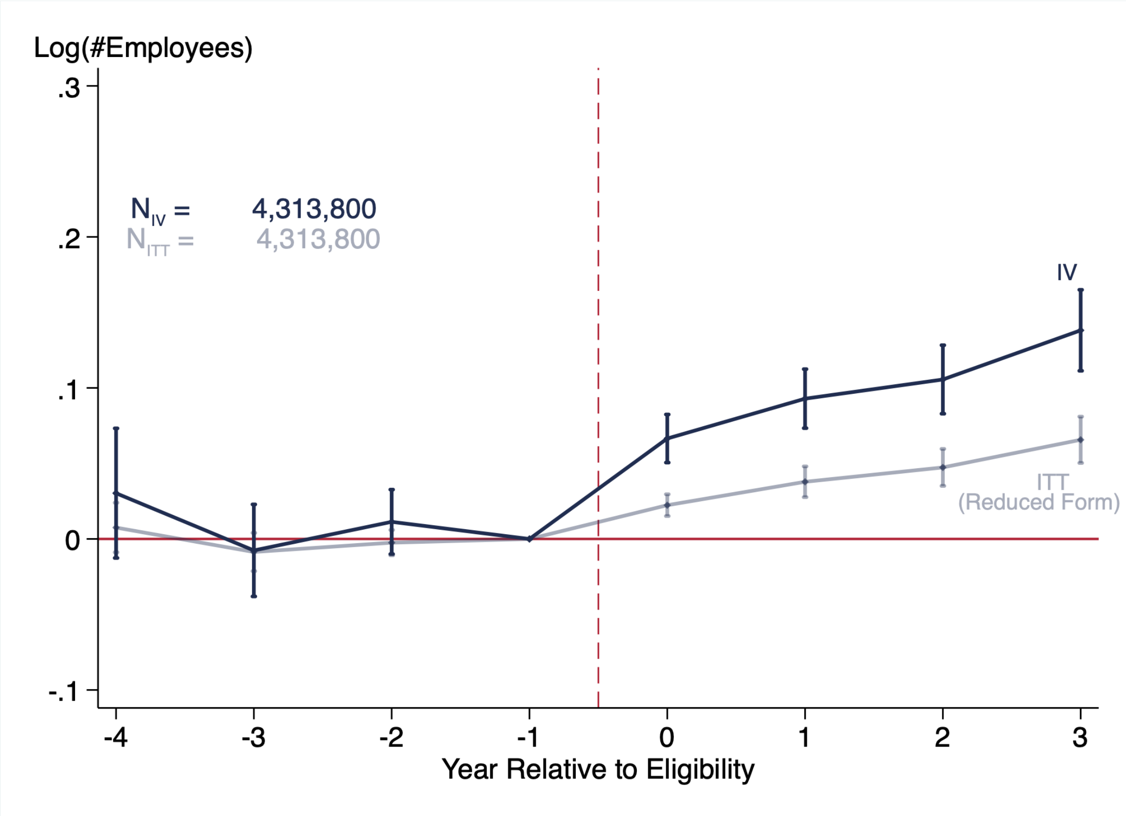
In this section, I report causal effects of the payroll tax reform on a comprehensive set of labor market outcomes. The findings indicate that employment and wages rise when the payroll tax is reduced. However, it takes time for workers' earnings to respond. These results are confirmed in several analysis, coupled with novel heterogeneity findings at both the firm and worker level.

3.2.1 Firm-level Analysis

The difference-in-difference estimation, from equations 3 and 4, finds that the payroll tax program causes a highly significant 9% increase (se = 0.03) in the number of employees. The event study analysis validates the parallel trends assumption and shows that there is an immediate employment response that is sustained and slightly increased over time, as shown in figure 2. The associated empirical elasticity of employment with respect to the labor cost ($\epsilon^{L}_{1+\tau}$) is -0.71[36]. These results remain qualitatively similar within the balanced sample of firms (refer to appendix C), which suggests that the employment effect is not governed by the dynamics of firm entry and exit. However, the average effect masks substantial heterogeneity, which I exploit next.

[36]This elasticity is derived from a policy induced labor cost variation of -0.133.

Figure 2: Event Study Estimates on Employment



Note: This figure presents the event study estimates for employment. The event is the year in which the firm enters treatment for the first time. I normalize the results with respect to one year prior to the event. The analysis spans four years prior to entering the payroll tax cut program and three years after. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Firm size. The richness of the data allows me to evaluate heterogeneity responses based on firm size, which is measured in the pre-reform period. Table 1, columns (2)-(4), reveal statistically significant differences in employment responses between small and large firms. These findings are consistent with the heterogeneity per market concentration, measured by the market share at the local labor market[37]. Figure 17 shows both employment and wage effects are greater for firms exhibiting less market concentration.

The firm size analysis is implemented by separately fitting the same specification for a sample of small, medium and large firms. One advantage of this approach is that by comparing small firms in treatment and control groups, it elimi-

[37]Based on job switcher patterns, and following Felix 2021, local labor markets are defined at the commuting zone x 2-digits occupation level

nates the *"mean reversal"* channel that could potentially explain the heterogenous pattern. Important to mention that this heterogeneity is not mechanically driven by differences in the first stage. Figure D.5 reassures that the reform impacts labor costs uniformly across the entire firm size distribution. Another potential explanation for the firm size heterogeneity could be financial constraints, which I discuss next.

Table 1: Firm Level Estimates

	Log(#Employees)	**Log(#Employees)**			**Log(#Employees)**	
	All Sample	Small	Medium	Large	New	Old
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: IV						
Diff-in-Diff	$.095^{***}$	$.251^{***}$	$.141^{***}$	$.058^{**}$	$.156^{***}$	$.072^{**}$
	(.03)	(.048)	(.031)	(.028)	(.036)	(.031)
Long Diff	$.138^{***}$	$.313^{***}$	$.198^{***}$	$.121^{***}$	$.155^{***}$	$.104^{***}$
	(.027)	(.045)	(.038)	(.037)	(.036)	(.033)
Panel B: ITT						
Diff-in-Diff	$.044^{***}$	$.095^{***}$	$.071^{***}$	$.034^{**}$	$.073^{***}$	$.036^{**}$
	(.014)	(.016)	(.016)	(.017)	(.017)	(.016)
Long Diff	$.066^{***}$	$.136^{***}$	$.101^{***}$	$.064^{***}$	$.085^{***}$	$.051^{***}$
	(.015)	(.02)	(.02)	(.023)	(.019)	(.019)
Controls	✓	✓	✓	✓	✓	✓
Worker FE	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓
Sector x Year FE	✓	✓	✓	✓	✓	✓
# Clusters	10, 720	8, 617	7, 068	6, 138	9, 008	8, 159
N	4, 313, 800	2, 649, 413	704, 782	472, 392	1, 533, 074	2, 306, 712

Note: This table presents IV and reduced form (ITT) estimates for the firm level sample. Difference-in-differences coefficient is estimated in equations 3 and 4, where there is only one post period. The long difference comes from the period t=+3, in the event study design. Panel A reports the IV coefficients, which adjust for the imperfect compliance and are interpreted as the local average treatment effect on compliers. Panel B reports the reduced form coefficients, which are interpreted as the intention to treat (ITT). The dependent variable is log of employment. Column (1) presents the average effect in the all sample. Columns (2-6) present heterogeneity based on pre-reform firm size and firm age. Firms are categorized as small if they had less than 9 workers in the pre-period. Medium if they had 10-49, and large if they had more than 50 workers. Firms are classified as new or old depending on thei position around the firm age median in the pre-period. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

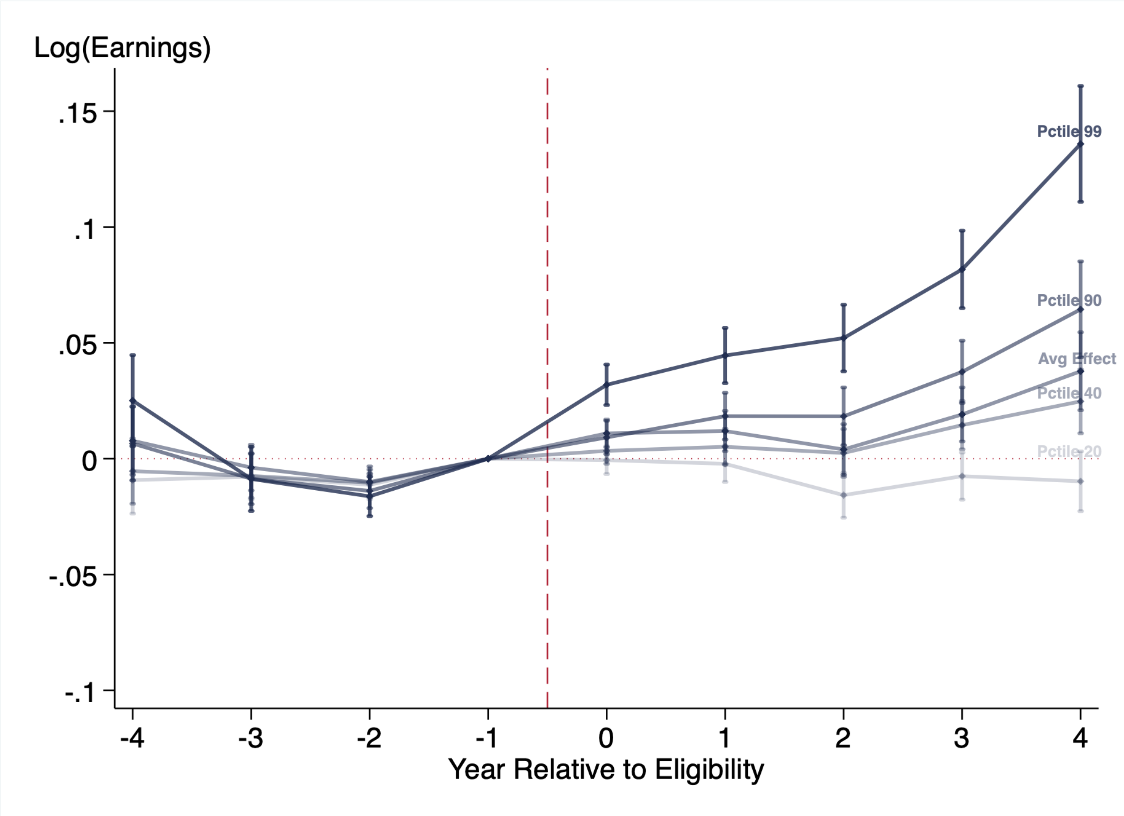
Liquidity constraints. The observed heterogeneity in firm size is consistent with previous studies (Zwick and Mahon 2017; Criscuolo et al. 2019; Howell 2017; Bronzini and Iachini 2014) which found that tax subsidies yield greater employment responses by small firms in different contexts. Due to lack of data and appropriate variation, the literature hasn't been able to rationalize the mechanism underlying this fact. Criscuolo et al. 2019 suggested that liquidity constraints might be a contributing factor. According to their argument, small firms, being financially constrained, may not be able to expand employment even when the marginal benefit exceeds the marginal cost. In this view, the payroll tax cut serves to alleviate the financial constraints of the firm, subsequently leading to an increase in employment.

To test this hypothesis, I utilize anonymized firm-level balance sheet data[38]. I define liquidity constraint as the ratio of short-term assets to short-term liabilities. An example of current assets is cash, whereas an example of current liabilities are short term bills, such as the wage bill. I divide the sample into firms that fall below and above the median based on the pre-reform measure of liquidity constraint. The employment effects for both groups turn out to be strikingly similar. The results are reported in Table 14. In Section 5, I estimate that small firms have less market power and then discuss how market power provides a suitable explanation for the greater employment response observed in smaller firms.

Earnings. Several key findings emerge when looking at the effect on workers' earnings. First, there is a positive average earnings increase that gradually builds over time. More strikingly, however, is the inequality on the wage pass-through across the within firm distribution. Figure 3 illustrates this by showing that the earnings effect across different percentiles of the within-firm distribution displays a monotonic pattern. The effect is null at the bottom, and raises up to more than 4% at the firm's 99^{th} percentile. The absence of pre-trends across the entire range of the earnings distribution further corroborates these findings. These results shed light to an important consequence of the tax policy, the within firm wage inequality. As the Government reduces payroll tax rates, wages for those that already had higher earnings, increase relatively more.

[38] Presented in section 3.4.

Figure 3: Earnings Effect Within Firm Wage Distribution



Note: This figure presents the event study estimates for wages at different percentiles of the within firm wage distribution. The event is the year in which the firm enters treatment for the first time. I normalize the results with respect to one year prior to the event. The analysis spans three years prior to entering the payroll tax cut program and three years after. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Composition. The interpretation of the earnings results in terms of pass-through could be compromised if, as a result of the policy, firms change the composition of their labor force. However, table 2 demonstrates that the tax reform does not significantly impact the composition of employed workers across various dimensions. The only exception is gender, where the reform induces a marginal but statistically significant effect of a 1 p.p in the share of male workers, from a baseline share of 60 percent (column 3). Columns (1) and (2) show that the effects on the share of workers with high school and college degree is undistinguishable from zero. Columns (4) and (5) presents evidence that the reform did not affect the share of employed white workers or the average employees' age.

Columns (6) and (7) follow the composition analysis from Kline et al. 2019. Column (6) investigates whether the reform influences firms to hire workers from different parts of the earnings distribution, finding no effect on the quality of new hires proxied by their pre-hiring earnings. Column (7) displays the impact on a quality index, computed through a Mincer regression of log earnings on a quartic in age fully interacted with gender and race, estimated annually with firm fixed effects as additional controls. All results in Table 2 suggest no evidence of skill-upgrading in response to the reform.

Finally, I explore whether the tax cut affects the types of occupations firms employ. I exploit the detailed CBO occupation data, which contains 2,300 occupation codes. These occupations are ranked based on pre-reform average earnings and grouped into percentiles. Then, for each firm, I calculate the average occupation percentile they employ from. Table 10 reveals a sharp zero effect of the reform on firms' average occupation percentile. This empirical fact implies that the tax reform expands employment within, rather than between occupations. This result reinforces the idea that the reform didn't change the composition of the employed workforce. Another interpretation is that the reform does not induce a structural change in technology, and consequently, in the type of labor demanded.

Table 2: Effect on Labor Composition

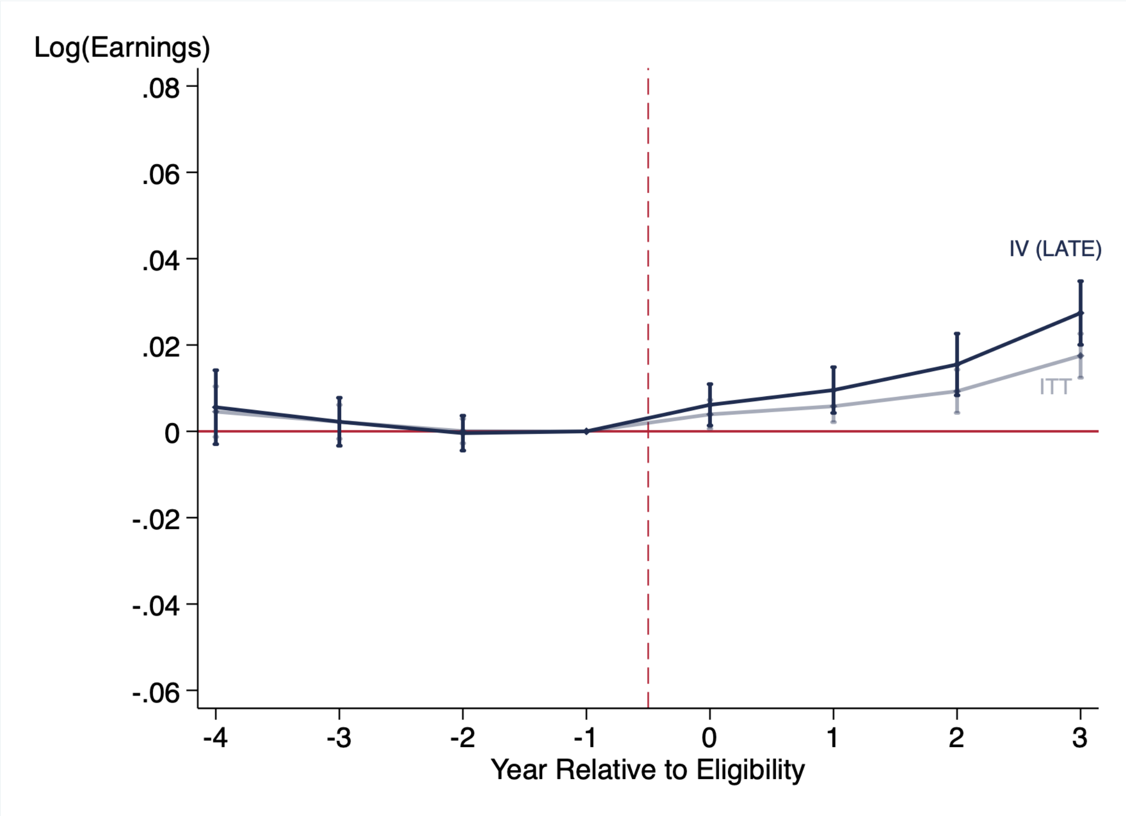
	(1) Share High School +	(2) Share College +	(3) Share Male	(4) Share White	(5) Avg Worker's Age	(6) Log Earnings New Hires (bf hired)	(7) Log Quality
Post × Treatment	.0091	.0099	.0132***	.0005	-.1343	-.0014	-.0005
	(.0085)	(.0061)	(.0034)	(.0045)	(.1497)	(.0116)	(.005)
Mean	.52	.11	.59	.67	39.72	7	1
Controls	×	×	×	×	×	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓	✓
Sector x Year FE	✓	✓	✓	✓	✓	✓	✓
# Clusters	7,925	7,925	7,930	7,924	7,930	6,924	7,561
N	2,494,842	2,494,842	2,521,030	2,491,523	2,521,030	604,988	1,739,827

Note: This table reports difference-in-differences coefficients to access the effect of the reform on the firm's labor composition. The empirical specification is the same as presented in equations 3 and 4. The regression is estimated in the balanced sample of firms to isolate any noise due to firm entry and exit. The goal is to depict the firm level compositional effect. In columns (1)-(5) controls are not included to avoid over-controlling given that in this regressions the outcome is part of the set of variables typically used as control. Column Standard errors are reported in paranthesis.

3.2.2 Worker-level Analysis

An alternative approach to evaluate the pass-through to workers' earnings involves tracking workers over time. This strategy offers two main advantages. First, it provides further validation that the firm-level earnings results were not influenced by shifts in labor force composition. Second, it yields insights into how the tax reform impacts workers' career paths, particularly among various types of workers. Important to clarify that throughout this analysis our focus is on the net earnings received by workers, after the deduction of payroll taxes. As shown in Figure 18, the reform causes leads to a sharp decrease in gross earnings paid by firms, mostly due to the mechanical reduction on payroll tax rates. However, figure 4 demonstrates that net earnings pocketed by workers (after payroll taxes have been deducted) actually increase by an average of 1.5%. This effect intensifies to a 3% increase three years following the tax cut.

Figure 4: Event Study Estimates on Workers' Earnings



Note: This figure presents the event study estimates for average earnings (net of payroll taxes) for stable workers. I normalize the results with respect to one year prior to the treatment event. The analysis spans four years prior to the payroll tax cut program and three years after. The blue markers report the IV estimates, while the gray markers are the intention-to-treat. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Occupation. Consistent with the within-firm earnings inequality result presented in the previous section, I show that workers at high skill and managing positions benefit relatively more from the reform. To implement this analysis, I rely on the CBO occupational code[39] to split employees into two occupation groups. Managers, directors and qualified technical positions are in the top bucket and comprise 15% of the sample, whereas the remaining 85% of lower positions are evaluated separately. Figure 19 shows that the pass-through to high skilled workers is 6%, and it almost zero to low skilled workers. In the appendix, I provide an extension of the model, including two types of labor, and able to rationalize the findings in a setting where low skilled workers have higher labor

[39]Classificação Brasileira de Ocupação (CBO) it is the legal norm to classify occupations in the Brazilian labor market. It was established on decree approval 397/2002.

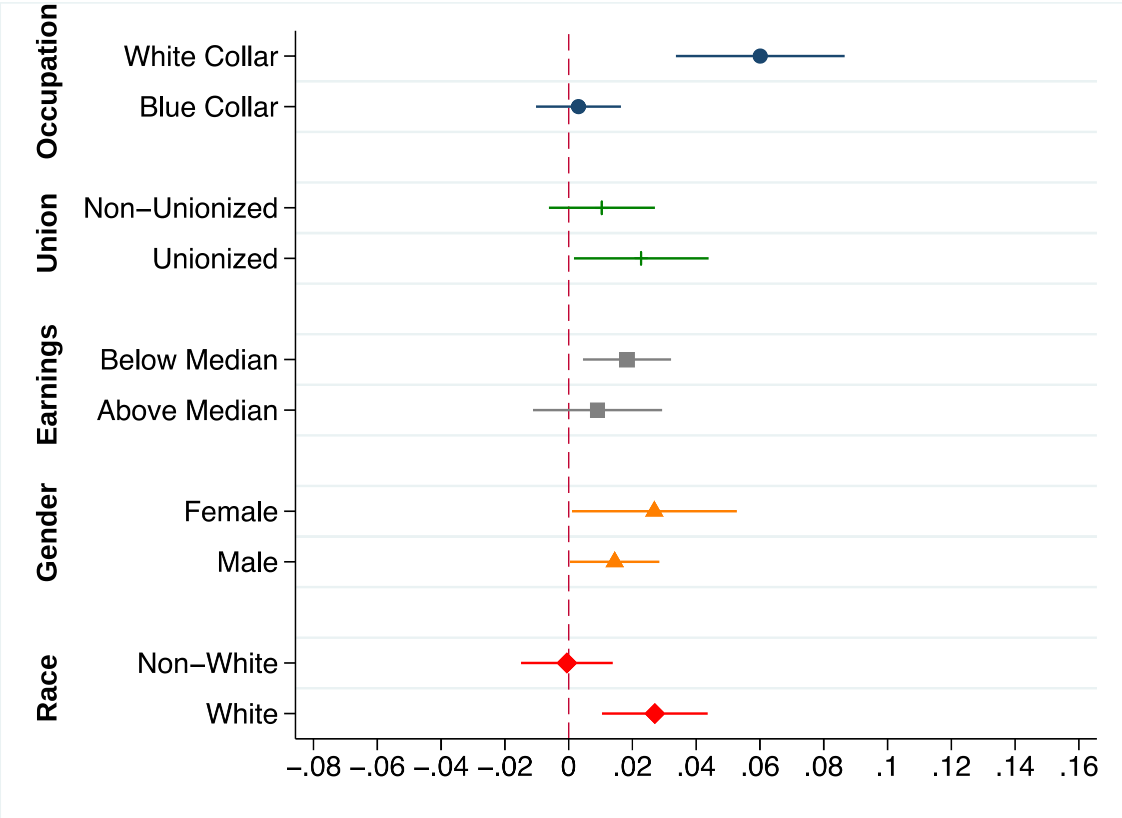
supply elasticity. This suggests that the lower skill labor market operates closer to perfect competition, as a commodity type of labor.

Minimum wage. One might wonder if the small earnings effect presented in figure 4 is driven by the minimum wage constraint. To address this question, I classify workers into the minimum wage categories based on the modal pre-reform minimum wage status. On average 20% of workers in the sample are constrained by the minimum wage. Figures D.6 and D.7 suggest that this is not the case, as there is no statistical difference between the earnings effect for workers below and above the minimum wage barrier. One rational for this result is that high informality rates, and low minimum wage levels are likely to bind even before the reform. Therefore, the labor demand expansion generates earnings response above the minimum constraint.

Racial wage gap. The payroll tax program does not distinguish workers based on background characteristics, such as race. It offers a flat 20 percentage point cut that remains constant across all income levels, suggesting no explicit intention to disadvantage workers of a specific race. However, this does not imply that the payroll tax cut did not intensify the existing racial wage gap. As discussed throughout this paper, equal tax cut can lead to a higher pass-through to high-wage occupations. If race correlates with occupation or other factors that determine unequal pass-through, the tax system can inadvertently widen the racial wage gap. A unique feature of the Brazilian data is its ability to identify workers and their race. I utilize the policy-induced tax variation to find that white workers benefit significantly more from the reform in comparison to non-white workers.

This intriguing result is estimated using regressions 5 and 6, which include firm fixed effects. Therefore, the expansion of the racial gap should not be attributed to firm sorting. Table 5 summarizes the heterogeneity analysis across multiple characteristics. By merging the dataset with the universe of *Collective Bargaining Agreement (CBA)* signed in the country, I find no statistically significant difference in the pass-through to workers in unionized versus non-unionized firms. There is also not a statistical difference across gender or baseline earnings.

Figure 5: Heterogeneities on the Earnings Pass-Through



Note: This figure presents the IV difference-in-differences coefficient for the earnings effect at the worker level sample, across many characteristics of interest, such as, occupation, unionization status, earnings, gender and race. The unionization variable is a firm level dummy that indicates if the firm had an unionization contract during the pre-reform period. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Unintended discrimination. This paper introduces another critical aspect to the public debate. Despite racial discrimination being a pressing social issue in modern society, it has not been incorporated into the tax literature[40]. This might be because modern tax codes do not contain any explicit elements of racial discrimination that could be classified as either statistical or taste-based discrimination. This study provides novel evidence that behavioral responses to tax changes can lead to unintended consequences on racial inequality, that must be addressed in future tax policy.

[40] A few exceptions are Brown 2022 and Holtzblatt et al. 2023, which study racial inequality in the context of couples' taxation in the US

3.3 Validity of Results

The identifying assumption is that eligibility, conditional on fixed effects, is uncorrelated with time-varying, unobserved factors influencing employment and wage growth. The validity of this assumption would be violated if Congress anticipated sector-specific trends during the definition of eligibility rules. Another issue could arise if firms strategically chose sectors after the reform was announced. In this section, I provide multiple tests to address both of these concerns related to selection on eligibility and sector choice manipulation.

3.3.1 Selection on Eligibility

Trends. The concern about selection on eligibility is whether firms that were granted the eligibility status might have been on different trends relative to those that weren't. To address this, it's common practice to evaluate pre-existing trends. Figures 2 and 4 depict event study coefficients that display reassuring pre-reform results, not statistically different from zero. This suggests that, in the absence of tax reform, the outcomes for both eligible and non-eligible firms (and workers) would have followed parallel trends in the post period, if the tax reform had not been enacted.

Baseline Levels. Besides parallel trends, firms and workers across various eligibility statuses also demonstrated a balance in baseline levels during the pre-reform period, as corroborated by Tables 8 and 9. One characteristic that did not exhibit a balanced distribution across groups was gender. Therefore, I included gender as a control variable in all specifications. Another method of illustrating the relationship between a firm's characteristics and its eligibility status is through a regression analysis conducted pre-reform:

$$X_{jt} = L_{s(j)} + \alpha_j + \gamma_t + \xi_{s1(j),t} + u_{jt}$$

where $Ls(j)$ is a dummy to indicate if the firm is eventually eligible; X_{jt} are characteristics of the firm; and the fixed effects are the same used in the main empirical specifications. Table D.3 reports results. In particular, it shows that under the TWFE model - identical to the one implemented in the main specification - there is a sharp zero difference across eligibility groups in the pre-reform period.

Anecdotal evidence. The striking similarities in trends and levels across eligibility groups are not surprising. The political political process that assigned eligibility was mostly based on the finest sector definition (7 digits), and have placed plausibly similar sectors on opposite side of the eligibility status. For instance, the sector of trains were eligible, but touristic trains were not. The sector of open television is eligible, but cable television is not. Table 6 provides a non-exhaustive list of numerous additional examples, further illustrating this point.

Alternative Identification. The reason why specific sectors was chosen was not disclosure, nor was there an objective criteria to determine eligibility. From an econometric stand point and potential identification concerns, it has been established that the sector choice did not seem to anticipate sector trends. To further investigate underlying criteria that determined eligibility, I fit a logit model on baseline firms' observable characteristics. I then use the propensity scores to break ties in a procedure, which matches firms based on pre-reform deciles on employment, wages and hires. In this matched sample, I conducted a difference-in-differences analysis as a robustness check. Interestingly, in this alternative empirical strategy, the identification assumption hinges on the Conditional Independence Assumption (CIA), validated by the balance tables in Appendix C.2. Importantly, this strategy does not make any assumptions about the political process that determines eligibility.

The results from both the primary empirical strategy and the alternative matching approach are qualitatively similar. A detailed analysis and the corresponding results are provided in Appendix C.2. To further substantiate the matching approach, I conducted additional robustness checks. I randomly assigned a placebo treatment and applied the same matching process to these placebo-treated firms. As anticipated, the placebo tests yielded zero effects on employment and wages, thereby providing evidence that the results are not influenced by any inconsistencies in the matching algorithm.

3.3.2 Manipulation on Sectoral Choice

Sector immobility. Given the seemingly arbitrary nature of eligibility assignment, one might wonder whether firms could manipulate their sector classification to move towards eligible sectors. In this scenario, the concern is that firms expecting employment growth could self-select into treatment, thereby compromising the causal interpretation. Fortunately, our panel data allows us to track firms and assess whether they changed sectors upon the reform's implementation. The data shows only a small number of firms changing sectors, and among these, there is not a trend of switching towards eligible sectors. This low manipulation response aligns with the bureaucratic challenges associated with changing sectors.

Bureaucratic process. Firms in the regular tax tiers (object of this study) face a long and costly process to change sectors. In order to do so, they need first to change their operating agreement, which requires proof that they are operating in a new industry. Subsequently, they must request new operational licenses from multiple administrative bodies, including city hall, state, federal tax authorities, and others. Additionally, they must obtain clearance from local tax authorities and civil registry offices. Failure at any of these steps can result in sanctions and fines.

Extra robustness. To further ensure that sector changes are not driving the results, I conducted several additional robustness checks. First, I assigned firms to eligibility based on their pre-reform sectors, and the results remained qualitatively the same. Similarly, when I restricted the sample to firms that have never changed sectors, the results were unchanged. All these tests taken together, indicate that sector manipulation is not an active margin of response, which reinforces the causal interpretation of the results.

3.3.3 Informality

The informality analysis is not as much intended to provide further validation of the empirical design, but aimed at exploring the economic interpretations of the results. Given the identified causal employment response to the tax cut, it's worth considering whether the increase in employment is due to the formalization of existing employees or the addition of new ones. I present several pieces of evidence indicating that informality is not driving the results.

Transition. The panel structure of the data allows to track previous employment spells for workers who held formal jobs in the past. Essentially, the data enables to determine for each new hire whether they transitioned from non-employment/informality or from another formal job. If the positive employment effect were due to the hiring of existing informal workers, we would expect to see a sharp increase in the proportion of new hires transitioning from non-employment or informality in treated firms after the reform. However, as Figure D.8 indicates, the proportion of new hires coming from non-employment and informality remains constant over time and across treatment status, suggesting that formalization is not a significant response margin.

Regional variation. Another approach to the informality question is to leverage the regional variation in informality rates. Brazil, a large and diverse developing economy, has local labor markets that range from those resembling developed economies to those similar to African countries. Two years prior to the payroll tax reform, the Brazilian Census Bureau conducted a national Census survey that provided rich regional informality data at the municipality level. As Figure D.9 shows, there is a wide range of informality rates across Brazil's 5,300 municipalities.

I exploit this variation to distinguish the effects of a payroll tax reform in settings with different degrees of exposure to informality. I divide regions into two groups, those below and above the median in terms of formalization rate. If the main employment response to the tax cut was driven by the mere formalization of informal workers, would reasonable to expect larger employment effects in regions with high informality. However, my findings indicate the opposite (Figures D.10 and D.11). One might still be concerned that the labor cost variation induced by the policy in low and high informality areas can be different. I show in figure D.12 that the first stage is uniform across informality status, reinforcing that formalization is not driving the results.

Workers' Education. As Ulyssea 2018 notes, informal employment is concentrated in firms with lower average education. The matched employer-employee data provides information on workers' educational level, allowing us to observe average education per firm. In Figure D.13, I show that the employment effects are larger in firms with higher shares of qualified workers, i.e., firms less likely to hire informally. This serves as additional evidence that the employment effect is not driven by informality.

Discussion. The empirical evidence suggesting that informality is not a primary driver of employment responses aligns with several factors highlighted in previous research. First, informality in Brazil is primarily driven by self-employment rather than informal employment in a formal firm (PNAD, 2012). This implies that informal workers are more similar to entrepreneurs than employees, and their decision to formalize involves other costs such as: licenses to operate, costs related to opening and maintaining a firm, other corporate taxes, legal liabilities, and sanitary and security regulations. Second, even though there is a reduction in labor cost, the worker's decision to formalize extends beyond a simple cost-benefit analysis[41], as there appears to be a segregation between for-

[41] Alcaraz, Chiquiar, and Salcedo 2015; Dalberto and Cirino 2018

mal and informal labor markets (see Perry 2007 for discussion).

3.4 Other Margins of Adjustment

We have empirically established that upon a sizable tax cut, firms present substantial employment growth and moderate increasing to workers' earnings. However, these two response dimensions alone don't provide a comprehensive understanding of the tax incidence. For instance, it isn't immediately clear whether the increase in employment arises from scale or substitution. These alternative channels influence prices and revenue differently, and thus have disparate impacts on the incidence to consumers. Furthermore, drawing conclusions on the incidence to firm owners, which requires knowledge of the causal effect on profits, is not feasible based on employment and earnings effects alone.

To measure these responses, we rely on firm level anonymized balance sheet information from tax records. Unfortunately, not all formal firms have to report balance sheet information. Only firms in the "Real Profit" tax tier[42] are obligated to file those information. Table 3 displays the results. Notably, within this tax tier, the treatment effects on labor market outcomes closely resemble those estimated for the entire universe of formal firms (refer to columns 2 and 4). Figure 20 further validates the parallel-trends assumption, demonstrating that the pre-reform coefficients are not statistically different from zero for all outcomes.

The payroll tax reform presents an unambiguous incentive for firms to expand employment from both the scale and substitution perspectives. Nevertheless, the usage of capital is subject to two counteracting forces. On one hand, reduced labor costs stimulate production, thus positively affecting capital demand. On the other hand, lower labor costs generate incentives to substitute labor for capital. Column (3) quantifies the net effect on capital, as well as the relative importance of these two channels. It shows that as payroll taxes decrease, firms expand employment and shift away from capital.

The identification of scale effects, based on the choice of inputs, serves as a crucial step towards determining price responses. This process is formally executed in Section 5, where I estimate demand elasticity, shedding light on the scale response. Combining scale and revenue pins down prices, which is instrumental in assessing the incidence passed onto consumers. The extremely close fit enhances the reliability of the structural estimations, as shown in section 5,

[42] "Real Profit" is a tax tier for larger firms with annual gross revenue above BRL $78 Million, approximately USD $15 Million at current exchange rate.

and the overidentification tests. Another advantage of observing many margins of responses is that it allows to evaluate the coherence among multiple channels and models' predictions. The revenue effect is presented in column (5).

Relying solely on employment responses requires the use of structural assumptions to map input choices to profit outcomes, an approach often employed in tax incidence literature (e.g., Suárez Serrato and Zidar 2016; Suarez Serrato and Zidar 2023). The comprehensive nature of the administrative data used in this study enables a clean empirical approach to document the tax incidence to firm owners (as shown in column 6). Given that profits can plausibly be negative or zero, I opted not to rely on logs for this specific outcome. Instead, I leverage the inverse hyperbolic sine transformation, which accommodates all observations. Results are further corroborated using the level counterparts expressed in Figure 20.

Table 3: Firms' Margins of Adjustment

	(1) Log Labor Cost $(1+\tau)$	(2) Log Employment	(3) Log Capital	(4) Log Earnings	(5) Log Revenue	(6) Asinh Ebit
Panel A: Diff-in-Diff						
Baseline	-.1321***	.1019***	-.0967***	.0251***	-.073***	.0399
	(.0063)	(.0206)	(.0335)	(.0063)	(.0232)	(.0417)
Small Firms	-.1352***	.2691***	-.0858	.0563***	-.0216	-.0376
	(.016)	(.0366)	(.0784)	(.016)	(.0516)	(.0448)
Large Firms	-.1308***	.0573**	-.0953***	.0152**	-.0831***	.0723
	(.0059)	(.0239)	(.0361)	(.0059)	(.0251)	(.057)
Panel B: Long-Diff (t + 3)						
Baseline	-.1247***	.1279***	-.0337	.0309***	.0495	.1223*
	(.0063)	(.0206)	(.0335)	(.0063)	(.0232)	(.0417)
Small Firms	-.1275***	.3461***	.0085	.0602**	.1864**	.1864
	(.016)	(.0366)	(.0784)	(.016)	(.0516)	(.0448)
Large Firms	-.1253***	.0969***	-.0312	.0228***	.0223	.1992**
	(.0059)	(.0239)	(.0361)	(.0059)	(.0251)	(.057)
Controls	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓
Sector x Year FE	✓	✓	✓	✓	✓	✓
N	449, 679	450, 387	345, 217	450, 387	374, 774	377, 619

Note: This table reports difference-in-differences and event study coefficients intrumented by sector eligibility. Each column reports different margins of adjustment, such as labor cost, employment, revenue, earnings, gross revenue, and profits. Results are presented for the baseline sample, and separately per firm size, which is defined with respect to the median in the pre-reform period. Panel A reports the difference-in-differences coefficients. Panel B reports the long-diff coefficients, which are the event study coefficients for t+3. Standard errors are reported in paranthesis.

3.5 Alternative Models of Labor Market Power

The positive earnings pass-through as a response to a firm-specific shock motivates the imperfectly competitive labor market. However, rejecting competitive markets doesn't necessarily imply that the wage setting process follows a standard monopsony model (as in Manning 2011; Card et al. 2018; Kroft et al. 2020). In fact, models of bargain (Hall and Milgrom 2008; Pissarides 2009; Jarosch, Nimczik, and Sorkin 2019), dynamic wage posting (Burdett and Mortensen 1998;

Postel–Vinay and Robin 2002), and static wage posting with recruiting costs (Kline et al. 2019; Manning 2006) also predict positive earnings effect, but they rely on different forces, which in turn, imply distinct policy implications. For example, while in the standard monopsony, the wage pass-through is driven by the employment expansion, in the bargain models, the underlying force is the surplus of the employee-employer match. Prior work, have faced identification challenges to disentangle these two forces.

Important to highlight that this paper doesn't have the ambition to discover an universal formula to explain how wages are set. There are several particularities on the way wages are negotiated in the real world. However, this subsection evaluates testable implications for each class of models, in order to find the most suitable framework to interpret the consequences of tax policy. I begin by discussing and falsifying potential candidates. Then, I provide supporting empirical evidence to the standard monopsony model.

Bargaining. In models such as those proposed by Hall and Milgrom 2008, Pissarides 2009, and Jarosch, Nimczik, and Sorkin 2019, wages are determined by the split of the surplus from the employer-employee match. Seen through this lens, the surplus generated by the Brazilian payroll tax reform exerts additional upward pressure on wages. Although it is not feasible to empirically disentangle the extent to which wage pass-through is attributable to surplus sharing versus employment growth, other empirical predictions can be tested. For instance, the pass-through argument via bargain operates during wage (re)negotiations. Given that such renegotiations occur less frequently for existing employees compared to new hires, the bargaining model would predict a higher wage pass-through for new hires relatively to incumbents. However, this hypothesis is empirically rejected, as reflected in figure D.14.

Dynamic Wage Posting. A popular framework to motivate firms' monopsonistic power hinges on search frictions, where firms are able to poach and negotiate over workers in other companies (Burdett and Mortensen 1998; Postel–Vinay and Robin 2002). In this model, high-productivity firms grow larger as they can attract and retain more workers, offering employees an increased likelihood of receiving matching counteroffers when presented with outside opportunities. Consequently, large firms tend to hire more via poaching, whereas small firms rely on hiring from non-employment. The Brazilian tax reform, by reducing labor costs, increases the surplus of the employment match, leading to several testable predictions. Small firms shouldn't exhibit a higher wage pass-through than larger ones. Large firms engaging in poaching might display higher wage pass-through as the poached firm, with some probability[43], have also experienced a tax cut and will bid back to retain the worker. This argument is formalized in Biro et al. 2022. My tests refute the hypothesis that small firms have greater wage pass-through than large firms, as shown in Figure 16, providing evidence against dynamic wage posting models.

Another implication is that when a small share of firms in the local labor market is shocked with a labor cost reduction, the outside option of the prospective worker is not affected, therefore we should see no treatment effect on new hires' wage.Conversely, treated firms, given their increased capacity to counterbid, may be more likely to retain poached workers, resulting in potentially higher wage pass-through for incumbent employees. However, the data does not corroborate this prediction. Figure D.14 shows that there is not a statistically significant difference between incumbent and new hire's wage effect, serving as an additional piece of evidence against dynamic wage posting.

Recruiting Cost. Several established papers have found empirical support for wage posting models with recruiting cost, in the context of technological firms in the US (Kline et al. 2019), and survey responses from large UK companies (Manning 2006). Applying these models to interpret the Brazilian tax reform, generates the prediction that incumbent workers would have a rent premium because they are expensive to be replaced. In Kline et al. 2019, the marginal cost of new hires is $c'(\frac{L_j}{I_j}) + w_j^m$, where $c(\dot{})$ is a convex cost function and w_j^m is the market wage. Figure D.14 tests the main prediction from this model, and rejects that incumbent workers experience a wage premium pass-through relative to incumbents.

Standard Monopsony. In its original form, wage posting models don't allow firms to negotiate wages differently with new hires and incumbent workers. For this reason, there should be no pass-through wedge across job tenure, which aligns with the empirical findings in this paper. It is also consistent with the fact that firms experiencing a greater employment effect due to treatment are likely to share more rents with inframarginal workers, thereby leading to greater wage responses. In order to fully rationalize the firm-level heterogeneities (as presented in Figure 16), one must take into consideration the potential for imperfect competition in the goods market, as well. This influences the incidence on the marginal revenue product of labor (MRPL) through the scale effect. When in-

[43] In the Brazilian case, this probability is small, because a small share of firms are affected.

tegrated, these considerations help to rationalize all the empirical findings from the Brazilian tax reform, bringing together two standard methods of modeling frictions in both the labor and product markets.

4 Model

In this section, I develop a simple model of factor demand with imperfect competition in the goods and labor market, intended to shed light on the role of market power to drive the efficiency and distribution of tax reforms. The labor market power is tailored to capture the empirical fact of positive wages pass-through after an idiosyncratic shock to the firm. Additionally, the combination of model and data allows me to estimate structural parameters, such as, wage markdown, capital-labor elasticity of substitution, and demand elasticity. Those are key elements to shed light on important mechanisms of adjustment to the tax policy, which inform policy makers on the design of alternative policies.

4.1 Preliminaries

The model has a single period, and firms operate in a partial equilibrium framework[44]. Firms are endowed with a CES technology with constant returns, which uses capital and labor as inputs,

$$f(L, K) = (s_L L^\rho + s_K K^\rho)^{\frac{1}{\rho}}$$

where, s_g are the inputs' cost share ($g \in \{L, K\}$). The capital market operates in perfect competition, which means that the marginal productivity of capital equals its cost. However, the labor market operates in imperfect competition, and the labor supply elasticity ϵ dictates the firm's ability to mark wages below the marginal revenue product of labor. Firms face an upward sloping labor supply curve, and cannot discriminate wages across incumbents and new hires, as in standard monopsony models (Card et al. 2018).

$$w_j = A_j L_j^{\frac{1}{\epsilon}}$$

The wage-setting rule suggests that if wages rise due to firm-specific shocks,

[44]This is motivated by the nature of the tax reform, which impacts individual firms but not the market as a whole.

both incumbents and new hires benefit equally - an observation supported by the data.[45] From a theoretical standpoint, the static labor supply curve can be micro-founded by an analogy to Industrial Organization's discrete choice models, which are employed to estimate demand with differentiated goods. In the labor market context, the "differentiation" arises from workers' preference for particular employers. This argument is formalized in appendix B, and the labor supply curve faced by the firm is derived according to workers' idiosyncratic preferences.

The output market operates in monopolistic competition, where firms choose quantity and face a constant price elasticity η (Hamermesh 1996; Criscuolo et al. 2019). The degree of monopolistic power is dictated by the parameter η, which is flexible to accommodate any market structure. Given the output choice, firms solve a cost minimization problem to decide on the input mix. The cost of labor is, $C_L = (1+\tau)wL$, where, τ is the average payroll tax rate. The government can manipulate labor cost through τ-perturbations, with the percentage variation on labor cost induced by the Brazilian policy denoted by ϕ_1.

4.2 Firm's Problems

Profit Maximization The firm chooses output to maximize profits, according to the following program:

$$\max_{Q} \underbrace{Q^{1-\frac{1}{\eta}}}_{\text{Revenue}} - \underbrace{A(1+\tau)L^{1+\frac{1}{\epsilon}} - rK}_{\text{Cost}}$$

At the optimum, firms choose quantity that equates marginal cost to marginal revenue,

$$\underbrace{\left(\frac{\eta-1}{\eta}\right)Q^{\frac{-1}{\eta}}}_{\text{Mg Revenue}} = \underbrace{\frac{\partial C(\tau, Q)}{\partial Q}}_{\text{Mg Cost}} \tag{9}$$

In contrast to a perfectly competitive environment, the marginal cost no longer operates as a linear function of the output level (see proof of lemma 1, in ap-

[45] The subscript j refers to a specific firm, but for ease of notation this subscript will be omitted in the rest of the paper.

pendix B). The intuition is that there is an increasing cost to expand plant size due to inframarginal wages. As a result, the imperfect labor competition limits the pass-through to employment. Mathematically, equation 9 reveals how output level influences labor demand by raising the marginal cost of scale expansion. This relationship is increasing in the firm's market power (decreasing in η). The employment effect is further determined in the cost minimization program, which I turn to next.

Cost Minimization Once the output quantity is fixed, firms decide on the input mix that minimizes cost. Formally,

$$\begin{aligned} \min_{K,L} \quad & A(1+\tau)L^{\frac{1}{\epsilon}+1} + rK \\ \text{s.t.} \quad & f(K,L) \geq Q \end{aligned}$$

At the optimal, the labor choice equates the marginal cost of labor to the marginal revenue product of labor,

$$\underbrace{\overbrace{\left(\frac{\epsilon+1}{\epsilon}\right)}^{\text{inverse mark down}} A(1+\tau)L^{\frac{1}{\epsilon}}}_{\text{MCL}} = \underbrace{\lambda f^{1-\rho} s_L L^{\rho-1}}_{\text{MRPL}}$$

Note that the marginal cost of labor decreases on the level of labor market competition, which guides the steepness of the labor supply curve. Putting together the optimal inputs choice and applying the envelope theorem, I can compute the cost function. The monopsony power in the labor market breaks the linear relationship between average and marginal cost:

$$\underbrace{\frac{\partial C}{\partial Q}}_{\text{Mg Cost}} = \underbrace{\frac{C}{Q}}_{\text{Avg Cost}} + \overbrace{\underbrace{\frac{C_L}{\epsilon}\frac{1}{Q}}_{\text{Avg Incumbent Rent}}}^{\textbf{New!}}$$

I denote this new term as the average incumbent's rent, because it relates to the wage increase perceived by inframarginal workers when the firm increases plant size. In particular, the rent converges to zero as we move to perfect competition ($\epsilon \to \infty$), similarly to traditional models (Hamermesh 1996). The nonlinearity in the cost function will be key to understanding the pass-through responses to payroll tax reforms.

4.3 Pass-Through

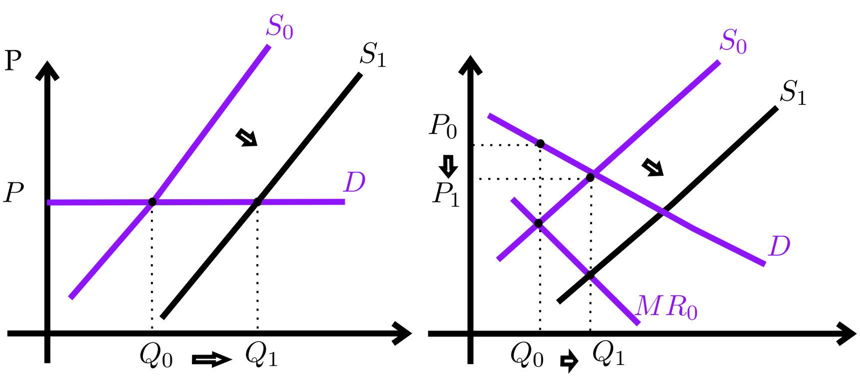
The pass-through of the payroll tax reform operates through two mechanisms. The output market, which determines the scale effect. And the input mix, which dictates the substitution across production inputs.

4.3.1 Output Market

In the output market, a payroll tax reduction shifts the supply of goods. The output consequences depend (i) on the magnitude of the shift; and (ii) the slope of the demand curve. Figure 6 illustrates how the price effect increases with market power. To quantify this effect, I totally differentiate the pass-through equations to compute the price elasticity with respect to labor cost:

$$\epsilon^{P}_{1+\tau} = \frac{-1}{\eta}\epsilon^{Q}_{1+\tau}$$

Figure 6: Conceptual Framework - Product Market



Note: This figure illustrates the pass-through in the product market from a firm-specific payroll tax cut. The left graph shows the intuition for the case of perfect competition in the output market. In this case, cost reduction shifts supply of goods, but doesn't affect prices as firms are price takers. On the right, the graph depicts the intuition for the monopolistic case. Compared to the perfectly competitive case, there is a smaller quantity (or scale) effect due to the price setting power.

The price elasticity depends not only on the market power, which is determined by the constant elasticity η, but also on the scale effect. Because of monopsony power, the scale effect cannot be evaluated solely based on the product market. Remember that now, the inframarginal rent affects the plant size decision. At the optimum, the effects of tax policy on marginal revenue should equate the effects on marginal cost:

$$\underbrace{-\frac{1}{\eta}\epsilon^{Q}_{1+\tau}}_{\substack{\text{Effect on mg revenue} \\ (\eta \text{ drives slope of D})}} = \underbrace{\overbrace{\epsilon^{\lambda}_{1+\tau}}^{\text{Direct}} + \overbrace{\epsilon^{\lambda}_{Q}\epsilon^{Q}_{1+\tau}}^{\text{Inframarginal}}}_{\substack{\text{Effect on mg cost} \\ \text{(shift in supply curve)}}} \tag{10}$$

Equation 10 sheds light in two mechanisms through which imperfect competition affects tax incidence. First, competition in the output market flattens the demand curve ($\frac{-1}{\eta}$), enhancing the scale effect. Second, competition increases the pass-through to the marginal cost, amplifying the shift in the supply curve and thereby the scale effect. It is important to highlight that the elasticities expressed in equation 10 are endogenous to the tax system, and the pass-through to the marginal cost ultimately depends on the level of competition in both the product and labor markets.

4.3.2 Labor Market

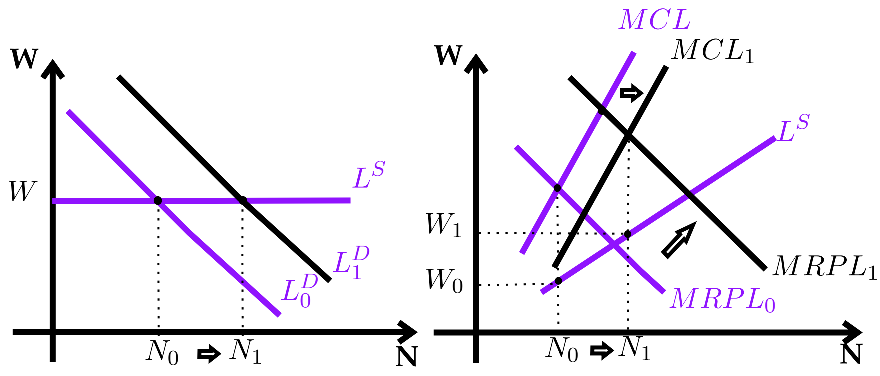
Labor market forces at work that determines tax pass-through are reflected on effects to the marginal cost of labor and marginal revenue product of labor. Figure 7 provides intuition on the interaction between imperfect labor competition and a firm-specific shock.

Equations 11 and 12 quantify the elasticity of the marginal cost of labor, and elasticity of marginal revenue product of labor with respect to labor cost, in the case of monopsonistic labor markets.

$$\frac{\partial \log MCL}{\partial \log(1+\tau)} = \underbrace{1}_{\text{Direct effect on MCL}} + \frac{\overbrace{\epsilon^{L}_{1+\tau}}^{\text{Inframarginal on MCL}}}{\epsilon} \tag{11}$$

$$\frac{\partial \log MRPL}{\partial \log(1+\tau)} = \underbrace{\epsilon^{\lambda}_{1+\tau} + \epsilon^{\lambda}_{Q}\epsilon^{Q}_{1+\tau}}_{\text{Direct + inframarginal on mg rev}} + \overbrace{(1-\rho)(\epsilon^{Q}_{1+\tau} - \epsilon^{L}_{1+\tau})}^{\text{Effect on MPL}} \tag{12}$$

Figure 7: Conceptual Framework - Labor Market



Note: This figure illustrates the pass-through in the labor market from a firm-specific payroll tax cut. The left graph shows the intuition for the case of perfect labor market competition. In this case, payroll tax cut shifts right the demand for labor, but doesn't affect wages since firms have no wage setting power. On the right, the graph depicts the intuition for the monopsonistic case. In this case, the employment effect is not as large, but as the tax reform expands labor demand, it provokes a wage increase.

Marginal Cost of Labor The pass-through to the marginal cost of labor (MCL) comprises two components. Like in a perfectly competitive labor market, the first component directly correlates MCL with variations in labor cost. The second component is unique to monopsonistic firms due to the upward slope of their labor supply curve. This component elevates the MCL through two channels: (i) the responsiveness of labor demand, which causes the firm to move further along the labor supply curve, and (ii) the firm's labor market power, which is reflected in the steepness of the labor supply curve.

Marginal Revenue Product of Labor As equation 12 suggests, the effect of the tax policy on the marginal revenue product of labor depends on the pass-through to the marginal product of labor (MPL), and to marginal revenue. The pass-through to the marginal product of labor is negatively related to the substitution across inputs ($\sigma_{KL} = \frac{1}{1-\rho}$), positively related to the scale effect, and negatively related to the employment effect. The pass-through to the marginal revenue, depends on the direct and inframarginal effects of the firm-specific labor cost variation. Note that the marginal revenue depends on the labor cost $(1 + \tau)$, and on the output level. Therefore, when the firm reacts to a labor cost reduction by increasing plant size, the scale effect inflates costs, offsetting part of the initial cost reduction. Equation 13 quantifies these mechanisms by relying on envelope arguments.

$$\overbrace{\frac{\partial \lambda(Q,\tau)}{\partial(1+\tau)}}^{\substack{\text{Effect on}\\ \text{mg cost}}} = \overbrace{\frac{AL^{1+\frac{1}{\epsilon}}}{Q}}^{\substack{\text{Effect on}\\ \text{avg cost}}} + \overbrace{\frac{AL^{1+\frac{1}{\epsilon}}}{Q\epsilon}}^{\substack{\text{Direct effect on}\\ \text{incumbent rent}}} + \overbrace{\frac{A(1+\tau)}{\epsilon}\left(\frac{\epsilon+1}{\epsilon}\right)\frac{L^{\frac{1}{\epsilon}}}{Q}\frac{\partial L}{\partial(1+\tau)}}^{\substack{\text{Indirect effect on incumbent}\\ \text{rent from L response}}} \tag{13}$$

The interaction between labor market power, and pass-through to the marginal cost is unambiguous. The higher the market power, the higher is the direct pass-through to the incumbent's rent, and it also amplifies the indirect effect on the incumbent's rent due to labor responses.

5 Structural Estimation

This section connects model and data. A clean structural estimation allows me to credibly estimate parameters of interest, understand mechanisms, and design counterfactual policies. To operationalize the structural estimation, first I derive the model's predictions for the effect of the Brazilian payroll tax reform on employment, capital, earnings and revenue. Then, I solve it for the structural parameters as a function of the reduced form estimates.

5.1 Parameters Estimate and Mechanisms

Model's Predictions. Following the derivation outlined in section 4 (and detailed in appendix B) I compute closed form solutions for the tax pass-through to employment, capital, earnings and revenue due the Brazilian payroll tax reform. To embrace all the elements of the Brazilian tax reform, I also take into account the revenue tax variation, which turns out to have muted effects due to the small rate variation on the revenue side, and the small share of firms subjected to this tax[46]. Once we account, for product and labor market power, the pass-through of the Brazilian tax reform to employment, capital, earnings and revenue can be expressed as a function of observables and three parameters to be estimated (ϵ, η, ρ):

$$\beta_L = \left(\frac{\epsilon}{1+\epsilon(1-\rho)}\right)\left(\Omega(\epsilon,\eta,\rho) - 1\right)\phi_1 \tag{14}$$

[46]Since the revenue tax has negligible effects, I will omit them in the main text. Careful derivation of the revenue tax perturbation can be found in appendix B.3.

$$\beta_K = \psi(\epsilon)\left[\left(\frac{1}{1-\rho}\right)\left(\frac{(\epsilon+2\epsilon^L_{1+\tau})(1-\eta)}{\epsilon+\epsilon^L_{1+\tau}}\right)\right]\phi_1 \tag{15}$$

$$\beta_{Rev} = \psi(\epsilon)\left[(1-\eta)\left(\frac{\epsilon+2\epsilon^L_{1+\tau}}{\epsilon+\epsilon^L_{1+\tau}}\right)\right]\phi_1 \tag{16}$$

$$\beta_W = \frac{\epsilon^L_{1+\tau}}{\epsilon}\phi_1 \tag{17}$$

where, $\Omega(\epsilon,\eta,\rho)$ and $\psi(\epsilon)$ simplify a combination of terms detailed in the Appendix B; ϕ_1 measures the first stage associated with the policy, i.e., the percentage variation on tax rates induced by the reform. Mathematically, $\phi_1 = \frac{\partial \log(1+\tau)}{\partial Reform}$. Using anonymized tax data, I precisely estimate ϕ_1.

The incidence model developed here is more general than prior work that assumed perfect labor market competition. In my framework, one can think of the perfectly competitive labor market as a particular case, where ϵ goes to infinity[47]. Taking the limit of pass-through expressions 14-17, I recover the exact same expressions derived in a standard Marshall-Hicks analysis and estimated by Curtis et al. 2021; Criscuolo et al. 2019; and Harasztosi and Lindner 2019. In the standard competitive case, substitution and scale effects are separable, as illustrated in the expressions below.

$$\lim_{\epsilon\to\infty}\beta_L = \left(\underbrace{\frac{-s_K}{1-\rho}}_{substitution} - \underbrace{s_L\eta}_{scale}\right)\phi_1 \qquad \lim_{\epsilon\to\infty}\beta_K = \left(\underbrace{\frac{s_L}{1-\rho}}_{substitution} - \underbrace{s_L\eta}_{scale}\right)\phi_1$$

Estimation Method. To operationalize the structural estimation, notice that the pass-through expressions can be solved for the structural parameters as a function of the reduced form responses estimated in the data from the quasi-experimental design. Equation 17 directly maps the labor supply elasticity faced by the firm to the reduced form elasticities estimated in the data. The intuition is that the ratio of the employment and earnings effect identifies the slope of the labor supply curve faced by firms.

$$\epsilon = \frac{\beta_L}{\beta_w} \tag{18}$$

[47]This limit case moves the economy to a perfectly competitive labor market, where the labor supply faced by the firm is perfectly elastic.

From the capital and labor responses, σ_{KL} is identified:

$$\sigma = \frac{\beta_K - \beta_L}{\beta_W + \phi_1} \tag{19}$$

The intutition is that σ is estimated from the tension between capital and labor substitution. Equation 19 depicts the intuition that as β_K decreases relative to β_L, it is an indication that firms are substituting capital for labor. Also interesting to notice that when the $\epsilon_{w\theta}$ elasticity goes to zero, σ is identified by an expression identical to the one used by previous studies that assumed perfectly competitive labor markets. Another angle to read equation 19 is that ignoring labor market power would generate a bias estimate for the capital-labor elasticity of substitution. Finally, I can identify the output demand elasticity using the capital and revenue responses:

$$\eta = \frac{\sigma\beta_R - \beta_K}{\beta_R - \beta_K} = \frac{-\beta_Q}{\beta_P} \tag{20}$$

Lastly, the economics behind equation 20 is that η can be identified based of the ratio between the scale and price responses to the tax reform. This ratio determines the slope of the demand curve in the product market. The point estimates for each parameter are directly estimated by the combination of reduced form effects outlined in equations 18-20. In the data, I jointly estimate equations 14-17 in order to obtain a covariance matrix, and therefore use the so called, *"Delta method"* to estimate standard errors for each structural parameter.

Parameters Estimate. Table 4 presents the estimates for the three parameters of interest. Column (1) presents the baseline results for all firms, and columns (2) and (3) breakdown the estimates based on firm size.[48] The labor-capital elasticity of substitution ($\sigma_{KL} = \frac{1}{1-\rho}$) is equal to 1.72 (SE 0.57) at the baseline. This result suggests that capital and labor are substitutes, as in Karabarbounis and Neiman 2014[49]. Interestingly, I find that capital and labor are more substitutes in small firms[50] (5.01, SE 2.95), as opposed to in large firms (1.25, SE 0.56). This result is valuable because most of the literature dedicated to capital-labor elasicities is

[48]There are several reasons to breakdown the estimates per firm size. First, they are highly correlated with measures of market power that we can observe, such as market share at the local labor market (see figure ??). Second, it is policy informative, given that firm size is a characteristic easy to target policy on. Third, there is a large literature finding that small firms react more to industrial policies, which contributes to a general interest in understanding small firms' behavior.

[49]Important to highlight that this is not a consensus in the literature. Other recent studies have found that capital and labor are complementaries (Raval 2019).

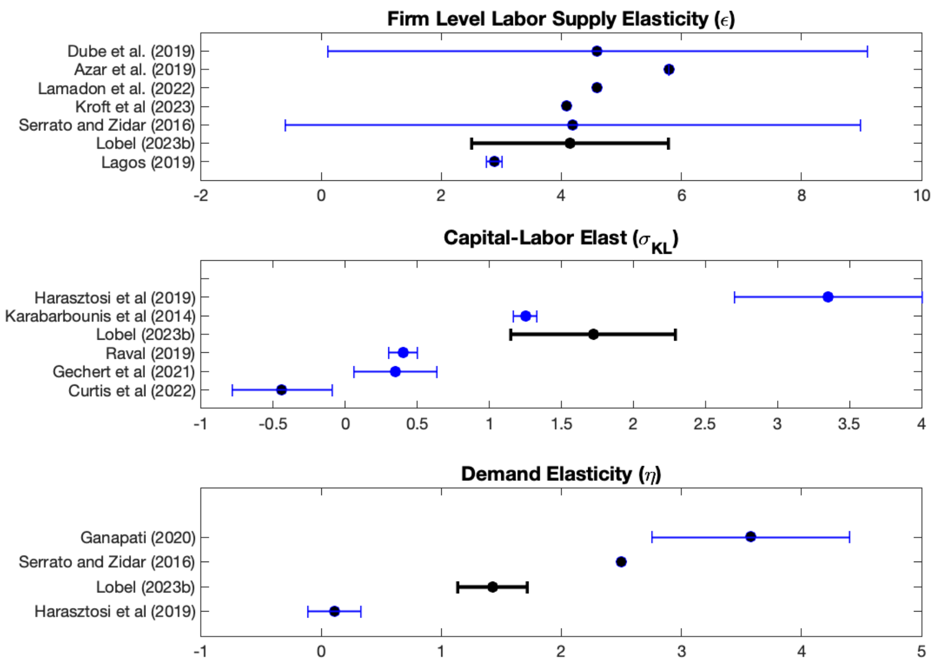
[50]Small firms are defined as the ones below the median in the pre-reform period.

focused on manufacturing and large firms. The fact that the data and the policy variation are comprehensive across a vast firm size range, allows the study to shed light on the capital-labor elasticity for small firms.

The labor supply elasticity faced by the firm ϵ is 4.15 (SE 1.63), which is more precisely estimated, but remarkably close to the 4.19 (SE 4.79) estimated by Suárez Serrato and Zidar 2016. Many other studies find similar labor supply elasticities in the order of 4.08 (Kroft et al. 2020), 4.0 (Card et al. 2018), 4.6 (Lamadon, Mogstad, and Setzler 2022). Figure 8 summarizes the literature, and points out several studies reporting labor supply elasticity between 3 and 5. My baseline estimate implies a wage markdown of 0.81 ($\mu = \frac{\epsilon}{1+\epsilon}$), suggesting that Brazilian workers leave on the table 19% of their marginal revenue product of labor. Columns (2) and (3) report that the labor supply elasticity for small and large firms are 5.75 (SE 2.65) and 4.25 (SE 2.23), respectively. This result is strinkly consistent with the increasing and monotonic relationship between labor market power and firm size demonstrated by Yeh, Macaluso, and Hershbein 2022.

The output demand elasticity with respect to price is 1.43 (SE 0.29), a value greater than one, which aligns with the conventional notion that monopolies operate on the elastic side of the demand curve. If a firm independently decides to raise prices, the quantity loss outweights the revenue gains from higher prices. Heterogeneous responses to the tax variation reveals that large firms have substantial more market power in the product market. The output demand elasticity for small and large firms is 5.21 (SE 2.93) and 1.10 (SE 0.22), respectively. These elasticities are key to examining the theoretical implications of market power on the scale response to a tax cut, phenomenon which I will focus next.

Figure 8: Literature Benchmark



Note: This figure places my estimates with respect to existing estimates in the literature. The parameters are outlined in the x-axis, and its respective estimates on the y-axis. The top panel refers to the labor supply elasticity faced by the firm. The middle panel reports the capital-labor elasticity of substitution. Finally, the bottom panel depicts the output elasticity with respect to price.

Mechanisms. Large firms, due to their market power, have relatively greater influence on prices. This power paradoxically limits their ability to scale up their plant size in response to tax relief. As Table 4 demonstrates, the scale effect for large firms is modest at 6%, accompanied by a similar 6% reduction in prices. Column (1) indicates that after a 15p.p drop in the labor cost, employment increases by 12%[51], with the scale margin accounting for 65% of this effect. The baseline result hides interesting heterogeneity. Column (2) points out that for small firms, the employment effect can be decomposed in 24% scale and 11% substitution away from capital. Column (3) reports that the empirically observed

[51]The empirical moments refer to the long-diff regressions, i.e., responses measured in t+3. This is the preferred timing because some responses such as workers' earnings take time to manifest. Therefore, it is logical to evaluate the firm's responses after allowing for an initial transition period. However, using the difference-in-differences coefficients don't cause abrupt changes to the estimates.

9% employment boost for large firms is 6% due to scale and only 3% substitution. The interplay between scale and substitution has unique implications for analyzing tax incidence and efficiency. A more prominent scaling effect leads to larger price reductions, which ultimately benefit consumers.

Table 4: Structural Estimation

Structural Estimates	(1) Baseline	(2) Small Firms	(3) Large Firms
Labor Supply Elasticity, ϵ	4.15	5.75	4.25
	(1.63)	(2.65)	(2.23)
Labor-Capital Elasticity, σ_{KL}	1.72	5.01	1.25
	(0.57)	(2.95)	(0.56)
Output Demand Elasticity, η	1.43	5.21	1.10
	(0.29)	(2.93)	(0.22)
Mechanisms			
Price effect, β_P	-0.06	-0.05	-0.06
Scale effect, β_Q	0.08	0.24	0.06
Scale / Employment, β_Q/β_L	0.65	0.70	0.64
Empirical Estimates			
Employment effect, β_L	0.12	0.35	0.09
Capital effect, β_K	0.07	0.23	0.03
Earnings effect, β_w	0.03	0.06	0.02
Revenue effect, β_R	0.05	0.18	0.02
Cost Shares			
Labor	0.80	0.80	0.80
Capital	0.20	0.20	0.20

Notes: This table presents the parameters estimated, according to the method presented in 5.1. Column (1) reports results for the baseline case, which includes all firms. Columns (2) and (3) restricts the analysis to small and large firms, respectively. In the "Mechanisms" section, the table reports effects on prices (β_P), scale (β_Q), and the share of employment effect that is explained by the scale response $\frac{\beta_Q}{\beta_L}$. In the empirical section, the table displays coefficients estimated in section 3, and used for the structural estimation. At the bottom, it displays the cost shares used to estimate the model.

5.2 Incidence and Efficiency Gains

In this subsection, I establish the incidence of payroll taxes on workers, firm owners, and consumers. The computation of tax incidence lays the groundwork for a welfare measure, which allows me to compute the efficiency gains from tax reductions or, from a different perspective, the deadweight loss associated with payroll taxation. The goal is to connect empirically observed policy responses to the incidence framework, in order to deliver three key insights. First, a novel payroll tax examination that accounts for the role of consumers in the tax pass-through. Second, the assessment of distortionary costs arising from payroll taxation. Third, the development of a credible framework able to measure the role of market power in shaping the efficiency and distributional consequences of payroll taxation.

Incidence framework. The tax base is determined by the total wage bill:

$$B = wL = AL^{1+\frac{1}{\epsilon}}$$

When payroll tax rates drop, there is a mechanical effect on tax collection,

$$dM = Bd\tau = B(\tau_1 - \tau_0)$$

where, τ_0 is the payroll tax rate in the pre-reform period, and τ_1 is the post-reform rate. Nonetheless, the empirical analysis in section 3 reveals substantial employment and wages responses to tax variation, which partially offset the mechanical tax loss. The resulting behavioral effect is:

$$dH = \tau_1 dB = \tau_1 B\left(\frac{\epsilon+1}{\epsilon}\right)\epsilon^L_{1+\tau}\overbrace{\phi_1}^{\equiv\frac{\partial\tau}{\tau_0}}$$

Putting all together, the impact of the reform on total tax collection is the mechanical effect net of the behavioral adjustments:

$$dR = dM + dH = B\left[(\tau_1 - \tau_0) + \tau_1 B\left(\frac{\epsilon+1}{\epsilon}\right)\epsilon^L_{1+\tau}\phi_1\right]$$

where, $\epsilon^L_{1+\tau}$ is the elasticity of employment with respect to the labor cost, and ϕ_1 measures the effect of the reform on labor cost, which has a negative sign in this case. Given that said, two things worth noting. First, the more employment responds to the reform, the smaller is the loss in tax collection. Second, for a fixed employment response, the larger the labor market power, the smaller the tax loss. The intuition is that in perfect competition wages do not respond to firm expansion.

For each dollar that is effectively lost in tax collection, it is possible to identify the associated gains. To avoid taking stances on the social welfare function, and to ensure comparability with existing literature, I rely on a money metric approach for welfare measurement. To access the tax incidence to firm owners, the causal effects of the tax reform on profits can be directly observed. The comprehensive dataset is an addition to related work that relied on structural models to calculate the tax incidence on firm owners in cases where profits were not observed (Suárez Serrato and Zidar 2016). Therefore, the welfare impact on firm owners is calculated as follows:

$$d\pi = \epsilon_{\pi\theta} B \frac{s_\pi}{s_L} \phi_1$$

where, $\epsilon^{\pi}_{1+\tau}$ is the elasticity of profits with respect to the labor cost, while s_L and s_π represent the labor and profit shares, respectively. I rearrange terms to write the effect on firm owners as a function of the total wage bill. The benefit of this approach is that it allows all individual welfare measures to be referenced to the same base, appropriately weighting the welfare attributed to each stakeholder.

The tax impact on a monopsonistic labor market equilibrium illustrates the implications for worker surplus, as follows

$$dB = w_1 L_1 - \int_0^{L_1} A L^{\frac{1}{\epsilon}} dL - \left(w_0 L_0 - \int_0^{L_0} A L^{\frac{1}{\epsilon}} dL \right) = B \beta_W$$

where, w_0, L_0, w_1, L_1 relates to the wage level and employment before and after the reform, respectively. The intuition is that the incidence borne by workers is dictated by the wage effect. Thus, in a perfectly competitive labor market - where all jobs offer equivalent compensation for a given skill set - the incidence to workers is null. This is because under perfect competition, employment at a specific firm provides no additional benefits, as workers have equally attractive opportunities elsewhere.

Analogously, the tax impact on a monopolistic product market equilibrium illuminates the welfare effects to consumers surplus,

$$dC = \int_0^{Q_1} Q^{\frac{-1}{\eta}} dQ - P_1 Q_1 - \left(\int_0^{Q_0} Q^{\frac{-1}{\eta}} dQ - P_0 Q_0 \right) = \frac{B}{s_L} \frac{\beta_R}{\eta - 1}$$

The intution is that the effect on consumers is driven by the output price reduction. The reform mitigates labor costs, and a portion of this cost reduction is transferred to the output price, thereby benefiting consumers. Despite prices not being directly observed, they can be inferred from the revenue and quantity effects. The estimated scale effect perfectly fits the employment and capital responses, in addition to pass the overidentification tests.

Efficiency gains. The incidence formulae serve as a foundation for two interrelated measures of efficiency. The first measure involves a direct comparison between the welfare gains and the revenue costs triggered by a tax reform. In mathematical terms, the efficiency gains derived from a tax cut can be written as

$$dW = dC + d\pi + dB + dR$$

The second measure is the *"Marginal Value of Public Funds" (MVPF)* metric, applied accross a variety of contexts to evaluate the willingness to pay in relation to the net cost of the policy (Mayshar 1990; Slemrod and Yitzhaki 2001; Kleven and Kreiner 2006; Hendren 2016; Bailey et al. 2020; Hendren and Sprung-Keyser 2020).

Estimates. Upon establishing the theoretical incidence and efficiency formulae as functions of the reduced-form coefficients, I proceed with the structural estimation, as shown in Table 5. Panel B reveals that consumers bear 75% of payroll taxes, while firm owners and workers bear 11% and 14% respectively. The aggregate welfare gains experienced by these stakeholders surpass the decrease in Government revenue, resulting in an efficiency gain of 44% as reflected in the MVPF value of 1.44.

Discussion. The analysis conducted herein reveals insightful findings for the tax incidence literature. A key takeaway is that payroll taxes are predominantly paid by consumers. This novel insight, although not yet thoroughly explored in the tax literature, aligns remarkably with minimum wage incidence studies (Harasztosi and Lindner 2019). Furthermore, the efficiency gain from a tax cut is inversely proportional to the distortionary effects of the tax. Essentially, a higher efficiency gain signifies a more distortionary tax. The substantial welfare gain calculated for Brazil underscores the prevailing notion that taxes exert partic-

Table 5: Structural Parameters and Incidence Estimation

	(1) Identified by	(2) Estimate
Panel A. Parameters Estimate		
Labor Supply Elasticity, ϵ	$\frac{\beta_L}{\beta_W}$	4.14
K-L Elasticity of Substituion, σ	$\frac{\beta_K - \beta_L}{\beta_W + \phi_1}$	1.72
Demand Elasticity, η	$\frac{\sigma\beta_R - \beta_K}{\beta_R - \beta_K}$	1.43
Panel B. Incidence		
Worker, dB	β_W	0.14
Firm Owner, $d\pi$	$\frac{\beta_\pi s_\pi}{s_L}$	0.11
Consumer, dp	$\frac{\beta_R}{s_L(\eta - 1)}$	0.75
Government, dT	$\Delta\tau + \tau_0\beta_L\frac{\epsilon+1}{\epsilon}$	-0.70
Welfare, dW	$\frac{dw + d\pi + dp + dT}{dT}$	0.44
MVPF	$\frac{dw + d\pi + dp}{dT}$	1.44

Note: This table bridges reduced form and structural estimation. Panel A identifies and estimates structural parameters. Panel B identifies and estimates payroll tax incidence to workers, firm owners and consumers. Panel B also reports the connection between welfare and MVPF measures to the reduced form estimates.

ularly distortionary effects in developing economies. This view is support by the MVPF calculated for Brazil, which falls in the upper end of the 0.5-2 range reviewed in Hendren and Sprung-Keyser 2020. These results enhance our understanding of tax incidence and efficiency, particularly in the context of developing economies.

5.3 Alternative Policies

From a policy perspective, having acknowledged the influence of market power on the efficiency and distributional outcomes of tax policy, immediate questions arise: How can we design better tax policies? Can we quantify the potential impacts of alternative policies? This subsection answers to these inquiries by proposing and evaluating two alternatives: First, improving the targeting of the tax subsidy only to small firms. Second, implementing labor market regulations aimed at eliminating monopsony power.

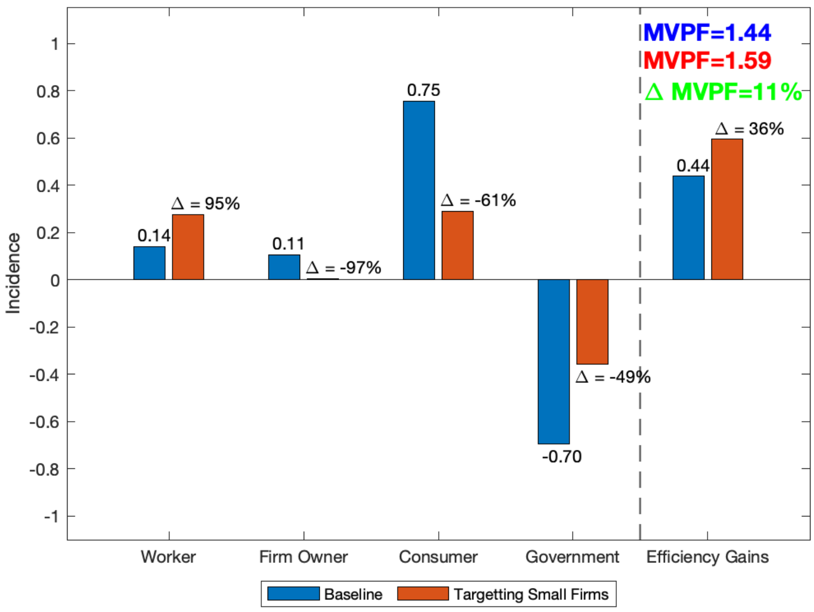
5.3.1 Targeting Small Firms

Targeting tax subsidies to small firms presents a logical proposition for two main reasons. First, firm size represents an easily observable characteristic, thereby enabling simple policy tagging. Second, section 5.1 have illustrated that small firms possess less power in both product and labor markets, which consequently makes taxation more burdensome for them. Thus, this subsection aims to quantify such potential gains and further evaluate potential winners and losers in this alternative scenario. To operationalize this exercise, I take the firm labor supply (ϵ) to infinity, and compute the incidence and efficiency using the empirical moments and structural parameters estimated for small firms.

Figure 9 reports that efficiency gains increase from 44 to 59%, which represents a 36% gain. Part of this efficiency boost is justified by the behavioral response on Government revenue, which offsets part of the subsidy cost. In simpler terms, smaller firms tend to expand employment relatively more, which makes it cheaper to subsidize them. Quantitatively, behavior responses counterbalance 49% of the revenue decline. This alternative policy stands out as more efficient, even in a revenue preserving scenario, which is evidenced by the 11% rise in MVPF.

In terms of distributional impacts, the alternative policy's effects on workers is not immediately clear. On one hand, smaller firms face a less steep labor supply curve; on the other, their lack of product market power allows for a greater scale effect. Ultimately, workers experience a 95% gain in the alternative scenario, suggesting that moving further along the supply curve more than compensates the less steep curve, thereby triggering a more pronounced wage response. Conversely, in the product market, shifting further along the demand curve does not compensate for the less steep demand curve. Therefore, the price decrease is relatively less pronounced for small firms. As a result, consumers find themselves worse off by 61% under the alternative targetting. Lastly, firm owners retain almost negligible tax benefits, primarily because small entrepreneurs lack the power to capture rents.

Figure 9: Baseline vs Targeting Small Firms



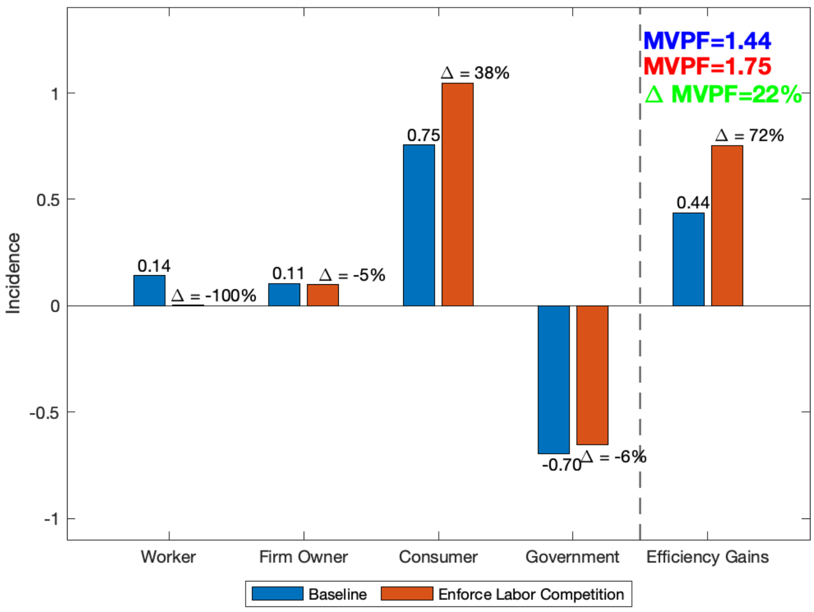
Note: This figure presents the baseline incidence analysis (blue bars), and counterfactual incidence in case of alternative targetting. The alternative policy, represented in the red bars, relies on a policy that targets only firms with low market power. The figure plots the differential incidence, and differential efficiency gains across the baseline and alternative policy.

5.3.2 Enforce Labor Market Competition

In the baseline evaluation, both labor and product market power shape firms' responses to tax reform. However, to disentangle the role of each of these competitive frictions, the counterfactual exercise envisions a policy capable of eliminating labor market power. Such a policy could take the form of regulatory sanctions, the dismantling of entry barriers, or more broadly, any initiatives that foster labor market competition. To operationalize this evaluation, I simulate the firm's pass-through to labor, capital, and revenue, when labor supply elasticity (ϵ) approaches infinity.

Figure 10 shows that when labor market power is assumed away, the efficiency of tax cuts increases remarkably by 72%. This efficiency gains come with winners and losers. Relative to baseline, workers lose their entire welfare gains because in the alternative scenario their labor supply is perfectly elastic. Competitive forces in the labor market stimulate firms to ramp up production, downward pressuring output prices, especially as firms retain product market power. Consequently, consumers are unambigously better off, and their welfare increase by 38%. Firm owners' welfare, ex-ante, could go in either directions. On one hand, perfectly competitive labor markets allow firms to expand without sharing rents with incumbent workers. On the other, greater expansion combined with product market power leads to a more salient price decrease. At the end of this horse race between cost and revenue effects, firm owners experience a minor loss of 5%.

Figure 10: Baseline vs Enforcing Labor Competition



Note: This figure presents the baseline incidence (blue bars), and counterfactual incidence when labor market competition is enforced. The alternative policy, represented in the red bars, assumes that the labor supply elasticity faced by the firm is infinity. The figure plots the differential incidence, and differential efficiency gains across the observed and alternative policies.

5.3.3 Revenue Maximizing Tax Rate

The payroll tax rate variation induces a mechanical change on Governments' revenue, measured by $dM = B(\tau_1 - \tau_0)$. This change further incites a behavioral response to tax revenue given the labor supply responses elicited by the policy: $H = \tau B(\frac{\epsilon+1}{\epsilon})\frac{\partial L}{\partial 1+\tau}\frac{1}{L}$. To compute the mechanical effect we rely on the elasticity of labor with respect to the labor cost that was empirically estimated from the quasi-experimental variation. Notice that this elasticity is fixed around the observed rate. To evaluate the counterfactual elasticity at hypothetical tax rates significantly diverging from the observed level, I undertake a Taylor expansion, with rates varying from τ_0 to τ_1.

$$\frac{\partial L}{\partial 1+\tau}(1+\tau_1) = \frac{\partial L}{\partial 1+\tau}(1+\tau)\Big|_{\tau=\tau_0} + \frac{\partial L}{\partial^2 1+\tau}(1+\tau)\Big|_{\tau=\tau_0}(\tau_1-\tau_0) + \frac{1}{2}\frac{\partial L}{\partial^3 1+\tau}(1+\tau)\Big|_{\tau=\tau_0}(\tau_1-\tau_0)^2 + \ldots \quad (21)$$

$$\frac{\partial L}{\partial 1+\tau}(1+\tau_1) = \frac{L\epsilon_{L,1+\tau}}{1+\tau}\Big|_{\tau=\tau_0}\left[1 + \frac{(\epsilon_{L,1+\tau}-1)}{1+\tau}\Big|_{\tau=\tau_0}(\tau_1-\tau_0) + \frac{1}{2(1+\tau)^2}\Big|_{\tau=\tau_0}(\epsilon_{L,1+\tau}(\epsilon_{L,1+\tau}-1)+2)(\tau_1-\tau_0)^2\right] \quad (22)$$

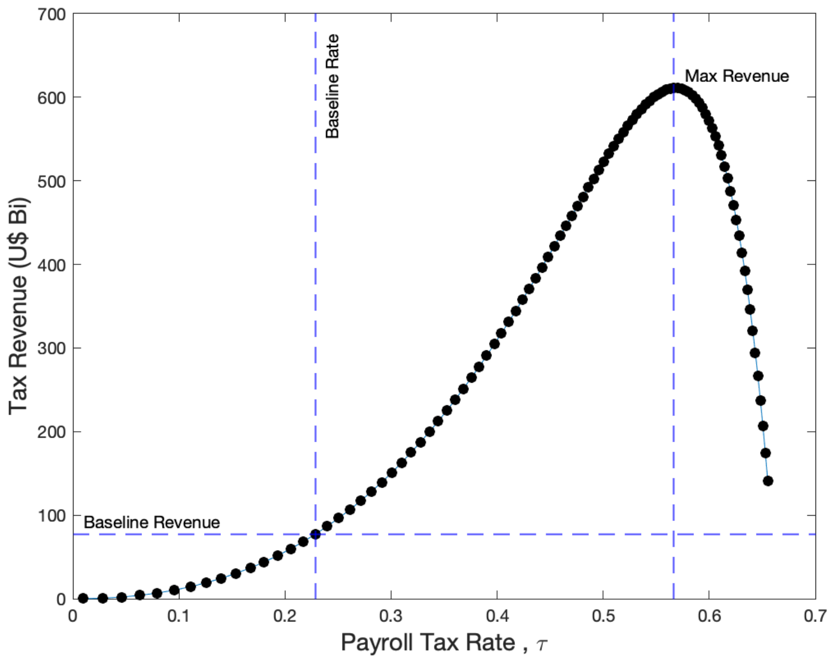
In counterfactual scenarios, where the payroll tax rate moves to τ_1, I compute the behavioral response by evaluating the mechanical effect H at the counterfactual elasticity delineated in 22. With this framework, I simulate the revenue impact from perturbing the labor tax rate. Figure 11 presents a shape similar to the so called, *Laffer curve*, and shows that the Brazilian tax revenue would be maximized if the labor tax rate were 56%[52].

One immediate takeaway from this exercise is that payroll tax rates in Brazil are fairly far from the revenue maximizing rate, which is indicative that existing average tax rates are on the "right side of the Laffer curve". The direct consequence is that Brazilian policymakers can increase the payroll tax without fearing a decline in tax revenue. In fact, this conclusion is further supported by the positive MVPF reported on table 5.

[52]Alternatively, this could be expressed as a firm's payroll tax rate of 130%. Throughout this paper, we refer to the payroll tax rate as it applies to the wage bill paid by firms. To establish a connection with prior research exploring revenue maximizing tax rates, it is helpful to express the tax rate that applies to workers' take-home payment. In this case, if the tax rate t=1, there is no incentive to work. A one-to-one relationship between the two aforementioned tax rates can be depicted as:

$$\underbrace{\frac{wL}{1-t}}_{\text{Received by worker}} = \underbrace{wL(1+\tau)}_{\text{Paid by firm}}$$

Figure 11: Laffer Curve for Payroll Taxation



Note: This figure plots

6 Conclusion

In this paper, I study a payroll tax reform that promoted unprecedentedly large tax reduction to a small share of firms in Brazil. The setting allowed fewer assumptions on general equilibrium effects, in order to estimate firm and worker level responses. While capital decreases, a payroll tax reduction causes an increase in employment, wages and profits. These effects exhibit heterogeneity across firms, with small firms responsding relatively more. Interpreted through the lenses of a novel model of factor demand that incorporates product and labor market power, the empirical estimates inform that consumers pay most of the payroll tax burden.

The efficiency cost of taxation hinges on the curvature of the product demand and labor supply curves, as well as the policy induced shift in product supply and labor demand. Therefore, the role of market power in determining the distortionary costs of taxation is not obvious. The theoretical framework I provide, combined with rich data, can elucidate the opaque mechanisms that underpin firms' responses to tax policy. This offers policy makers insights into potential avenues for enhancing the efficiency and distribution of tax policies, considering the economic forces at play, such as product and labor market power.

Utilizing this model, I assess an alternative policy that directs tax subsidies towards smaller firms, identified to have less market power. The results suggest that taxing firms with greater market power decreases the distortionary costs of taxation, lending support to proposals advocating for relatively higher tax rates for larger firms. The aforementioned efficiency gain is followed by a distributional change: market power shifts the incidence away from workers and towards firm owners and consumers.

Industrial policies that target specific sectors and firms are pervasive in the developing world. In addition to providing deep understanding about firm-specific tax policies, this work shines light on the underlying dynamics at work in market-wide policies, such as the role of firms' market power. Yet, an intrinsic limitation is the inability to explore the interplay between market power and general equilibrium effects due to the nature of the examined policy. To bridge this gap, I am partnering with the United States Internal Revenue Services (IRS) to study the Tax Cuts and Jobs Act (TCJA), and incorporate the role of general equilibrium effects.

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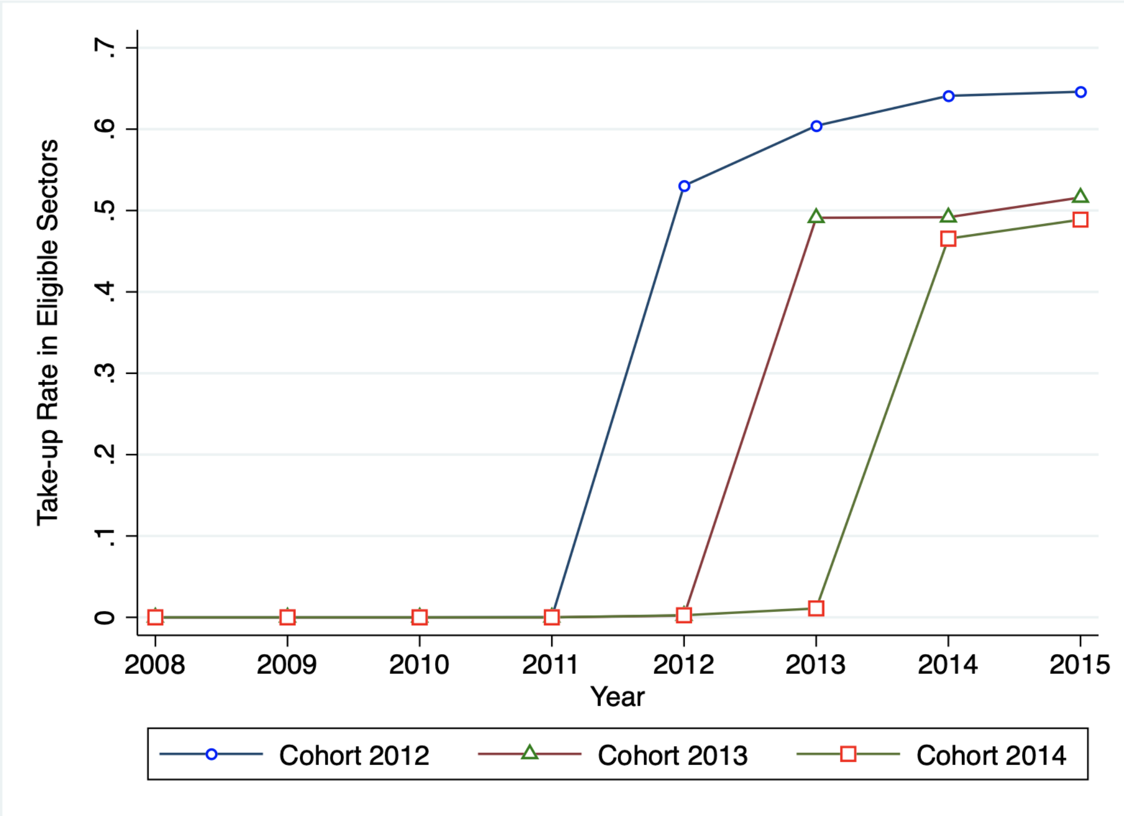
Figure 12: Take-up



Note: This figure plots cohort specific take-up rates among eligible firms. Eligibility is based on the firm's 7-digit sector, and their observed tax tier. Firms in the "Simples" tax regime are not eligible for the reform, even if they belong to eligible sectors. Firms that have ever participated in the "Simples" regime are not included in this analysis. As expected, take-up rates are zero in years prior to the implementation of the reform to each cohort.

Figure 13: Tax Forms Information

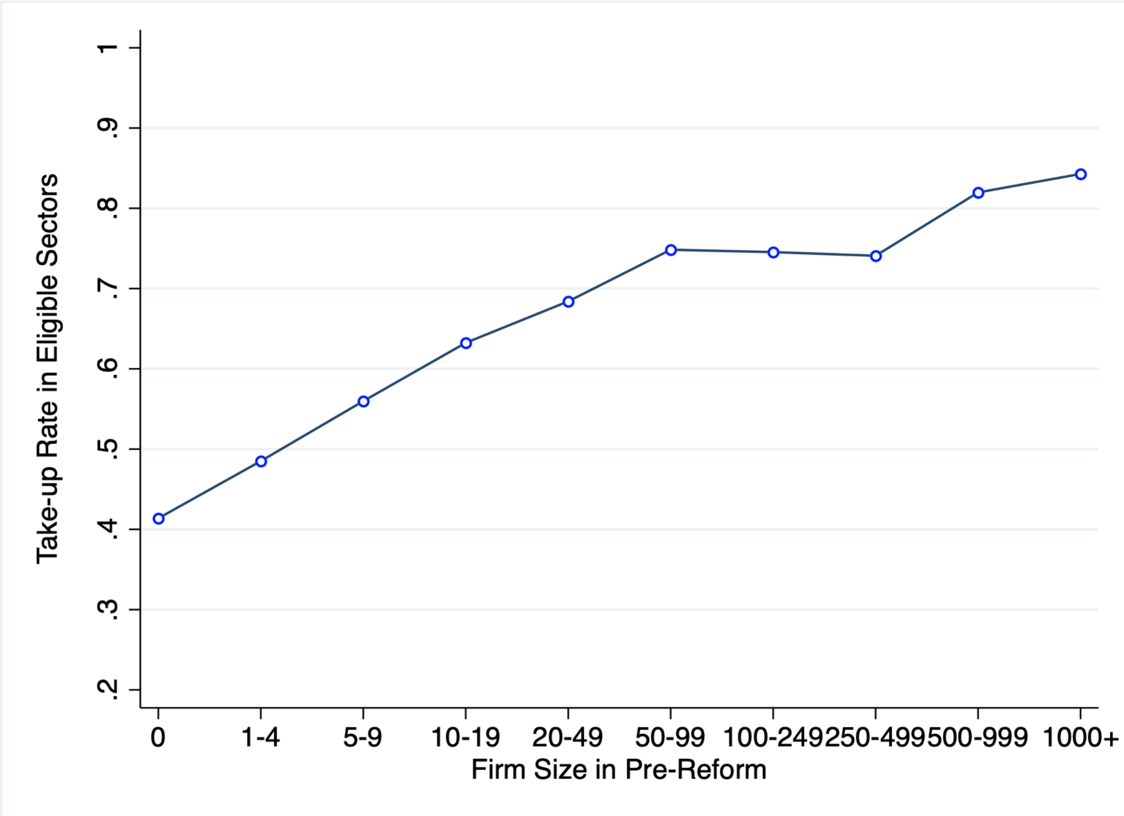
Nº	Campo	Descrição	Tipo	Tam	Dec	Obrig
01	REG	Texto fixo contendo “0145”.	C	004*	-	S
02	COD_INC_TRIB	Código indicador da incidência tributária no período: 1 – Contribuição Previdenciária apurada no período, exclusivamente com base na Receita Bruta; 2 – Contribuição Previdenciária apurada no período, com base na Receita Bruta e com base nas Remunerações pagas, na forma dos nos incisos I e III do art. 22 da Lei nº 8.212, de 1991.	N	001*	-	S
03	VL_REC_TOT	Valor da Receita Bruta Total da Pessoa Jurídica no Período	N	-	02	S
04	VL_REC_ATIV	Valor da Receita Bruta da(s) Atividade(s) Sujeita(s) à Contribuição Previdenciária sobre a Receita Bruta	N	-	02	S
05	VL_REC_DEMAIS_ATIV	Valor da Receita Bruta da(s) Atividade(s) não Sujeita(s) à Contribuição Previdenciária sobre a Receita Bruta	N	-	02	N
06	INFO_COMPL	Informação complementar	C	-	-	N

Nº	Campo	Descrição	Tipo	Tam	Dec	Obrig
01	REG	Texto fixo contendo "P100"	C	004*	-	S
02	DT_INI	Data inicial a que a apuração se refere	C	008*	-	S
03	DT_FIN	Data final a que a apuração se refere	C	008*	-	S
04	VL_REC_TOT_EST	Valor da Receita Bruta Total do Estabelecimento no Período	N	-	02	S
05	COD_ATIV_ECON	Código indicador correspondente à atividade sujeita a incidência da Contribuição Previdenciária sobre a Receita Bruta, conforme Tabela 5.1.1.	C	008*	-	S
06	VL_REC_ATIV_ESTAB	Valor da Receita Bruta do Estabelecimento, correspondente às atividades/produtos referidos no Campo 05 (COD_ATIV_ECON)	N	-	02	S

Nº	Campo	Descrição	Tipo	Tam	Dec	Obrig
01	REG	Texto fixo contendo "P100"	C	004*	-	S
07	VL_EXC	Valor das Exclusões da Receita Bruta informada no Campo 06	N	-	02	N
08	VL_BC_CONT	Valor da Base de Cálculo da Contribuição Previdenciária sobre a Receita Bruta (Campo 08 = Campo 06 – Campo 07)	N	-	02	S
09	ALIQ_CONT	Alíquota da Contribuição Previdenciária sobre a Receita B ruta	N	008	04	S
10	VL_CONT_APU	Valor da Contribuição Previdenciária Apurada sobre a Receita Bruta	N	-	02	S
11	COD_CTA	Código da conta analítica contábil referente à Contribuição Previdenciária sobre a Receita Bruta	C	255	-	N
12	INFO_COMPL	Informação complementar do registro	C	-	-	N

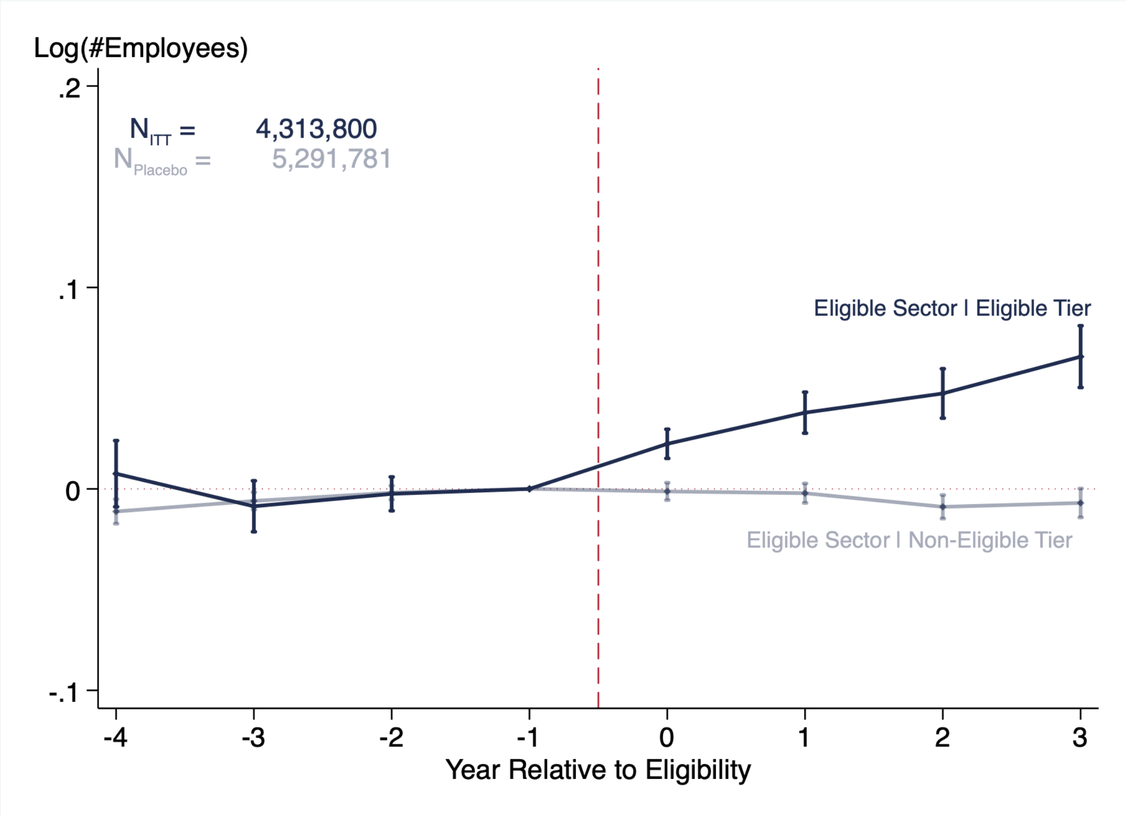
Note: This figure shows instructions for eligible firms to request the payroll tax benefit. It describes detailed information to be provided in Tax Administration software, in order to substitute part of the payroll tax by revenue taxes.

Figure 14: Take-up per Firm Size



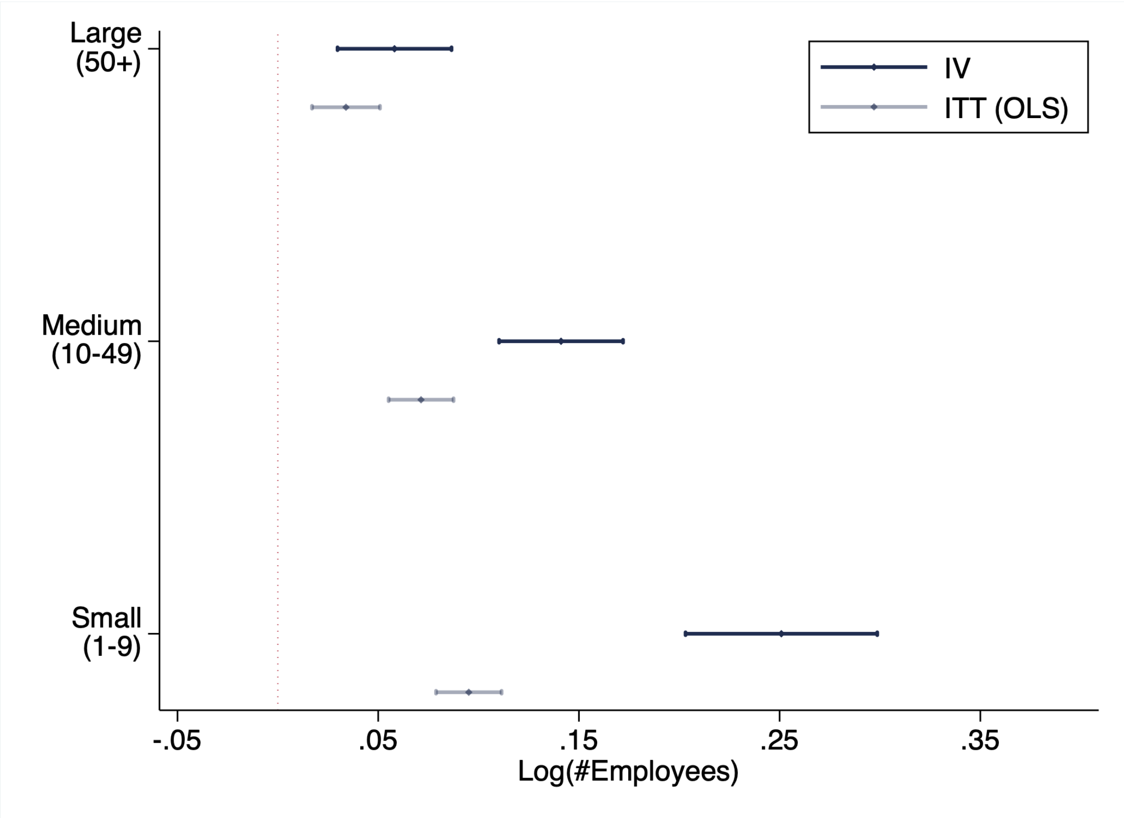
Note: This figure plots cohort specific take-up rates among eligible firms. Eligibility is based on the firm's 7-digit sector, and their observed tax tier. Firms in the "Simples" tax regime are not eligible for the reform, even if they belong to eligible sectors. Firms that have ever participated in the "Simples" regime are not included in this analysis. The figure is computed in the year 2015, after all cohorts have gained eligibility. Firm size buckets are constructed in the pre-reform years. As it can be seen, eligibility is monotonically increasing with firm size.

Figure 15: Spillover Test



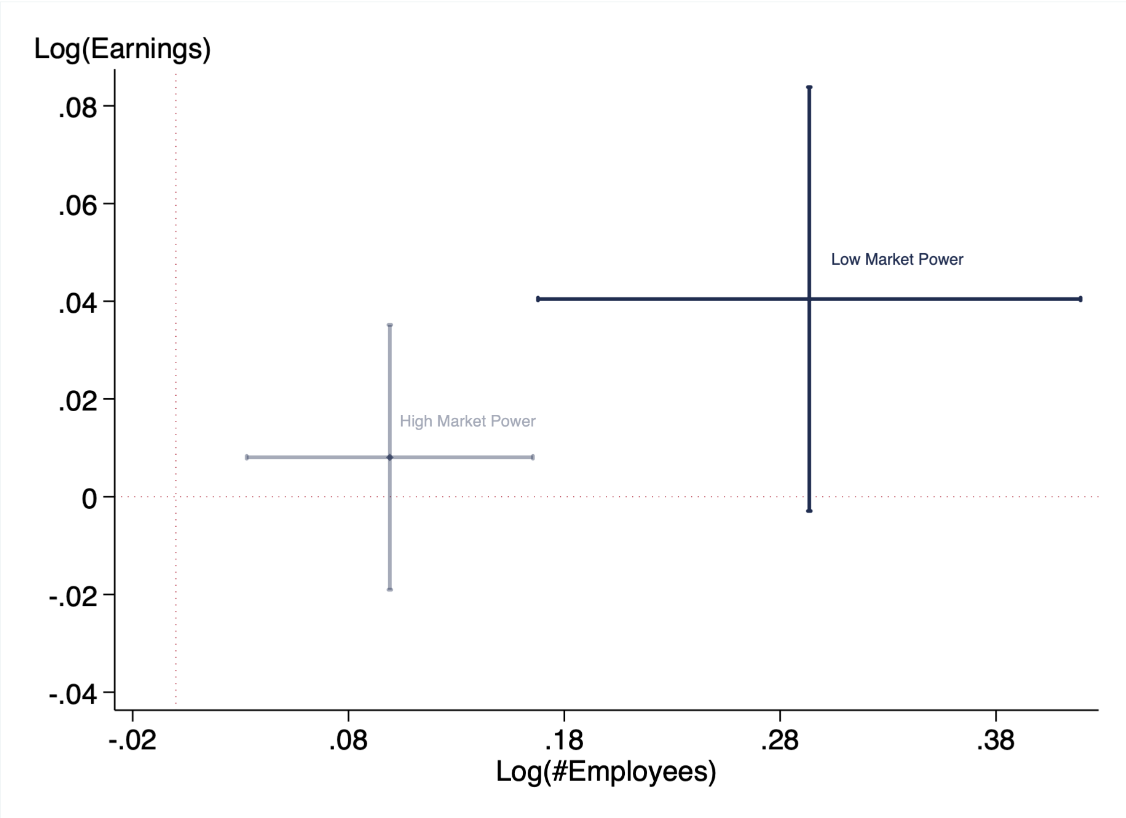
Note: The gray line plots event study coefficients that show non-statistically significant spillover effect to firms in eligible sectors, but ineligible tax tiers. The gray line is estimated on a sample that is restricted to firms in non-eligible tax tiers ("Simples" regime) and depicts a comparison between firms in eligible and non-eligible sectors. To avoid concerns about tier changes, this analysis is restricted to firms that have never changed tiers. The blue line is estimated on a sample that is restricted to firms in eligible tax tiers ("non-Simples" regime). It reports the intention to treat (ITT), i.e., compares eligible firms in eligible vs non-eligible sectors. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure 16: Earnings and Employment per Firm Size



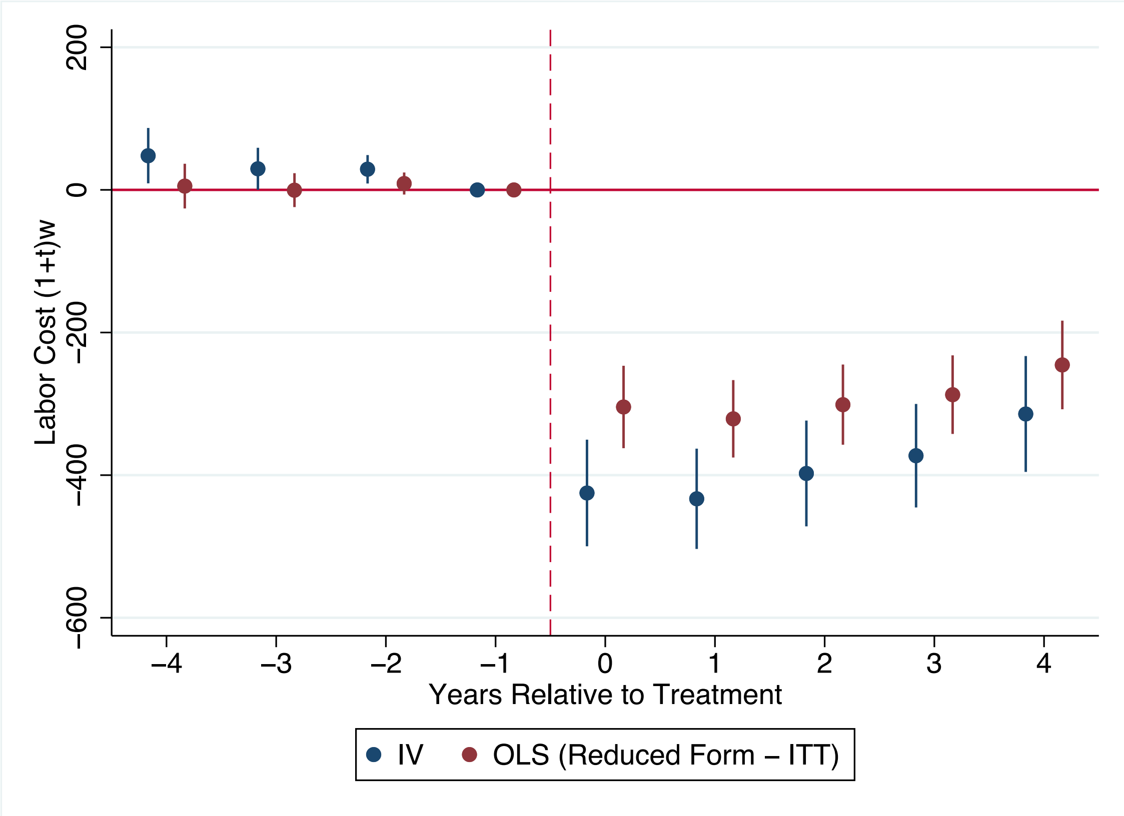
Note: This figure presents the event study estimates for the firm level estimates, for three firm size groups (small, medium and large firms). Size categories are defined in the pre-reform period. Firms are classified as small if they had less than nine employees, medium if they had between 10 and 49 workers, and large if they had more than 50 workers. The blue marks plot the employment difference-in-differences coefficient, while the gray markers plot the firm level earnings effect. Standard errors are reported and conservatively clustered at the 5-digit industry-by-state level.

Figure 17: Earnings and Employment per Market Power



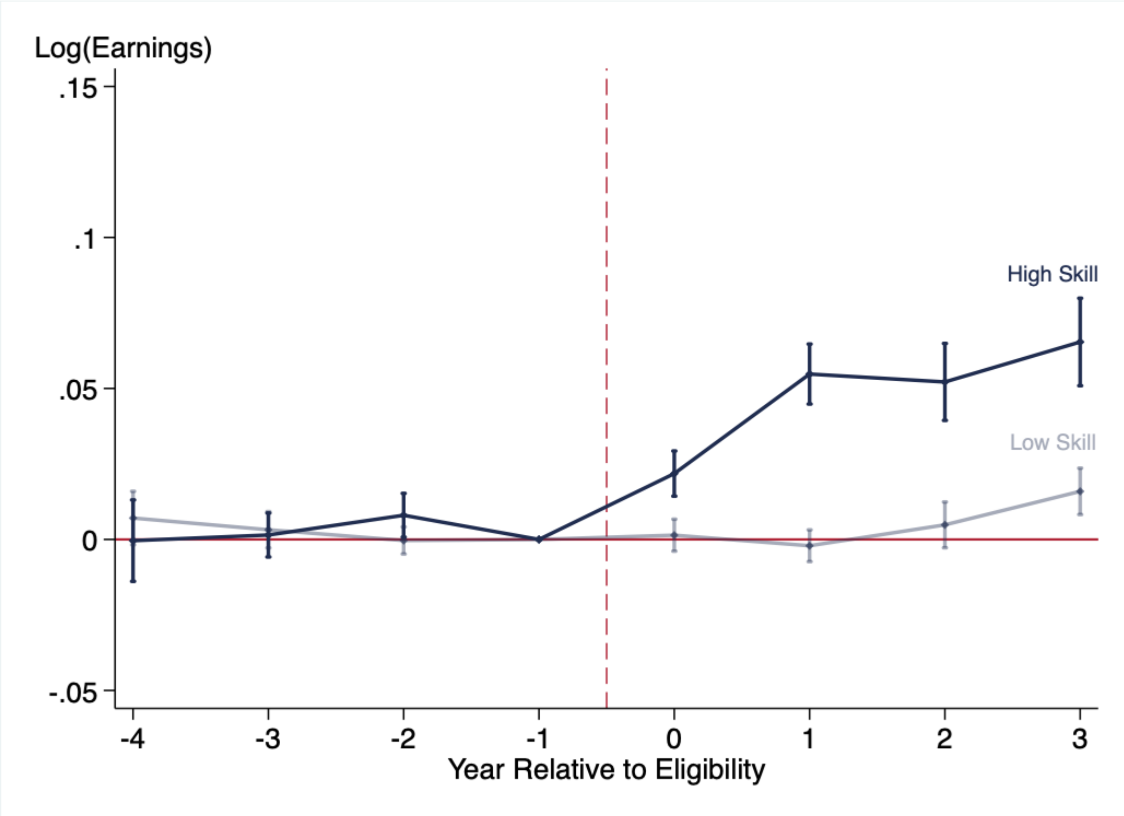
Note: This figure presents firm level IV difference-in-differences coefficients for above and below the median on pre-reform employment market share within each local labor market. The outcomes are employment and earnings. The blue marker plots the effect for firms below the median (low market power), whereas the gray marker plots the effect for firms with high market power. Horizontal and vertical lines plot the confidence intervals for the employment and earnings estimates, respectively. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure 18: Worker Level: Gross Earnings Effect



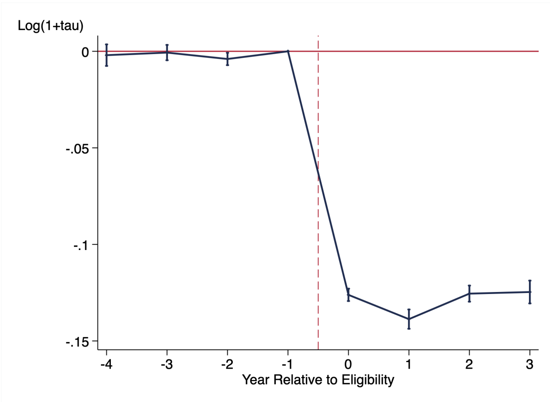
Note: This figure presents the event study estimates for average gross earnings paid workers that were employed for at least three years in the same firm during the pre-reform period. The labor cost is computed using firm level tax data, and worker level earnings data. I apply the firm payroll tax rate in year t, to all employees in that firm in year t. I normalize the results with respect to one year prior to the treatment event. The analysis spans four years prior to the payroll tax cut program and three years after. The plot shows an average decrease of \$400 on the gross earnings, which has an approximate average of \$2,300 during the pre-reform period. The blue markers depict IV coefficients, and the red markers intention-to-treat. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure 19: Worker Level: Earnings per Occupation

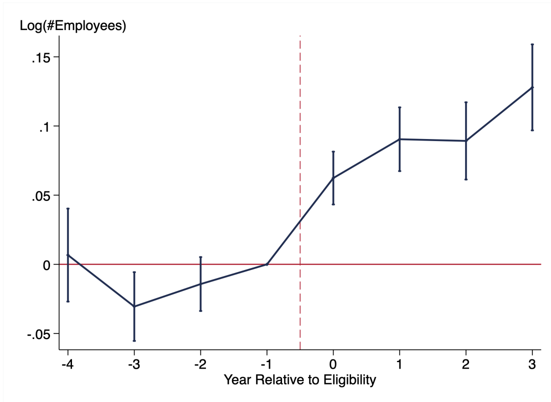


Note: This figure presents the event study estimates for the log of earnings per occupation group, at the worker level based on pre-reform occcupations. Leaders are directors, managers and qualified technical positions according to the CBO classification. While leaders experience high pass-through to earnings of 6%, low skilled occupation didn't see any significant earnings increase. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

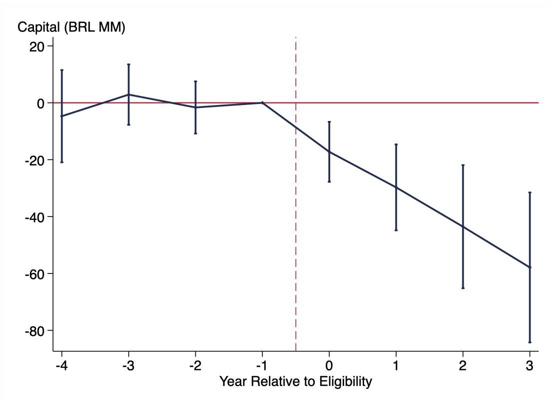
Figure 20: Firms' Margins of Adjustment



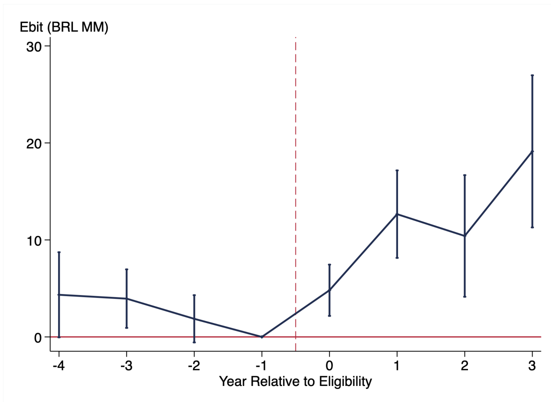
(a) Effects on payroll tax rates



(b) Effects on employment



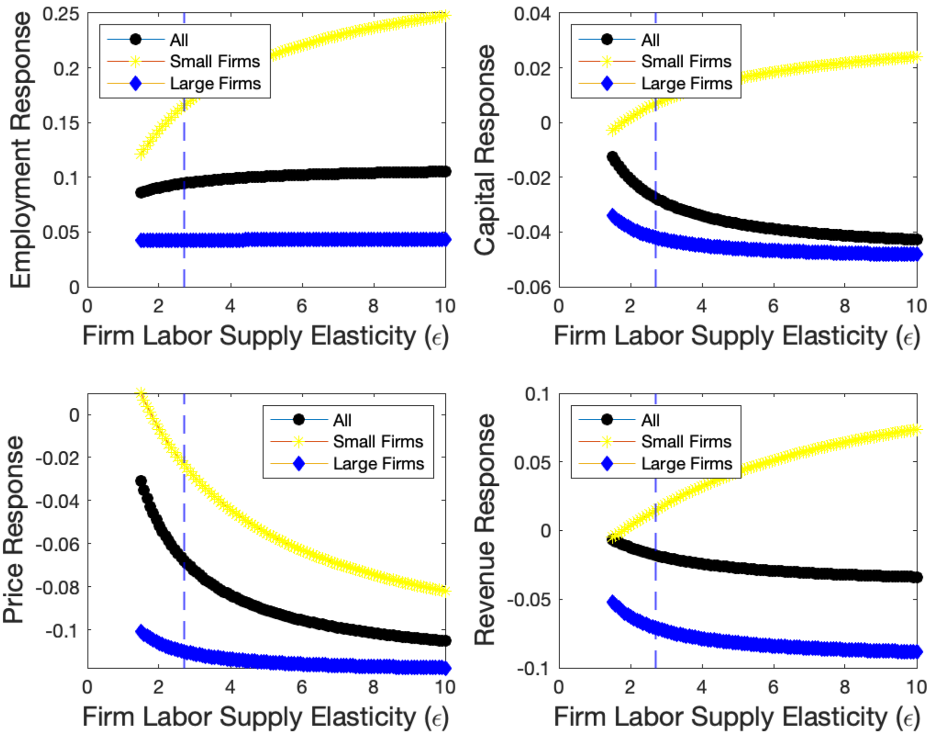
(c) Effects on capital



(d) Effects on profits

Note: This figure plots event study coefficients for multiple of the firms' margins of adjustment after the payroll tax cut. First, at the top left plot it shows the first stage, i.e., the reform induced a reduction on payroll tax liability. On the top right plot, it depicts the employment increase that has already been documented. The two bottom graphs shed light on other business outcomes, such as, capital and profit.

Figure 21: Simulations



Note: This figure simulates the model in several dimensions. In each plot, the x-axis represents the labor supply elasticity faced by the firm. As we move from the left to the right, we are moving to more competitive labor markets. The horizontal dashed line depicts the labor supply elasticity estimated in the data. The black line simulates the effects for the overall sample. This would be analogous to fixing the targeting of the policy and perturbing the degree of labor market competition. The yellow line represents the effects to small firms, which is useful to evaluate the effects of an alternative policy that focuses its targeting on firms with low market power. The blue line represents the effects to large firms, which is useful to evaluate the effects of an alternative policy that focuses its targeting on firms with high market power. The four plots represents the counterfactual responses across the margins of employment, capital, price, and revenue.

Table 6: Eligible vs Non-Eligible Sectors

Eligible	Not Eligible
Hotels	Motels
Open television	Cable television
Public bus transportation	School bus and taxi
Eletronic games manufacturing	Toys and other recreative games manufacturing
Internet portals and content providers	News agencies
Trains	Touristic trains
Newspaper, magazine and book printing	Other periodic printing
Maintenance aircraft and vessels	Maintenance aircraft and other transportation modes

Note: This table presents a list of sectors that are displayed in the tax bills as eligible to the payroll tax cut, and compares it with another list of similar sectors that are not included in the tax reform. This is an anedoctal evidence that the Governement was not anticipating sector trends when determining eligibility.

Table 7: Macro Relevance of the Reform

	2012	2013	2014
# Sectors	10	81	124
Share	0.0076	0.0617	0.0944
# Firms	20,865	33,705	49,253
Share	0.0079	0.0121	0.0170
# Workers	2,950,925	5,028,078	6,113,091
Share	0.0304	0.0513	0.0618

Note: This table shows the comprehensiveness of the policy rollout over the years that new sectors gained eligibility (2012-2014). In the first part of the table it shows the number of 7-digit sectors eligible for the tax reform, and their representativeness computed as the share of existing sectors in the Brazilian economy. The second part of the table shows the number of formal firms in the final sample that were treated in each year. To adjust for informal firms that does not appear in my sample, I multiply the share by 0.55, which is the average formalization rate in Brazil, according to PNAD (official survey administred by the Brazilian Census Bureau, IBGE). In the last rows, the table reports the number of workers employed in treated firms. I compute the share of treated workers dividing # of workers by the universe of Brazilian workers according to PNAD-C.

Table 8: Descriptive Statistics (Worker Level Sample)

	(1) Non-Eligible (pre)	(2) Eligible (pre)	(3) Avg (pre)
Descriptive Statistics			
Earnings	2,326.54 (3,060.70)	2,166.05 (2,837.04)	2,315.46 (3,046.06)
Employees Age	39.30 (10.82)	37.20 (10.51)	39.16 (10.81)
Share White	0.67 (0.47)	0.65 (0.48)	0.67 (0.47)
Gender	0.54 (0.50)	0.79 (0.41)	0.55 (0.50)
High School +	0.70 (0.46)	0.62 (0.49)	0.70 (0.46)
College +	0.27 (0.45)	0.18 (0.38)	0.27 (0.44)
N	54,373,025	4,031,321	58,404,346

Note: This table presents descriptive statistics of the baseline worker level sample in the pre-reform period (2008 to 2011). Each observation is a worker x year, restricted by the sample restriction of stable workers. Column (1) and (2) reports the pre-reform values for non-eligible and eligible workers, respectively. Column (3) pool this two groups together. The variable "Earnings" reports monthly earnings in the month of December for employed workers. The currency used is the Brazilian Reais (BRL). "Employees Age" reports age of employed workers in the sample, on December of each year. "Share White" reports share of white workers in the sample. "High School +" reports the share of workers that achieved high school education or higher. "College +" reports the share of workers that achieved college education or higher. Standard deviations are presented in paranthesis.

Table 9: Descriptive Statistics (Firm Level Sample)

	(1) Non-Eligible	(2) Eligible	(3) Pooled	(4) Take-up
Pre-Reform Characteristics				
Employment	54.86	57.68	55.03	108.74
	(1,055.81)	(339.52)	(1,026.82)	(493.98)
Payroll Tax Rate	31.79	31.63	31.78	34.84
	(13.18)	(14.55)	(13.26)	(9.97)
Share Male	0.55	0.77	0.56	0.78
	(0.40)	(0.29)	(0.40)	(0.24)
Age	37.34	36.16	37.27	35.35
	(8.95)	(7.85)	(8.90)	(6.24)
Share High School +	0.52	0.58	0.52	0.60
	(0.41)	(0.38)	(0.41)	(0.34)
Share White	0.68	0.74	0.68	0.76
	(0.37)	(0.33)	(0.37)	(0.29)
Share Blue Collar	0.89	0.85	0.89	0.83
	(0.24)	(0.27)	(0.24)	(0.27)
N	1,775,601	114,153	1,889,754	47,315

Note: This table presents descriptive statistics of the baseline firm level sample in the pre-reform period (2008 to 2011). Each observation is a firm x year. The descriptive statistics are presented for different groups of interest. Column (1) and (2) reports the pre-reform values for non-eligible and eligible firms, respectively. Column (3) pool this two groups together. Column (4) reports the value for eligible firms that eventually take-up the treatment. The variable "Payroll Tax Rate" informs the average payroll tax rates in (%). The variable "High School" reports the share of workers that achieved high school education or higher. The variable "Gender Composition" reports the share of male workers. The variable "Share White" informs the average share of white workers per firm. Standard deviations are presented in paranthesis.

Table 10: Composition Effect (Between Occupations)

	(1) Occup Pctile Firm Level
Post x Treatment	0.000584
	(0.00235)
Observations	4,312,325
Firm FE	Yes
Sector (1 digit) x Year FE	Yes

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table presents IV difference-in-differences estimate for the average effect on the occupation percentile that firms employ from. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Table 11: Worker Level Estimates

Worker Level	Log(Earnings)	Log(Earnings)	
	All Sample (1)	Blue Collar (2)	White Collar (3)
Panel B: IV			
Diff-in-Diff	.015** (.008)	.003 (.007)	.06*** (.013)
Long Diff	.027*** (.007)	.016** (.008)	.065*** (.014)
Panel A: OLS			
Diff-in-Diff	.01** (.004)	.002 (.004)	.035*** (.008)
Long Diff	.018*** (.005)	.01* (.005)	.045*** (.011)
Controls	✓	✓	✓
Worker FE	✓	✓	✓
Firm FE	✓	✓	✓
Sector x Year FE	✓	✓	✓
# Clusters	10,212	10,195	8,851
N	110,991,027	83,007,473	25,152,631

Note: This table presents IV and reduced form (ITT) estimates for the worker level sample. Difference-in-differences coefficient is estimated in equations 3 and 4, where there is only one post period. The long difference comes from the period t=+3, in the event study design. Panel A reports the IV coefficients, which adjust for the imperfect compliance and are interpreted as the local average treatment effect on compliers. Panel B reports the reduced form coefficients, which are interpreted as the intention to treat (ITT). The dependent variable is log of workers' earnings. Column (1) presents the average effect in the all sample. Columns (2-6) present heterogeneity based on pre-reform occupation. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Table 12: First Stage: Cost of Labor (1 + payroll tax rates)

	(1) Log(Labor Cost) (1 + Payroll Tax)	(2) Log(Labor Cost) (1 + Payroll Tax)
Treatment x Post	-0.133***	
	(0.00247)	
Eligibility x Post		-0.0675***
		(0.00248)
Age	-0.00148***	-0.00157***
	(0.000228)	(0.000260)
Gender	-0.00664***	-0.00916***
	(0.00133)	(0.00148)
High School	0.000890	-0.00131
	(0.00119)	(0.00110)
College	-0.000554	-0.00297*
	(0.00130)	(0.00134)
Observations	3,962,064	3,962,064
IV	Yes	Reduced
Firm FE	Yes	Yes
Sector (1 digit) x Year FE	Yes	Yes
Worker FE	No	No

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports the first stage impact of the reform, i.e., how much payroll tax rates were affected by the reform. Column (1) presents the IV results, which adjust eligibility by the take up rates. Column (2) displays the payroll tax changes in eligible firms due to the reform. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Table 13: Earnings Estimates (Firm Level)

	(1) Log(Earnings) firm (99p)	(2) Log(Earnings) firm (90p)	(3) Log(Earnings) firm (40p)	(4) Log(Earnings) firm (20p)
Currently Treated	0.0420**	0.0212	0.00835	-0.00421
	(0.0162)	(0.0136)	(0.0109)	(0.00997)
Observations	4,207,381	4,205,884	4,149,217	4,071,896
Firm FE	Yes	Yes	Yes	Yes
Sector (1 digit) x Year FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table presents IV estimates for the causal impacts of the reform on outcomes labeled on each column. The instrument is the sector eligibility. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Table 14: Heterogeneity Across Liquidity Constraints

	(1) Employment Low Liquidity	(2) Employment High Liquidity
Currently Treated	0.107***	0.109***
	(0.0283)	(0.0289)
Observations	228,087	233,691
Firm FE	Yes	Yes
Sector (1 digit) x Year FE	Yes	Yes
Worker FE	No	No

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports IV difference-in-differences coefficients for firms below/above the median on liquidity constraint, during the pre-reform period. Liquidity constraint is defined as the ratio of current assets over current liabilities. An example of current assets is cash, whereas an example of current liabilities are short term bills, such as the wage bill. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Appendix

A More Details on the Empirical Model

A.1 Derivation of the Reduced Form Equations

Given the set of k first stage equations, the reader might not be able to see immediately the reduce form equation. Starting with the firm level design, we obtain the reduced form by substituting all first stage equations into the second stage,

$$Y_{jt} = \sum_{k=-4,\neq -1}^{3} \beta_k \left[\sum_{l=-4,\neq -1}^{3} \pi_{kl} \times \mathbb{I}(t = e_{s(j)} + l) \times L_{s(j)} + \alpha_j + \xi_{s1(j),t} + X'_{jt}\delta_k + \eta_{jt} \right] + X'_{jt}\gamma + \alpha_j + \xi_{s1(j),t} + \epsilon_{jt}$$

where, $D^k_{jt} = 1$, if $t = e_j + k$; e_j is the year when firm j enters treatment; $L_{s(j)}$ indicates if firm j's sector is eventually eligible; $e_{s(j)}$ is the date when firm j's sector becomes eligible; X_{jt} set of controls such as education, race, age and its square; $\xi_{s1(j),t}$ is 1-digit sector x year fixed effect; α_j is the firm fixed effect; η_{jt} and ϵ_{jt} are residuals. Standard errors are conservatively clustered at the 5-digit industry-by-state level. Reorganizing terms,

$$Y_{jt} = \sum_{l=-4,\neq -1}^{3} \left[\sum_{k=-4,\neq -1}^{3} \beta_k \pi_{kl} \times \mathbb{I}(t = e_{s(j)} + l) \times L_{s(j)} \right] + X'_{jt} \left[\gamma + \sum_{k=-4,\neq -1}^{3} \beta_k \delta_k \right] + \\ + (\alpha_j + \xi_{s1(j),t}) \left[1 + \sum_{k=-4,\neq -1}^{3} \beta_k \delta_k \right] + \left[\epsilon_{jt} + \sum_{k=-4,\neq -1}^{3} \beta_k \eta_{jt} \right]$$

Thus, the reduced form coefficient is,

$$\rho_l = \sum_{k=-4,\neq -1}^{3} \beta_k \pi_{kl}$$

Notice that if K=L and diagonal is such that $\pi_{kl} = 0$ (when k $\neq$ l), then $\rho_l = \beta_l \pi_{ll}$, and $\beta_l = \frac{\rho_l}{\pi_{ll}}$. However, if K$< L$ then the system $\rho_l = \sum_{k=-4,\neq -1}^{3} \beta_k$ for l=1,...., L is a system of L equations in K<L unknowns and generally cannot be solved. The off diagonal coefficients estimated in equation 2 are small and not statistically different than zero, which makes the interpretation of the reduced form coefficients equal to the one dimensional case, i.e., $\rho_l = \beta_l \pi_{ll}$. At the worker level, the algebra to obtain the reduced form coefficient is analogous to the firm level computations presented in this appendix.

A.2 Validity of the IV Design

A long strand of empirical work (Hoxby and Murarka 2009; Angrist et al. 2010; 2012; Davis and Heller 2019; Cohodes, Setren, and Walters 2021 to cite a few) on education has used the random lottery assingment to schools' admissions in order to estimate the causal effect of charter schools on academic outcomes. Simply regressing outcomes of interest on school enrollment does not capture the causal school effect, because enrollment is an endogeneous decision that can be driven by selected non-observable that are correlated with the outcomes. To exemplify this issue, consider the case of constant effects in Angrist, Hull, and Walters 2022. In this case, the mean comparison of outcomes based on enrollment generates a selection bias that can be computed by,

$$\mathbb{E}(Y_i|D_i=1)-\mathbb{E}(Y_i|D_i=0)=\beta+\underbrace{\mathbb{E}(\epsilon_i|D_i=1)-\mathbb{E}(\epsilon_i|D_i=0)}_{S\equiv \textit{Selection Bias}}\neq\beta$$

The reason why the selection bias (S) is generally different than zero is because enrollment and achievement tend to be correlated through unobserved student's characteristis such as motivation and family support. Therefore, $\mathbb{E}(\epsilon_i|D_i = 1) - \mathbb{E}(\epsilon_i|D_i = 0) \neq 0$. The IV uses randomized lottery assignments to deal with the source of selection and estimate the local average treatment effect on compliers (LATE). Considering heterogeneous potential outcomes, IV identifies:

$$\beta_{IV}=\mathbb{E}[Y_i(1)-Y_i(0)|D(1)>D(0)]$$

The underlying identification assumptions are:

Assumption 1. *Independence/ exclusion:* $(Y_i(1), Y_i(0), D_i(1), D_i(0)) \perp Z_i$

Assumption 2. *First Stage:* $\mathbb{E}[D_i|Z_i = 1] > \mathbb{E}[D_i|Z_i = 0]$

Assumption 3. *Monotonicity:* $D_i(1) \geq D_i(0)\forall i$

I borrow from the empirical literature that studies the causal effect of school lottery to identify the causal effect of the Brazilian tax reform. Similarly to the education case, the mean difference of labor market outcomes according to payroll tax variation would lead to biased causal estimates because in additional to the formal eligibility, the implementation of the reform required an endogeneous take-up decision. The sector eligibility to the tax reform does not depend on any of the firm's decision, and I show that it satisfies the three assumptions outlined above.

Assumption 1 is plausible because sector eligibility seems to be uncorrelated to unobserved firms' characteristics. Section 3.3 provides a set of robustness checks showing that (i) eligible and non-eligible firms follow similar trends in the pre-reform period; (ii) in addition to similar trends, they are also similar in levels across many labor market covariates; (iii) similar sectors' description across eligibility status reinforces suggestive evidence of arbitrariness on the eligibility rules; and (iv) the effects are qualitatively similar to alternative identification strategy that relies on propensity score and matching difference-in-differences. Assumptions 2 and 3 are plausible because being eligible does not discourage firms from taking-up the benefit. Even though, there are non-eligible firms that take-up the benefit through the NCM criteria, and there is imperfect take-up among eligible firms, eligibility is a strong predictor for payroll tax reduction.

A.3 Characterizing Compliers

Section 3.1 stresses that the causal interpretation for the LATE is restricted to the set of compliers. Often times, compliers are not representative of the population, therefore it is useful to have deeper understanding on who the compliers are. The challenge is that differently from *always-takers* and *never-takers* compliers' characteristics are not observationally identified. Even though, it is observable if an eligible firm took up treatment, it is not observable if the take-up decision is because the firm is an *always-taker* or *complier*. This comes from the fact that the counterfactual decision (what an eligible firm would do if it were not to be eligible) is not observable in the data.

Abadie Abadie 2002 proposes a 2SLS approach to detect compliers. This method relies on the fact that *never-takers* (eligible firms that don't take-up) and *always-takers* (ineligible firms that take-up) are observable. Concretely, it estimates the pair of regressions:

$$X_{jt} \times \mathbb{I}_{D_j=d} = \alpha_d + \gamma_d \mathbb{I}_{D_j=d} + \nu_{jtd} \tag{23}$$

$$\mathbb{I}_{D_j=d} = \zeta_d + \pi_d L_{s(j)} + \eta_{jtd} \tag{24}$$

where, X_{jt} is a vector of firm's characteristics at the baseline; d ={0,1} indicates if $L_{s(j)}$ is instrumeting eventual treatment or never treatment; and α_d, ζ_d are constants. The IV coefficients for d ={0,1} recovers average characteristics for never and eventually treated compliers, respectively. To obtain baseline characteristics for *never-takers* I regress $X_{jt}(1-D_j)L_{s(j)}$ on $(1-D_j)L_{s(j)}$. Finally, the characterization of *always-takers* come from regressing $X_{jt}D_j(1-L_{s(j)})$ on $D_j(1-L_{s(j)})$. Table A.1 reports results for the same regressions when we incorporate the 1-digit sector x year dummies and set of controls that are included in the main specification[53]. The table shows that covariates' means for treated and untreated compliers are not statistically distinguishable between each other, except for share of male workers, variable which we control for in all specifications. As Angrist, Hull, and Walters 2022 points out, the balance check accross compliers is equivalent to the hidden complier RCT embedded in the treatment assignment with imperfect compliance. Comparisons to the remaining columns showcase that

[53]The interpretation of coefficients are compliers' weighted average characteristics within sector x year cells.

always-takers are larger firms, and *never-takers* are smaller firms compared to compliers.

Table A.1: Compliers' Characteristics

	Compliers			
	Untreated (1)	Treated (2)	Always-Takers (3)	Never-Takers (4)
Employment	108.54 (29.592)	103.69 (16.237)	188.94 (16.86)	27.1 (2.692)
Payroll Tax Rate	.33 (.011)	.35 (.002)	.35 (.002)	.29 (.004)
Share Male	.73 (.02)	.73 (.02)	.71 (.017)	.74 (.022)
Age	35.17 (.326)	35.17 (.326)	33.36 (.087)	36.68 (.354)
High School +	.58 (.026)	.6 (.022)	.57 (.007)	.58 (.022)
White	.75 (.025)	.75 (.025)	.76 (.007)	.71 (.027)
Blue Collar	.8 (.018)	.8 (.018)	.92 (.003)	.86 (.016)
Sector x Year FE	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Note: This table reports baseline estimates characteristics of compliers, always-takers and never-takers in the context of the Brazilian tax reform. Values for each covariate are computed in the pre-reform period at the firm x year level, and the regressions include 1-digit sector x year fixed effects and set of controls considered in the main specification (section 3.1). Standard errors are reported in paranthesis and conservatively clustered at the 5-digit industry-by-state level.

B Model

In this appendix, I present the model derivation. For didactic purposes, I start by analyzing the revenue and payroll taxes, separately. At the end, I put both taxes together to map the structural equations to the reduced form estimates. I start by studying the shape of the labor supply curve faced by the firm, based on workers' choice.

B.1 Microfounding the Labor Supply

As in Card et al. 2018, workers exhibit idiosyncratic preferences for employers. These preferences can be understood through non-pecuniary match factors such as corporate culture and commuting distance. Unlike traditional search models, this approach posits that wage-posting behavior induce firms to pay identical wages to all workers of the same quality. Upon meeting the requisite quality standards, a firm hires any worker willing to accept the posted wage. In this scenario, worker i is fully knowledgeable of available job opportunities, and derive the following utility from working at firm j:

$$u_{ij} = \epsilon \ln(w_j - b) + a_j + \nu_{ij}$$

where, w_j is the wage level paid by firm j, b is the competitive wage level defined by the workers' outside option, a_j is a firm-specific amenity, and ν_{ij} is the idiosyncratic preference for worker i to be at firm j. Assuming that ν_{ij} comes from an extreme type I distribution, I follow McFadden et al. 1973 to compute the logit probabilities to work at firm j:

$$p_j = \frac{\exp(\epsilon \ln(w_j - b) + a_j)}{\sum_{k=1}^{J} \exp(\epsilon \ln(w_k - b) + a_k)}$$

If the total number of firms J is large enough, the logit probabilities can be approximated by exponential probabilities of the form,

$$p_j = \lambda \exp(\epsilon \ln(w_j - b) + a_j)$$

where λ is a constant common to all firms in the market. Therefore, for large J, we can write the firm-specific supply function as:

$$\ln L_j(w_j) = \ln \mathbb{L}\lambda + \epsilon \ln(w_j - b) + a_j$$

where $\mathbb{L}$ represents the total number of workers in the market. Taking exponential transformations on both sides, we can compute the labor supply function:

$$L_j = \exp(\epsilon \ln(w-b)) \exp(a_j) \exp(\lambda \mathbb{L}) \iff L_j^{\frac{1}{\epsilon}} \underbrace{\exp\left(\frac{-L\lambda - a_j}{\epsilon}\right)}_{\equiv A_j} = (w_j - b)$$

As b $\rightarrow 0$, then

$$w_j = A_j L_j^{\frac{1}{\epsilon}} \tag{25}$$

In this case, ϵ is the constant labor supply elasticity faced by the firm.

B.2 Effects of Payroll Taxation

The labor supply function gives rise to the cost function faced by firms,

$$C = A((1+\tau))L^{\frac{1}{\epsilon}+1} + rK$$

Production function exhibits constant returns to scale, and the firm faces demand at the product market given by, $P = Q^{\frac{-1}{\eta}}$. The firm solves two related problems. First, it chooses plant size to maximize profit. Second for a given plant size (Q), it chooses inputs of production (L and K) to minimize costs, according to the following program:

$$\begin{aligned} \min_{K,L} \quad & A(1+\tau)L^{\frac{1}{\epsilon}+1} + rK \\ \text{s.t.} \quad & f(K,L) \geq Q \end{aligned} \tag{26}$$

Summing and rearranging the optimality conditions, I obtain the cost function:

$$C = \underbrace{\lambda(w,r,Q)}_{=\frac{\partial C}{\partial Q}} \underbrace{(Lf_L + Kf_K)}_{=Q} - \underbrace{A(1+\tau)\frac{1}{\epsilon}L^{1+\frac{1}{\epsilon}}}_{new} \tag{27}$$

Differently from the perfectly competitive labor market, under monopsony average and marginal cost no longer align. Lemma 1 proves this point.

Lemma 1. *In a perfectly competitive labor market, the marginal cost of production is constant in the quantity Q.*

Proof. From FOC,

$$C(w,r,Q) = \lambda(w,r,Q)Q \iff C(w,r,\alpha Q) = \lambda(w,r,\alpha Q)\alpha Q$$

From constant returns,

$$\begin{aligned} C(w,r,\alpha Q) = \alpha C(w,r,Q) = \alpha\lambda(w,r,Q)Q \\ \lambda(w,r,\alpha Q)\alpha Q = \lambda(w,r,Q)\alpha Q \Rightarrow \lambda(w,r,Q) = \lambda(w,r) \end{aligned}$$

□

The profit maximizing firm chooses output Q,

$$\max_{Q} P(Q)Q - c(Q,\tau)Q + \frac{1}{\epsilon}A(1+\tau)L^{1+\frac{1}{\epsilon}}$$

At the optimal, marginal cost and marginal revenue are equated:

$$\left(\frac{\eta-1}{\eta}\right)Q^{\frac{-1}{\eta}} = \lambda(Q,\tau) \tag{28}$$

To evaluate the effect on labor costs $(1+\tau)$ induced by the payroll tax variation, I take logs and differentiate with respect to $(1+\tau)$,

$$\epsilon^{Q}_{1+\tau} = \frac{-\epsilon^{\lambda}_{1+\tau}}{\left(\frac{1}{\eta}+\epsilon^{\lambda}_{Q}\right)} \tag{29}$$

Also notice that from 28,

$$P\left(\frac{\eta-1}{\eta}\right) = \lambda \iff \frac{\partial \log P}{\partial \log(1+\tau)} = \frac{\partial \log \lambda}{\partial \log(1+\tau)} = \epsilon^{\lambda}_{1+\tau} + \epsilon^{\lambda}_{Q}\epsilon^{Q}_{1+\tau}$$

$$\frac{\partial \log Q}{\partial \log(1+\tau)} = \underbrace{\frac{\partial \log Q}{\partial \log P}}_{-\eta}\overbrace{\frac{\partial \log P}{\partial \log(1+\tau)}}^{\epsilon^{\lambda}_{1+\tau}+\epsilon^{\lambda}_{Q}\epsilon^{Q}_{1+\tau}}$$

$$\frac{\partial \log Rev}{\partial \log(1+\tau)} = \frac{\partial \log PQ}{\partial \log(1+\tau)} = (1-\eta)(\epsilon^{\lambda}_{1+\tau} + \epsilon^{\lambda}_{Q}\epsilon^{Q}_{1+\tau}) \tag{30}$$

Applying the envelope theorem to derive equation 27 with respect to $(1+\tau)$,

$$AL^{\frac{1}{\epsilon}+1} = \lambda_{1+\tau}Q - \frac{AL^{1+\frac{1}{\epsilon}}}{\epsilon} - \frac{A(1+\tau)}{\epsilon}\left(\frac{1+\epsilon}{\epsilon}\right)L^{\frac{1}{\epsilon}}\frac{\partial L}{\partial(1+\tau)}$$

$$\frac{\partial \log \lambda}{\partial \log(1+\tau)} = \frac{(1+\tau)AL^{1+\frac{1}{\epsilon}}}{\lambda Q}\left(\frac{\epsilon+1}{\epsilon}\right)\left(1+\frac{(1+\tau)}{\epsilon L}\frac{\partial L}{\partial(1+\tau)}\right) \tag{31}$$

Equation 31 refers to the the elasticity of the marginal cost with respect to the labor cost, which is a key aspect of the incidence analysis. Taking this expression to the data is challenging because we don't oberserve neither λ, nor Q. However, manipulating equation 27 and dividing both sides by the total wage bill we obtain,

$$\frac{\lambda Q}{(1+\tau)AL^{1+\frac{1}{\epsilon}}} = \frac{C + (1+\tau)AL^{1+\frac{1}{\epsilon}}(\frac{1}{\epsilon})}{\underbrace{(1+\tau)AL^{1+\frac{1}{\epsilon}}}_{wL}} = \frac{1}{s_L} + \frac{1}{\epsilon} \tag{32}$$

The right hand side of equation 32 depends on s_L and ϵ. It turns out that we do observe labor share (s_L), and we can estimate ϵ. Plugging 32 in 31,

$$\epsilon^{\lambda}_{1+\tau} = \left(\frac{1}{\frac{1}{s_L} + \frac{1}{\epsilon}}\right)\left(\frac{\epsilon+1}{\epsilon}\right)\left(1 + \frac{\epsilon^L_{1+\tau}}{\epsilon}\right) \tag{33}$$

Equation 33 shows that the effect of the labor cost on the marginal cost depends on three components. First, is the monopsony adjusted labor share. The more relevant is the labor share, means that reducing labor cost will have greater impact on the marginal cost. Second, is the inverse mark down. The intuition for this term is that as labor market power increases, there is more rents to be shared with incumbents workers when the firm expands plant size. Finally, the last term says that the pass-through to marginal cost is directly affected by the pass-through to the marginal cost of labor. Differentiating both sides of equation 27 by Q, after some manipulation I obtain,

$$\epsilon^{\lambda}_{Q} = \left(\frac{\epsilon+1}{\epsilon}\right)\left(\frac{\epsilon^L_Q}{\epsilon}\right)\left(\frac{1}{\frac{1}{s_L} + \frac{1}{\epsilon}}\right) \tag{34}$$

Notice that,

$$\epsilon^L_{1+\tau} = \frac{\partial \log L}{\partial \log(1+\tau)} = \frac{\partial \log L}{\partial \log Q}\frac{\partial \log Q}{\partial \log(1+\tau)}$$

$$\epsilon^L_Q = \frac{\epsilon^L_{1+\tau}}{\epsilon^Q_{1+\tau}} \tag{35}$$

Using 29 in 35,

$$\epsilon^L_Q = \frac{-\epsilon^L_{1+\tau}(\frac{1}{\eta} + \epsilon^{\lambda}_Q)}{\epsilon^{\lambda}_{1+\tau}} \tag{36}$$

Now, 36 and 33 in 34,

$$\epsilon^{\lambda}_Q = \frac{-\epsilon^L_{1+\tau}}{\eta(2\epsilon^L_{1+\tau} + \epsilon)} \tag{37}$$

To compute $\epsilon^{Q}_{1+\tau}$ substitute 33 and 37 in 29,

$$\epsilon^{Q}_{1+\tau} = -\left(\frac{\epsilon+1}{\epsilon}\right)\left(1+\frac{\epsilon^{L}_{1+\tau}}{\epsilon}\right)\left(\frac{1}{\frac{1}{s_L}+\frac{1}{\epsilon}}\right)\left(\frac{\eta(\epsilon+2\epsilon^{L}_{1+\tau})}{\epsilon+\epsilon^{L}_{1+\tau}}\right) \tag{38}$$

To compute the tax reduction pass-through to employment and capital, I can differentiate optimal choices in 26 with respect to the labor cost ($(1+\tau)$):

$$\epsilon^{L}_{1+\tau} = \frac{\epsilon}{1-\epsilon\rho+\epsilon}(\epsilon^{\lambda}_{1+\tau}+\epsilon^{\lambda}_{Q}\epsilon^{Q}_{1+\tau}-1)+\left(\frac{(1-\rho)\epsilon}{1-\epsilon\rho+\epsilon}\right)\epsilon^{Q}_{1+\tau}$$

Plugging 33, 37 and 38, I obtain the model's prediction for the pass-though to employment, in terms of observables and parameters to be estimated:

$$\epsilon^{L}_{1+\tau} = \left(\frac{\epsilon}{1+\epsilon(1-\rho)}\right)\left(\Omega(\epsilon,\eta,\rho)-1\right) \tag{39}$$

where,

$$\Omega(\epsilon,\eta,\rho) = \left(\frac{(\epsilon+2\epsilon^{L}_{1+\tau})(\sigma-\eta)}{\sigma\epsilon}\right)\left(\frac{\epsilon+1}{\epsilon}\right)\left(\frac{1}{\frac{1}{s_L}+\frac{1}{\epsilon}}\right) \tag{40}$$

Recall, that the elasticity of employment with respect to labor cost $\epsilon^{L}_{1+\tau}$ I empirically estimate in the reduced form analysis. The remaining structural parameters are jointly estimated in section 5. Similarly, I can find equations for the pass-through to capital, and revenue.

$$\epsilon^{K}_{1+\tau} = \left(\frac{\epsilon+1}{\epsilon}\right)\left(\frac{1}{\frac{1}{s_L}+\frac{1}{\epsilon}}\right)\left(\frac{\epsilon+2\epsilon^{L}_{1+\tau}}{\epsilon}\right)\left(\frac{1}{1-\rho}-\eta\right) \tag{41}$$

$$\epsilon^{R}_{1+\tau} = (1-\eta)\left[\left(\frac{\epsilon+1}{\epsilon}\right)\left(\frac{\epsilon+\epsilon^{L}_{1+\tau}}{\epsilon}\right)\left(\frac{1}{\frac{1}{s_L}+\frac{1}{\epsilon}}\right)\right]\left(\frac{\epsilon+2\epsilon^{L}_{1+\tau}}{\epsilon+\epsilon^{L}_{1+\tau}}\right) \tag{42}$$

B.3 Effects of Revenue Taxation

Under revenue taxation (τ_r), the firm solves the following program in the product market:

$$\max_{Q} P(Q)Q - \frac{C(Q)}{1-\tau_r}$$

The firm equates marginal revenue to marginal cost,

$$\left(\frac{\eta-1}{\eta}\right)Q^{\frac{-1}{\eta}}=\frac{\lambda(Q)}{1-\tau_r}$$

where, the right hand side is a direct application of the envelope theorem on the cost minimization problem. The plant size has direct implication on prices through demand, so if we take logs and differentiate with respect to $\log \tau_r$,

$$\frac{\partial \log P}{\partial \log \tau_r}=\frac{\tau_r}{1-\tau_r}$$

I know the relationship between the elasticity of prices and quantity with respect to revenue taxes,

$$\frac{\partial \log P}{\partial \log Q}\frac{\partial \log Q}{\partial \log \tau_r}=\frac{\partial \log P}{\partial \log \tau_r} \iff \frac{\partial \log Q}{\partial \log \tau_r}=-\frac{\tau_r}{1-\tau_r}\eta \tag{43}$$

where, the $\frac{\partial \log P}{\partial \log Q}=\frac{-1}{\eta}$ is known based on the iso-elastic demand function. The price and quantity responses allow me to compute the effect of revenue taxes on revenue,

$$\epsilon^R_{1+\tau_r}=\frac{\tau_r}{1-\tau_r}(1-\eta)$$

Once firms, choose the plant size, they will choose the inputs mix to minimize cost,

$$C(Q)=\min_{K,L} \quad AL^{\frac{1}{\epsilon}+1}+rK$$
$$\text{s.t.} \quad (s_L L^\rho+s_K K^\rho)^{\frac{1}{\rho}}\geq Q$$

The optimal choices of capital and labor are:

$$L=\left[\left(\frac{\epsilon}{\epsilon+1}\right)\frac{s_L}{A}\lambda(Q)\right]^{\frac{\epsilon}{1-\epsilon\rho+\epsilon}}Q^{\frac{(1-\rho)\epsilon}{1-\epsilon\rho+\epsilon}} \qquad K=\left(\frac{r}{\lambda(Q)s_K}\right)^{\frac{1}{\rho-1}}Q$$

Taking logs and differentiating with respect to $\log \tau_r$, we obtain the revenue tax pass-through to employment and wages,

$$\frac{\partial \log L}{\partial \log \tau_r}=\frac{-\epsilon}{1-\epsilon\rho+\epsilon}+\left(\frac{(1-\rho)\epsilon}{1-\epsilon\rho+\epsilon}\right)\frac{\partial \log Q}{\partial \log \tau_r}+\frac{\epsilon}{1-\epsilon\rho+\epsilon}\left(\frac{\partial \log \lambda(Q)}{\partial \log Q}\frac{\partial \log Q}{\partial \log \tau_r}\right) \tag{44}$$

$$\frac{\partial \log K}{\partial \log \tau_r}=\frac{\partial \log Q}{\partial \log \tau_r}-\left(\frac{1}{\rho-1}\right)\left[\frac{\partial \log \lambda(Q)}{\partial \log Q}\frac{\partial \log Q}{\partial \log \tau_r}\right] \tag{45}$$

To obtain closed form solution for the pass-through expressions we need to compute the elasticity of marginal cost with respect to quantity ϵ^λ_Q, which we can pin down by differentiating the cost function with respect to Q,

$$\epsilon^\lambda_Q=\left(\frac{1}{\frac{1}{s_L}+\frac{1}{\epsilon}}\right)\left(\frac{\epsilon+1}{\epsilon}\right)\frac{\epsilon^L_Q}{\epsilon} \tag{46}$$

Notice that,

$$\epsilon_Q^L = \frac{\epsilon_{L\tau_r}}{\epsilon_{\tau_r}^Q} \iff \epsilon_Q^L = \frac{-\epsilon_{\tau_r}^L(1-\tau_r)}{\tau_r \eta} \tag{47}$$

Plugging 47 in 46,

$$\epsilon_{\tau_r}^L = -\left(\frac{1}{\frac{1}{s_L}+\frac{1}{\epsilon}}\right)\left(\frac{\epsilon+1}{\epsilon}\right)\frac{\epsilon_{\tau_r}^L(1-\tau_r)}{\tau_r \eta}$$

Plugging ϵ_Q^λ and $\epsilon_{\tau_r}^Q$ in 44 and 45, we obtain the closed form pass-through expressions for the revenue taxation,

$$\epsilon_{\tau_r}^L = \frac{-(1-\rho)\epsilon}{1+\epsilon(1-\rho-\chi(\epsilon,s_L))}\frac{\tau_r}{1-\tau_r}\eta \quad (48) \quad , \quad \epsilon_{\tau_r}^K = \frac{\tau_r\eta}{1-\tau_r}\left(\frac{-\chi(\epsilon,s_L)\epsilon}{1+\epsilon(1-\rho-\chi(\epsilon,s_L))}-1\right) \tag{49}$$

where, I denote $\chi(\epsilon, s_L) = \left(\frac{1}{\frac{1}{s_L}+\frac{1}{\epsilon}}\right)\left(\frac{\epsilon+1}{\epsilon}\right)$ to simplify notation. The elasticity η makes the model versatile to accommodate different degrees of competition in the product market. As η increases we move to a more competitive product market. At first, we will be agnostic about its value, and let η be determined by the data. For the specific case of the Brazilian tax reform, the revenue tax rate is small (around 1.5%). For this reason, the effects depicted on equations 48 and 49 are negligible. This result makes intuitive sense, as most of the action in this reform is on the payroll tax side.

C Robustness Checks

[TO BE INCLUDED]

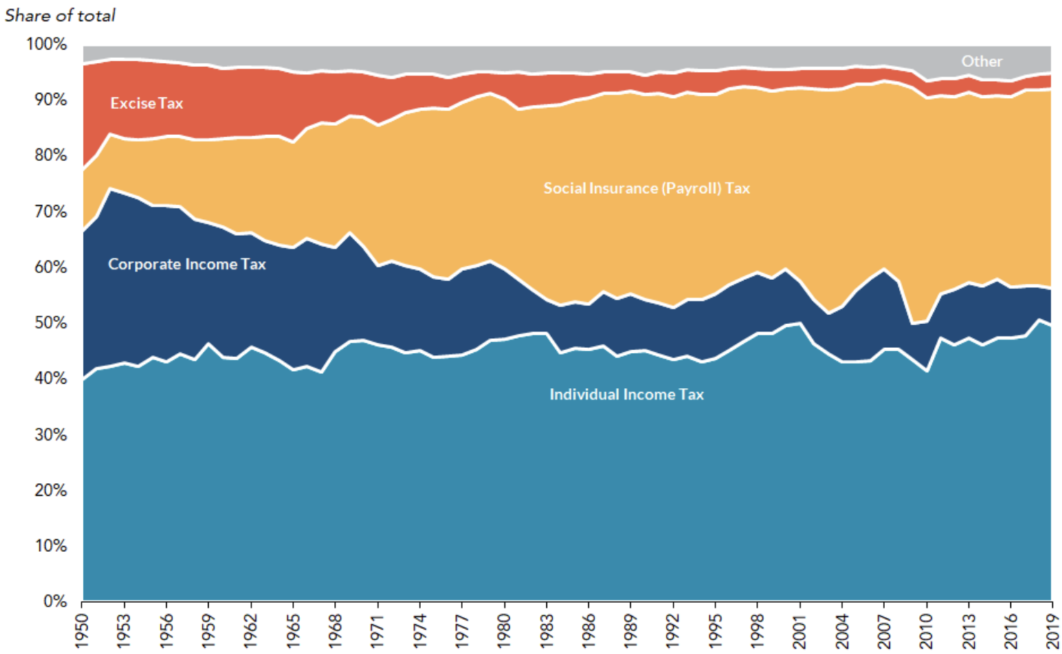
C.1 Balanced Sample

C.2 Matched Sample

I follow extensive theoretical (Cochran and Rubin 1973; Rosenbaum and Rubin 1984; Ho et al. 2007) and applied (Campos and Kearns 2022) literature that propose matching methods to deal with potential imbalances at baseline.

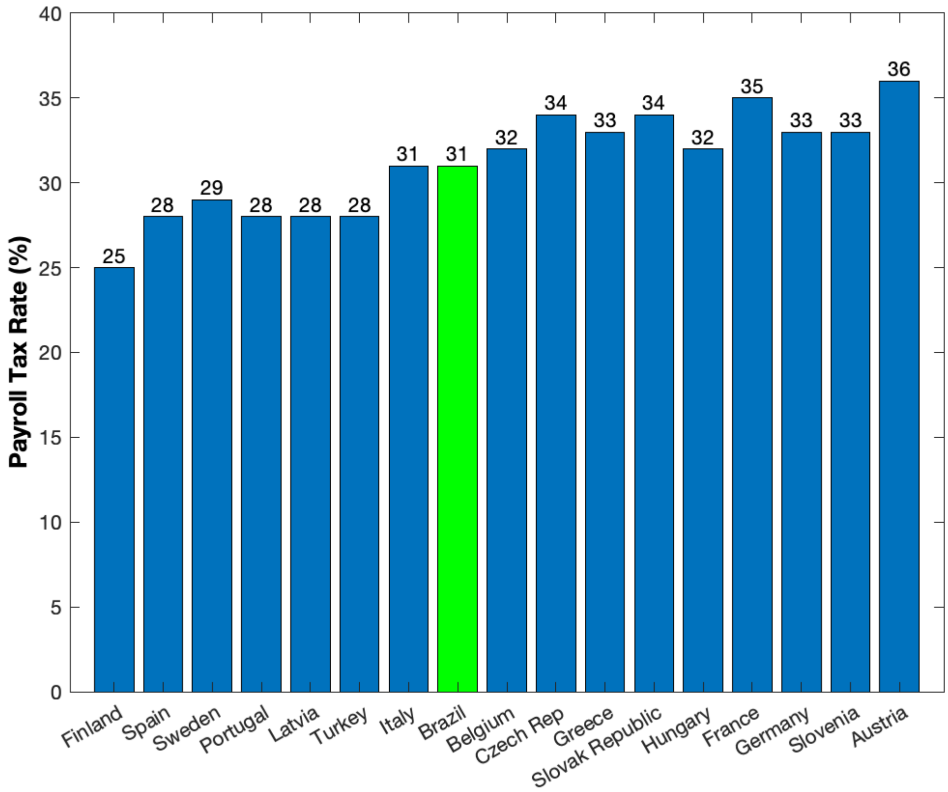
D Additional Figures and Tables

Figure D.1: Sources of Government Revenue



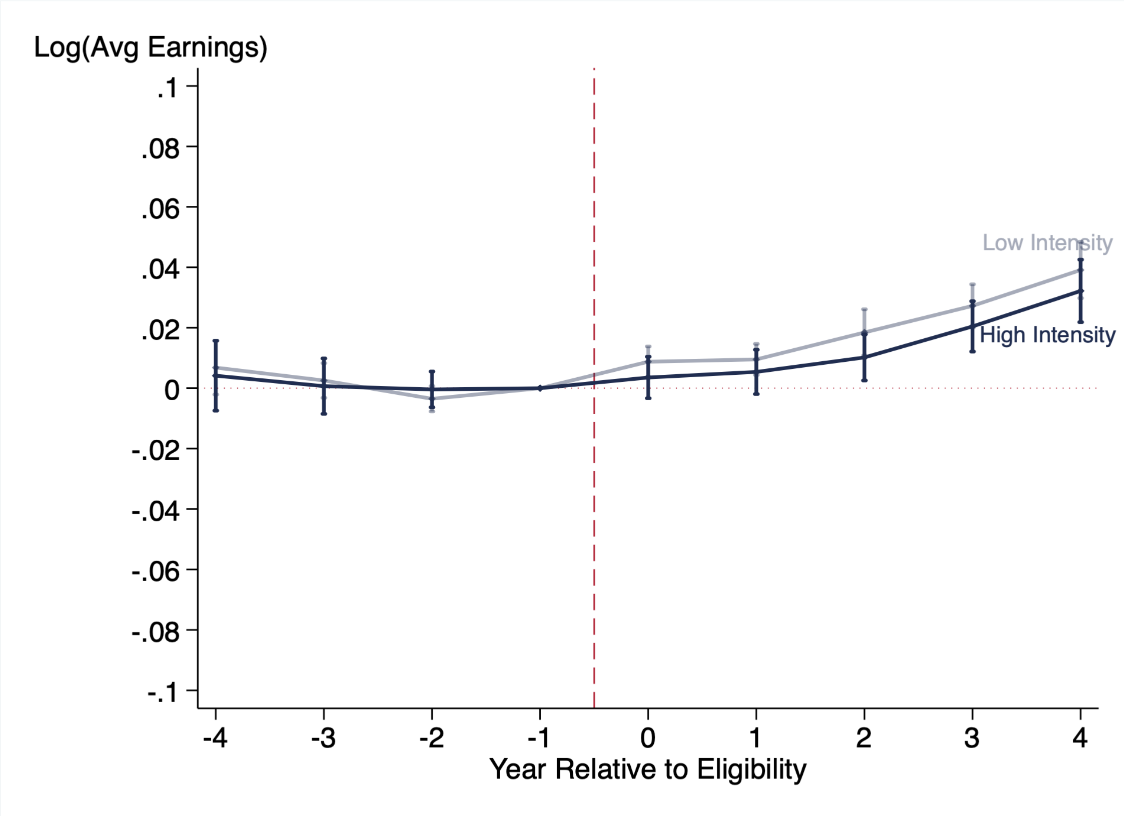
Note: This figure shows growing importance of payroll taxes compared to other souces of revenue for the US Federal Government. Currently, payroll taxes are the second most important source, accounting for more than 30% of total revenue.
Source: Office of Management and Budget. Historical Tables. Table 2.1, "Receipts by Source: 1950-2025"

Figure D.2: Payroll Tax Rates Around the World



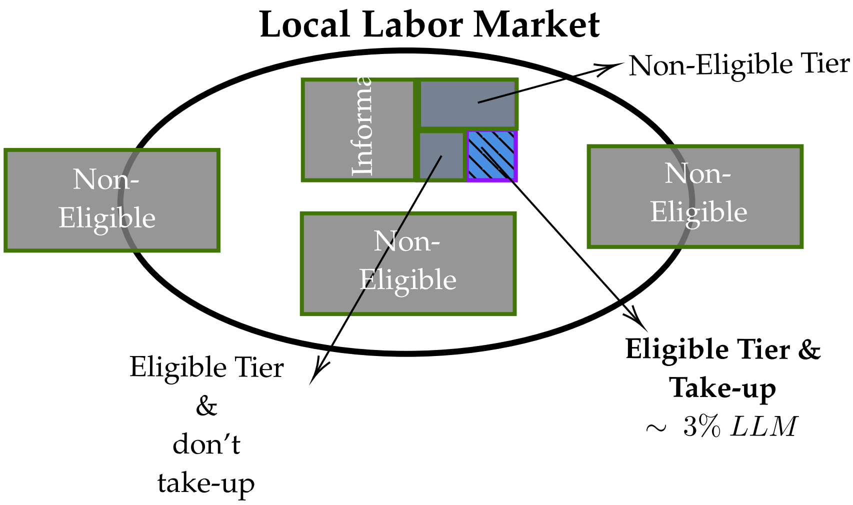
Note: This figure reports payroll tax rates around the world. The payroll tax rate is composed by the sum of employer and employee's contributions.
Source: Elaborated by author, based on information from OECD 2019.

Figure D.3: Firm vs Market Level Shock



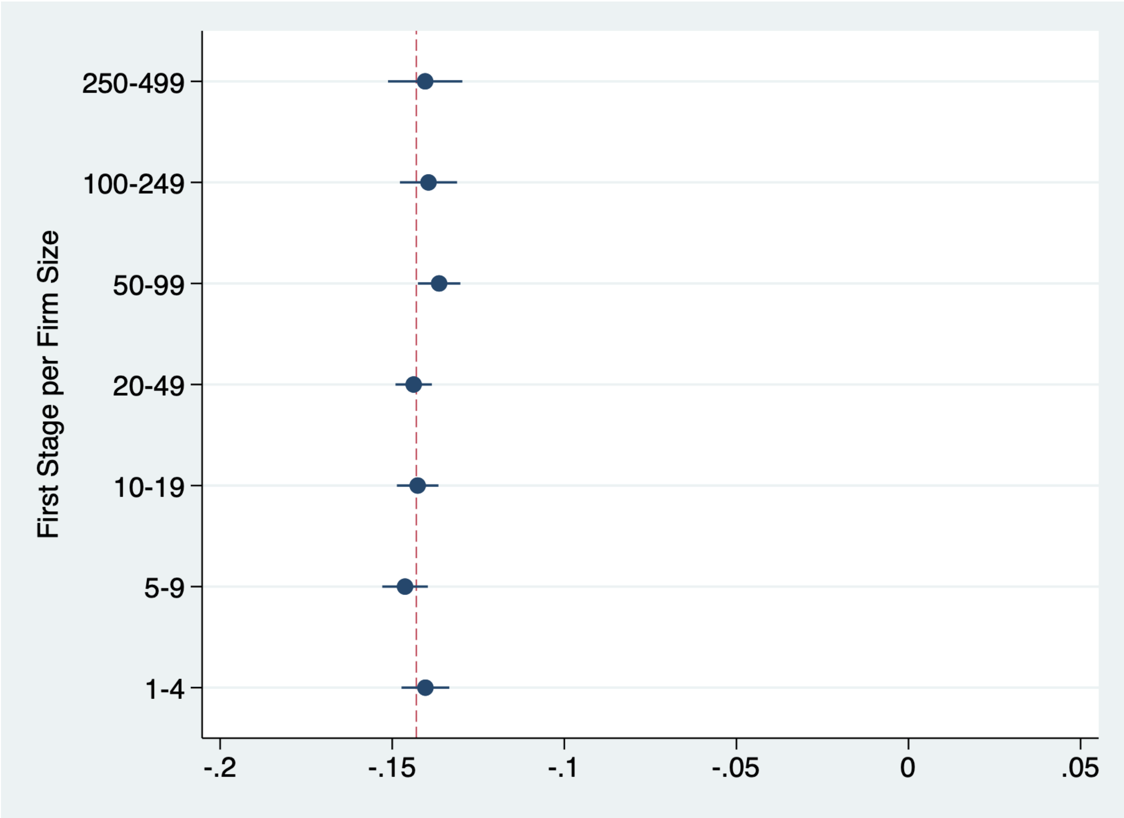
Note: This figure provides additional test on the spillover effect. It compares the worker's earnings effect for high and low intensitively treated markets. To measure market treatment intensity I compute the share of treated workers in each labor market, which are defined by the occupation x region cells. Then it separately estimates equations 7 and 8, for workers in markets below and above the median in market intensity. Standard errors are conservatively clustered at the 5-digit industry-by-state level. If the driving force for the earnings increase was a bump on workers' outside option through market spillover, we would expect to see more pass-through on high intensity markets. The figure shows no significant difference across market intensity.

Figure D.4: Illustration of Treatment Coverage in LLM



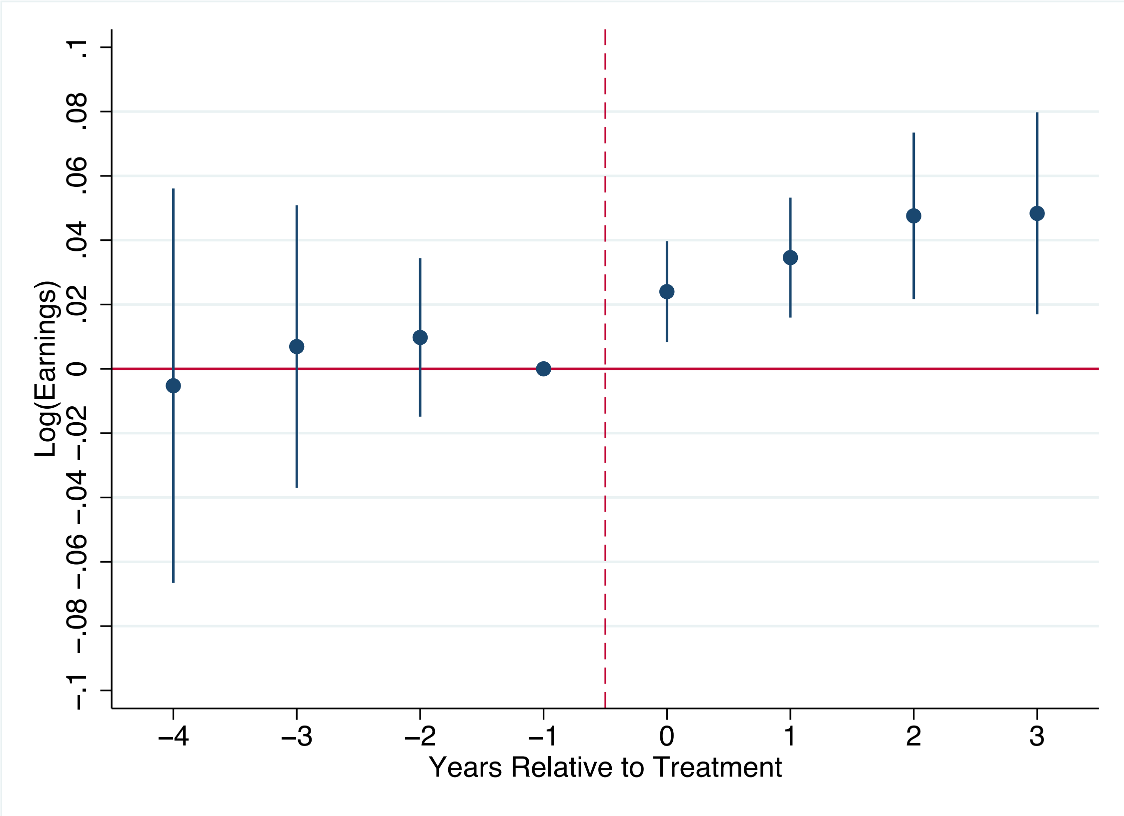
Note: This figure provides illustration of the policy coverage within local labor markets (LLM), which are defined as occupation x region cells according to job switching patterns. The figure shows that within LLM there are eligible and non-eligible sectors. In eligible sectors, there is ineligibility due to informality, non-eligible tax tier ("Simples"), and imperfect take-up. All together, the figure illustrates that conditional on existing an eligible sector in a local labor market, the share of treated firms in the LLM is approximately 3%.

Figure D.5: First Stage per Firm Size



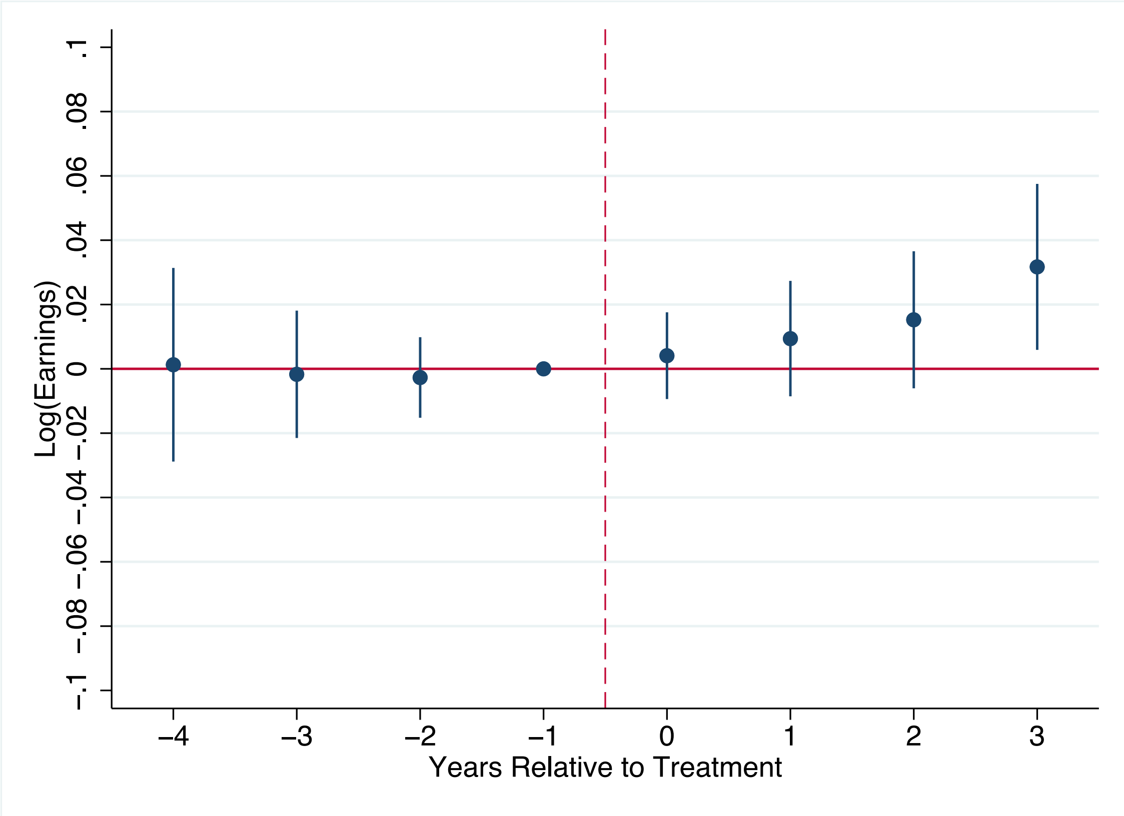
Note: This figure presents the difference-in-differences IV estimate for the effect of the tax treatment on the log of labor cost. Firms are categorized on size bins based on pre-reform levels. It shows that the effect of the reform on labor cost is similar for all size bins. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure D.6: Earnings Effect for MW Workers



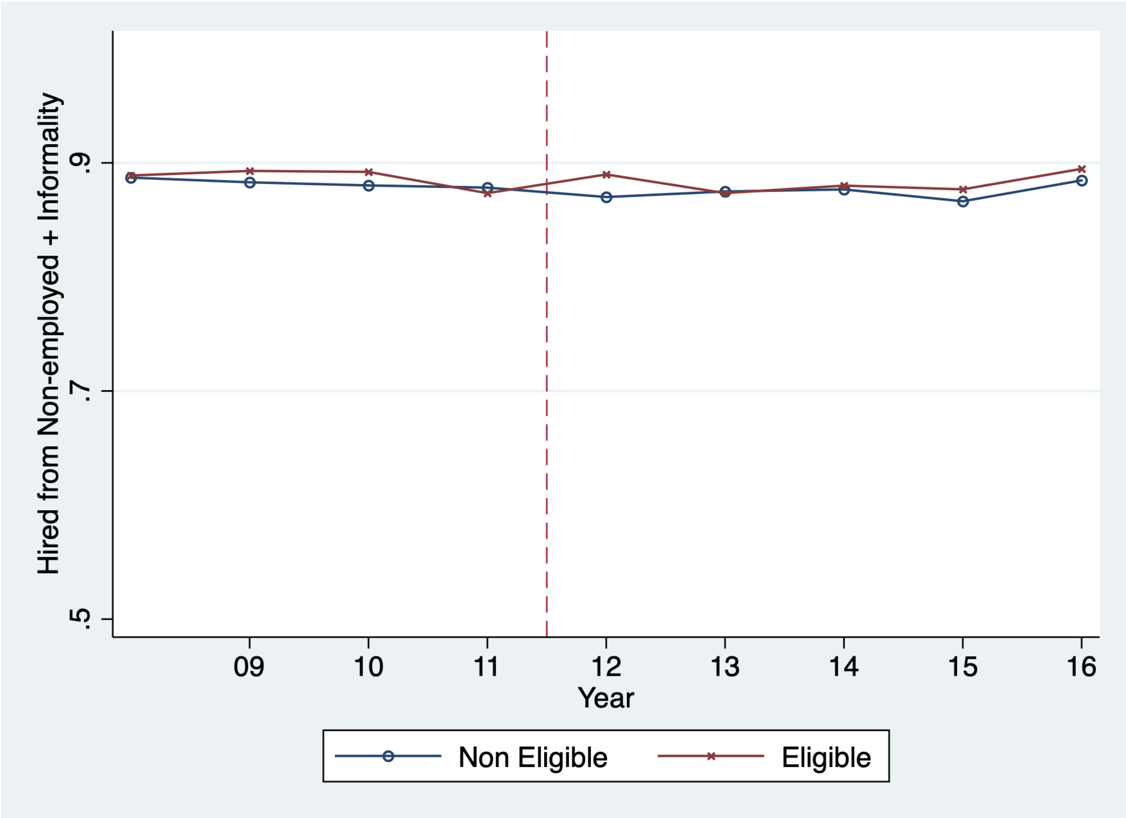
Note: This figure presents the event study estimates for workers that were constrained by the minimum wage in the pre-period. The plot shows the average earnings effect for workers that were employed for at least three years in the same firm during the pre-reform period. I normalize the results with respect to one year prior to the treatment event. The analysis spans four years prior to entering the payroll tax cut program and three years after. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure D.7: Earnings Effect for Non-MW Workers



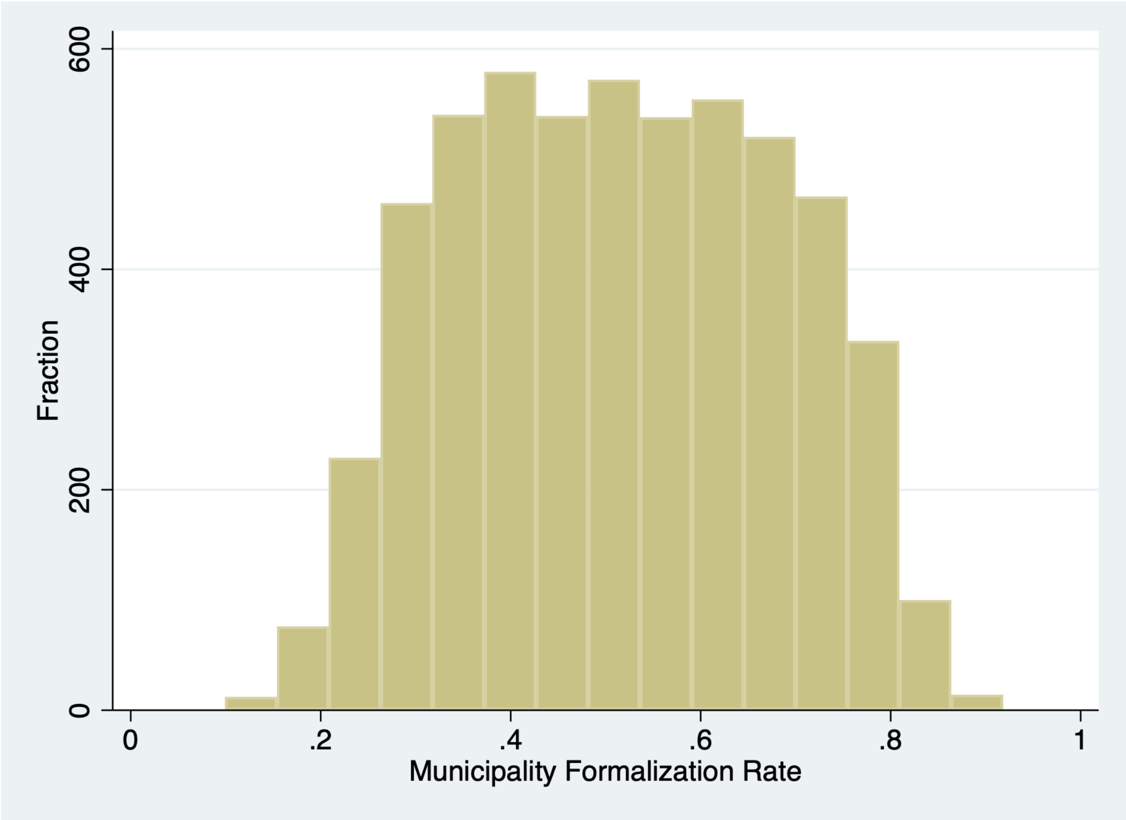
Note: This figure presents the event study estimates for workers that were not constrained by the minimum wage in the pre-period. The plot shows the average earnings effect for workers that were employed for at least three years in the same firm during the pre-reform period. I normalize the results with respect to one year prior to the treatment event. The analysis spans four years prior to entering the payroll tax cut program and three years after. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure D.8: New Hires Origin by Eligibility Status



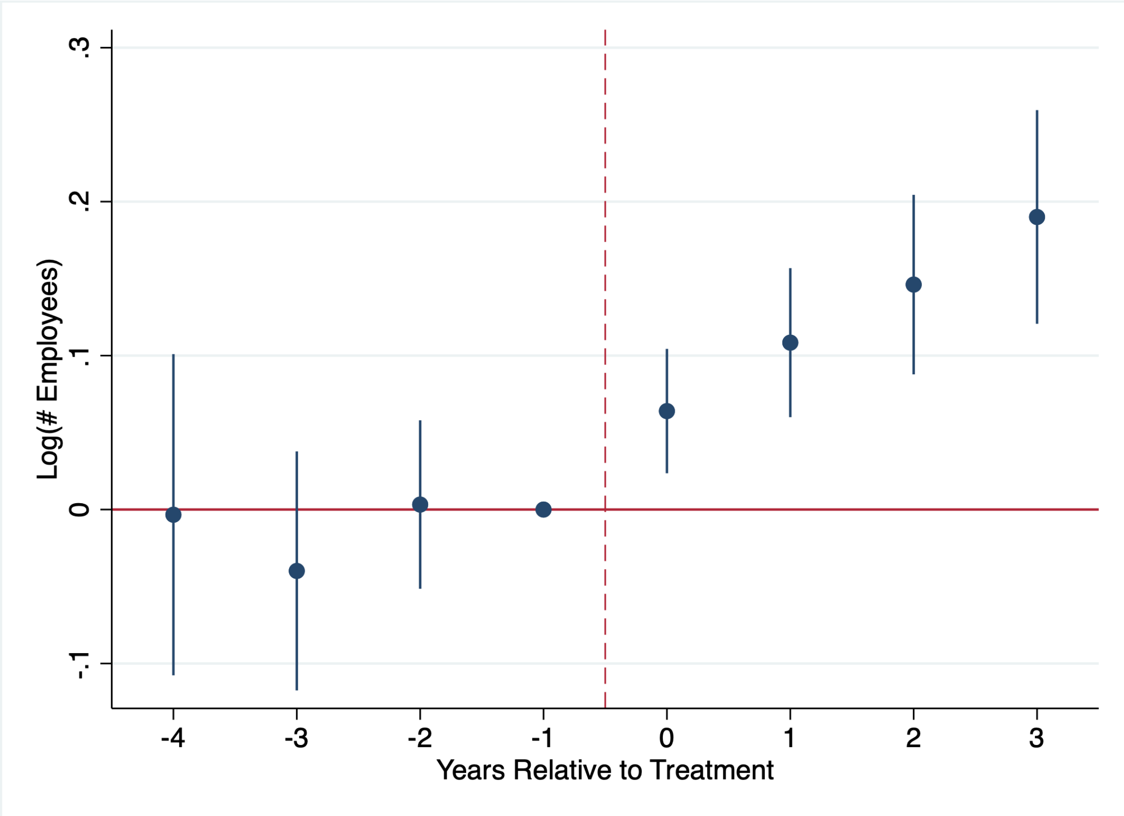
Note: This figure plots the share of new hires coming from non-employment or informality. A new hire is classified as previously informal or non-employed if she was not holding a formal job in the three months prior to being hired. Eligibility is defined based on the sector of employment.

Figure D.9: Formalization Rates per Municipality



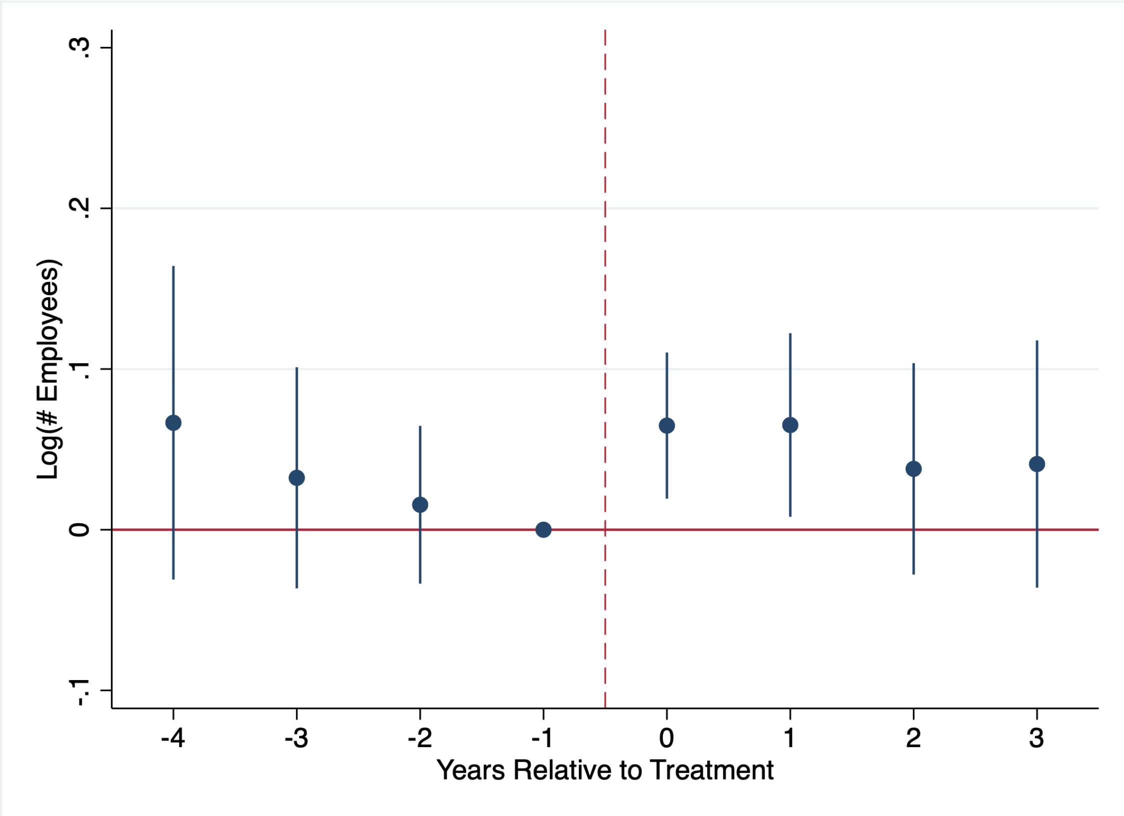
Note: This figure presents the distribution of formalization rates per municipalities in Brazil, according to the 2010 Census. There are 5,300 municipalities with heterogeneous informality rates.

Figure D.10: Firm Level: Employment Effect in High Formalized Regions



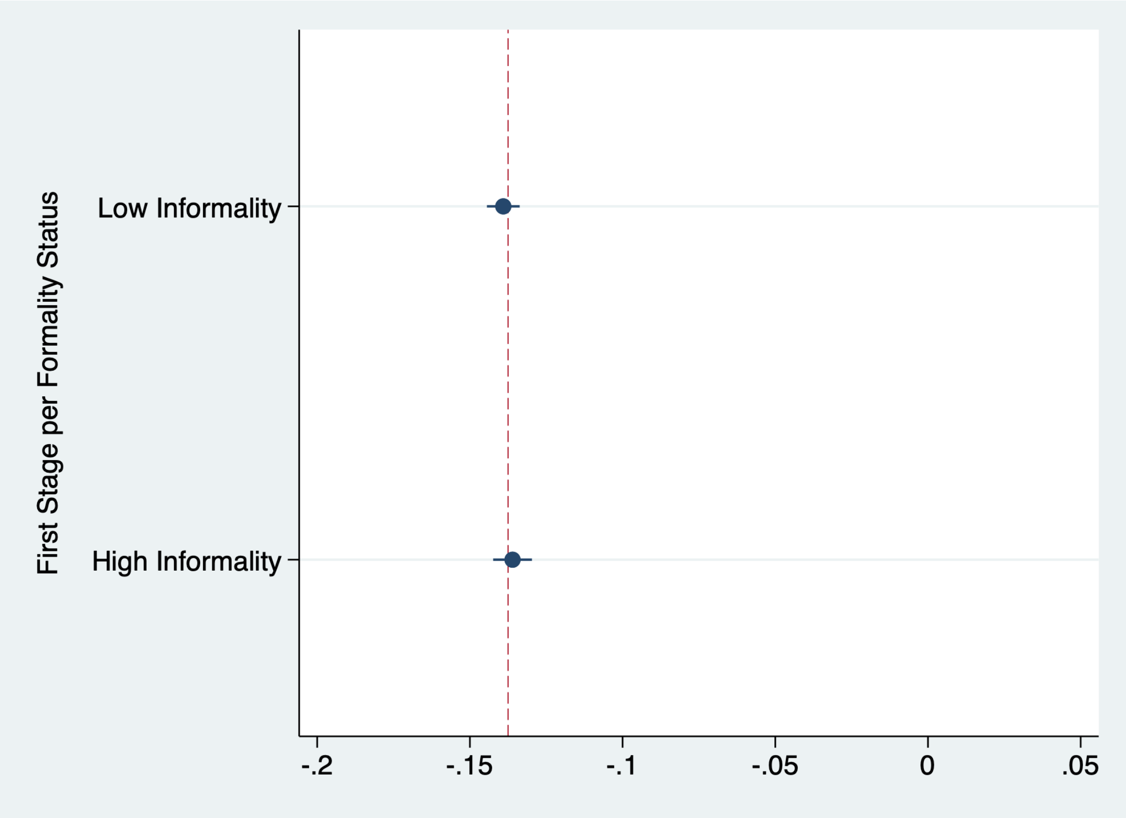
Note: This figure plots the event study presented in equations 2 and 1 estimated for firms in informality municipalities. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure D.11: Firm Level: Employment Effect in Low Formalized Regions



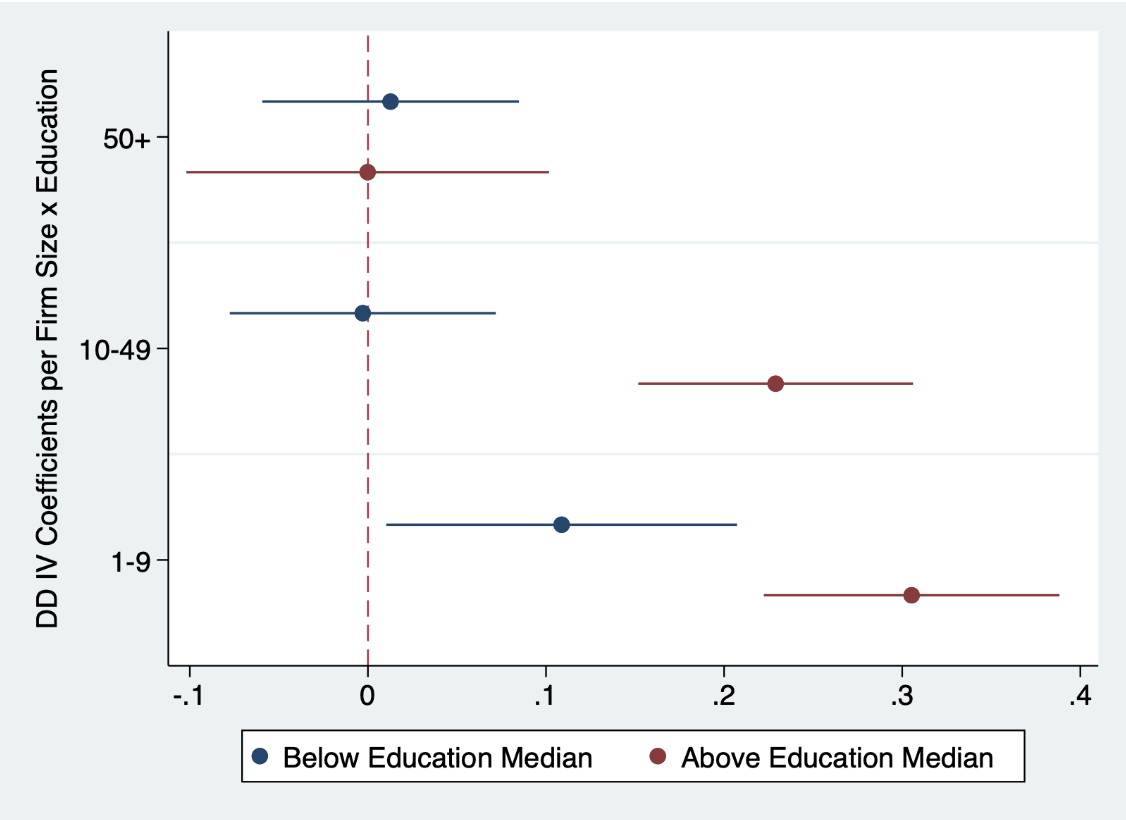
Note: This figure plots the event study presented in equations 2 and 1 estimated for firms in high informality municipalities. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure D.12: Tax Variation per Informality Status



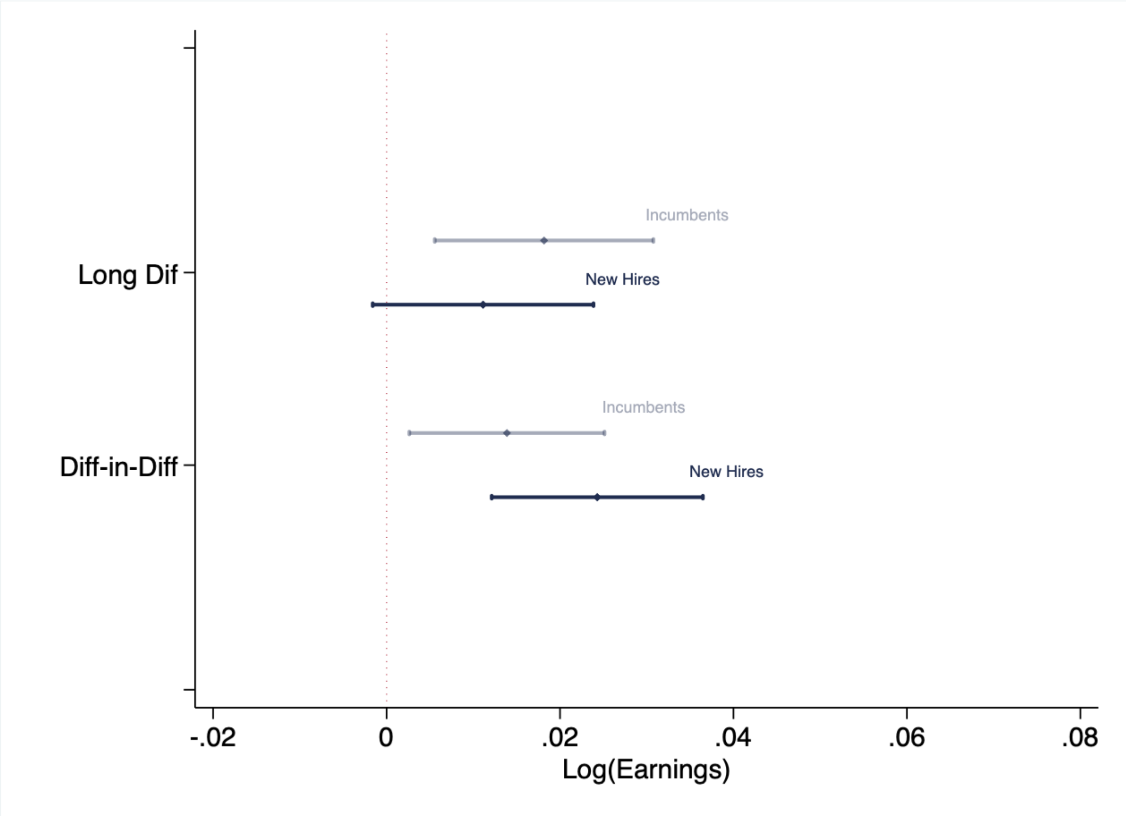
Note: This figure plots the labor cost variation induced by the policy across formalization status. The reduction in labor cost observed in the data is consistent with the statutory tax cut. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure D.13: Firm Level: Employment Effect by Avg Education and Size



Note: This figure plots the employment effect per firm size and average share of workers with high school degree. Firm size bins and share of workers with high school are computed in the pre-reform period. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Figure D.14: Pass-Through to New Hires and Incumbent Workers



Note: This figure presents firm level difference-in-differences coefficient, and event study coefficients in t+3. The blue dots represent estimates for new hires, whereas the gray bars are incumbent's response. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Table D.1: Descriptives on Market Level Treatment

		Average
	Share of Treated Firms in a Treated Market (1)*(2)*(3)*(4)	**0.03**
(1)	Share of Treated Sectors per Market — At Least One	0.211
(2)	Formality Rate	0.550
(3)	Share of Eligible Tax Tier	0.520
(4)	Take Up Within Eligible	0.517

Note: This table breaks down the calculation of treatment share per local labor market. Row (1) reports the local labor market (LLM) share of eligible sectors, conditional on existing at least one eligible sector on the given LLM. Row (2) reports worker's average formality rate in Brazil; row (3) reports the share of firms in the eligile tax tier; row (4) reports the take-up rate within eligible firms. The product of these 4 rows gives the share of treated firms in a treated market.

Table D.2: List of Eligible Sectors

Description (7-Digit Sector)	Industry	Year
Development and licensing of customizable computer programs	IT	2012
Technical support, maintenance and other information technology services	IT	2012
Data processing, application service providers and internet hosting services	IT	2012
Development of customed computer programs	IT	2012
Web design	IT	2012
Consulting in information technology	IT	2012
Development and licensing of computer programs	IT	2012
Call center activities	Call Center	2012
Hotel	Lodging	2012
Manufacturing of eletronic games	Manufacturing	2012
Manufacturing of eletronic components	Manufacturing	2012
Aircraft maintenance and repair, except runway	Maintanance	2013
Aircraft maintenance on the runway	Maintanance	2013
Maritime support navigation	Transportation	2013
Port support navigation	Transportation	2013
Maintenance of vessels and floating structures	Transportation	2013
Maintenance and repair of boats for sport and leisure	Transportation	2013
Construction of buildings	Construction	2013
Electrical installation and maintenance	Construction	2013
Waterproofing in civil engineering works	Construction	2013
Installation of doors, windows, ceilings, partitions and built-in cabinets of any material	Construction	2013
Plaster and stucco finishing works	Construction	2013
Building painting services in general	Construction	2013
Application of coatings and resins	Construction	2013
Other construction finishing works	Construction	2013
Foundational construction work	Construction	2013
Bakery and confectionery with predominance of resale	Retail and Motovehicles	2013
Retail trade of dairy and cold products	Retail and Motovehicles	2013
Retail sale of sweets, candies, bonbons and the like	Retail and Motovehicles	2013
Specialized retail trade of computer equipment	Retail and Motovehicles	2013
Refilling cartridges for computer equipment	Retail and Motovehicles	2013
Specialized retail trade of telephony	Retail and Motovehicles	2013
Specialized retail trade of home appliances	Retail and Motovehicles	2013
Furniture retail trade	Retail and Motovehicles	2013
Fabric retail trade	Retail and Motovehicles	2013
Retail sale of haberdashery items	Retail and Motovehicles	2013
Retail trade of bed, table and bath articles	Retail and Motovehicles	2013
Retail sale of upholstery, curtains and blinds	Retail and Motovehicles	2013
Retail sale of other household articles	Retail and Motovehicles	2013
Book retail trade	Retail and Motovehicles	2013

Retail trade of newspapers and magazines	Retail and Motovehicles	2013
Stationery retail trade	Retail and Motovehicles	2013
Retail sale of records, CDs, DVDs and tapes	Retail and Motovehicles	2013
Retail sale of toys and recreational items	Retail and Motovehicles	2013
Retail trade of sporting goods	Retail and Motovehicles	2013
Retail sale of cosmetics and perfumery	Retail and Motovehicles	2013
Retail sale of clothing and accessories	Retail and Motovehicles	2013
Footwear retail trade	Retail and Motovehicles	2013
Travel goods retail trade	Retail and Motovehicles	2013
Retail sale of household cleaning products	Retail and Motovehicles	2013
Retail sale of photographic and filming articles	Retail and Motovehicles	2013
Retail sale of pharmaceutical products	Retail and Motovehicles	2013
Construction of roads and railways	Construction	2014
Painting for signs on highways and airports	Construction	2014
Construction of special works of art	Construction	2014
Urbanization works - streets, squares and sidewalks	Construction	2014
Construction of dams for power generation	Construction	2014
Construction of stations and electricity networks	Construction	2014
Maintenance of electricity distribution networks	Construction	2014
Construction of stations and telecommunications	Construction	2014
Maintenance of stations and telecommunications	Construction	2014
Construction of water supply networks	Construction	2014
irrigation works	Construction	2014
Construction of pipeline transport networks, except for water and sewage	Construction	2014
Port, maritime and river works	Construction	2014
Assembly of metal structures	Construction	2014
Industrial assembly works	Construction	2014
Demolition of buildings and other structures	Construction	2014
Site preparation and land clearing	Construction	2014
Drilling and soundings	Construction	2014
Earthworks	Construction	2014
Land preparation services not otherwise specified	Construction	2014
Collective road transport of passengers (fixed route)	Transportation	2013
Collective road transport (fixed itinerary)	Transportation	2013
Collective road transport (metropolitan region)	Transportation	2013
Collective road transport of passengers (interstate)	Transportation	2013
Public passenger transport by road (international)	Transportation	2013
Transport by inland freight navigation	Transportation	2013
Transport by inland cargo navigation	Transportation	2013
Maritime cabotage transport - Cargo	Transportation	2013
Cabotage maritime transport - Passengers	Transportation	2013
Long haul maritime transport - Cargo	Transportation	2013
Long haul maritime transport - Passengers	Transportation	2013
Regular passenger air transport	Transportation	2013
air freight transport	Transportation	2013
Transport by inland navigation (municipal)	Transportation	2013
Transport by inland navigation (interstate)	Transportation	2013
Intercity and interstate passenger rail transport	Transportation	2014

Railway passenger transport in the city	Transportation	2014
Metro transport	Transportation	2014
Loading and unloading	Transportation	2014
Port infrastructure management	Transportation	2014
Port Operator Activities	Transportation	2014
Management of waterway terminals	Transportation	2014
Road freight transport (municipal)	Transportation	2014
Cargo road transport (interstate)	Transportation	2014
Road transport of dangerous goods	Transportation	2014
Road transport of removals	Transportation	2014
Rail freight transport	Transportation	2014
Newspaper printing	Media	2014
Printing of books, magazines and other periodicals	Media	2014
Book editing	Media	2014
Editing of daily newspapers	Media	2014
Editing of non-daily newspapers	Media	2014
Magazine editing	Media	2014
Integrated edition to print daily newspapers	Media	2014
Integrated edition to the printing of non-daily newspapers	Media	2014
Editing integrated with magazine printing	Media	2014
Radio activities	Media	2014
Open television activities	Media	2014
Portals, content providers and information services	Media	2014

Note: This table list the eligible sectors at its most granular sector definition (7-digits). The sector definition used in the tax bills are the CNAE classification administred by the Brazilian Census Bureau (IBGE). The table also reports the broader 1-digit industry for each eligible sector, and the year in which they gained eligibility.

Table D.3: Balance Test (Firm Level)

	OLS	**TWFE**
Employment	14.59	1.09
	$[-11.34, 40.52]$	$[-4.52, 6.69]$
Age	-1.24	0.07
	$[-2.72, 0.23]$	$[-0.09, 0.23]$
College +	0.02	-0.00
	$[-0.03, 0.07]$	$[-0.01, 0.00]$
Race	0.06	-0.00
	$[-0.00, 0.12]$	$[-0.01, 0.01]$
Gender	0.22	0.00
	$[0.13, 0.30]$	$[-0.00, 0.01]$
High School +	0.07	-0.00
	$[-0.03, 0.17]$	$[-0.01, 0.01]$
Firm FE	×	✓
Sector x Year FE	×	✓
N	1889754	1776214
Clusters	10516	9925

Note: This table reports the results of balance test for the firm level sample, which consists on regressing firms' characteristics on the time invariant eligibility dummy. The model is fitted in the pre-period (2008-2011), and the unit of observation is firm x year. The basline model is: $X_{jt} = L_{s(j)} + u_{jt}$, and the TWFE model: $X_{jt} = L_{s(j)} + \alpha_j + \gamma_t + \xi_{s1(j),t} + u_{jt}$, where $Ls(j)$ is a dummy to indicate if the firm was ever eligible; X_{jt} are firms' characteristics and the fixed effects are the same used in all firm level specifications presented before.The first column displays the results for the baseline (OLS) model. The second column reports the values for the TWFE. Standard errors are conservatively clustered at the 5-digit industry-by-state level.

Table D.4: Wage Effect for Poached vs Hired from Non-Employment

Log(Earnings)	**Poached**	**Non Employment**
All Sample	.032** (.013)	.021 (.013)
Controls	✓	✓
Firm FE	✓	✓
Sector x Year FE	✓	✓
# Clusters	7306	7678
N	890715	1282670

Note: This table reports the results from firm level regressions on earnings. First column reports the effect of average earnings for poached workers. Second column plots the effect to workers hired from non-employment. Hired from poaching or non-employment is defined based on the transition that can be observed at the worker level. The regressions are fit on a balanced sample of firms, and each observation is a firm x year. Standard errors are conservatively clustered at the 5-digit industry-by-state level.